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Wearable facial movement tracking devices

Abstract

This technology provides systems and methods for tracking facial movements and reconstructing facial expressions by learning skin deformation patterns and facial features. Frontal view images of a user making a variety of facial expressions are acquired to create a data training set for use in a machine-learning process. Head-mounted or neck-mounted wearable devices are equipped with one or more camera(s) or acoustic device(s) in communication with a data processing system. The cameras capture images of contours of the users face from either the cheekbone or the chin profile of the user. The acoustic devices transmit and receive signals to calculate a representation of the skin deformation. A data processing system uses the images, the profile of the contours, or skin deformation to track facial movement or to reconstruct facial expressions of the user based on the data training set.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC

8922481	12/2013	Kauffmann et al.	N/A	N/A
9672416	12/2016	Zhang et al.	N/A	N/A
10045737	12/2017	Tzvieli et al.	N/A	N/A
10078441	12/2017	Lee et al.	N/A	N/A
10217261	12/2018	Li et al.	N/A	N/A
10248852	12/2018	Kim et al.	N/A	N/A
10254842	12/2018	Kato et al.	N/A	N/A
10376153	12/2018	Tzvieli et al.	N/A	N/A
10614295	12/2019	Kim et al.	N/A	N/A
2012/0062719	12/2011	Debevec	348/E5.024	H04N 23/56
2015/0310263	12/2014	Zhang et al.	N/A	N/A
2017/0098122	12/2016	el Kaliouby et al.	N/A	N/A
2017/0351909	12/2016	Kaehler	N/A	G06V 40/171
2018/0004478	12/2017	Chen	N/A	A63F 13/26
2018/0027307	12/2017	Ni	345/419	H04N 21/44218
2018/0092549	12/2017	Tzvieli et al.	N/A	N/A
2018/0107275	12/2017	Chen et al.	N/A	N/A
2018/0158246	12/2017	Grau et al.	N/A	N/A
2018/0365484	12/2017	Yu	N/A	G06V 40/176
2019/0012528	12/2018	Wilson	N/A	G06F 3/012
2019/0285881	12/2018	Ilic et al.	N/A	N/A
2020/0322518	12/2019	Nagata	N/A	N/A
2021/0289134	12/2020	Athreya	N/A	G06T 3/4038
2024/0212388	12/2023	Li et al.	N/A	N/A

OTHER PUBLICATIONS

Extended European Search Report of European Patent Application Serial No. EP 21804664.7, Apr. 10, 2024, 11 pages. cited by applicant
Suzuki, et al., “Recognition and mapping of facial expressions to avatar by embedded photo reflective sensors in head mounted display,” 2017 IEEE Virtual R
pp. 177-185, doi: 10.1109/VR.2017.7892245. cited by applicant
A. Bulling et al., “It's in Your Eyes Towards Context—Awareness and Mobile HCI Using Wearable EOG Goggles,” Proceedings of the 10th International Con
Ubiquitous Computing, Sep. 2008, pp. 84-93. cited by applicant
A. Bulling et al., “Wearable EOG Goggles: Eye-Based Interaction in Everyday Environments,” CHI '09 Extended Abstracts on Human Factors in Computing
2009, pp. 3259-3264. cited by applicant
Cogain, “Eye Tracker Accuracy Terms and Definitions—working copy,” https://old.cogain.org/forums/eye-tracker-accuracy-and-precision-general-discussion/
accuracy-terms-and-definiti.html, Accessed Mar. 5, 2024, 1 page. cited by applicant
C. W. Cho et al., “Gaze Detection by Wearable Eye-Tracking and NIR LED-Based Head-Tracking Device Based on SVR,” ETRI Journal, vol. 34, No. 4, Aug.
552. cited by applicant
Github, “Analog Devices AI,” https://github.com/MaximIntegratedAI, Accessed Mar. 5, 2024, 3 pages. cited by applicant
Bitfoic Electronics, “OWR-05049T-38D Data Sheet,” https://www.bitfoic.com/detail/owr05049t38d-14578, Accessed Mar. 5, 2024, 5 pages. cited by applican
S. Dong et al., “GazBy: Gaze-Based BERT Model to Incorporate Human Attention in Neural Information Retrieval,” Proceedings of the 2022 ACM SIGIR In
Conference on Theory of Information Retrieval, Jul. 2022, 11 pages. cited by applicant
Hexagon, “Ergoneers Dikablis Glasses 3 Eye Tracker,” https://hexagondownloads.blob.core.windows.net/public/AutonomouStuff/wp-content/uploads/2019/1
glasses-3.pdf, Accessed Mar. 5, 2024, 2 pages. cited by applicant
CDC, “Niosh Sound Level Meter App,” https://www.cdc.gov/niosh/topics/noise/app.html, Jan. 25, 2023, 5 pages. cited by applicant
Gazerecorder, “Online Eye Tracking Software,” https://gazerecorder.com/, Accessed Mar. 5, 2024, 13 pages. cited by applicant
A. Golard et al., “Ultrasound for Gaze Estimation—A Modeling and Empirical Study,” Sensors, vol. 21, No. 4502, Jun. 30, 2021, 13 pages. cited by applicant
Brown HCI Group, “WebGazer.js: Democratizing Webcam Eye Tracking on the Browser” https://webgazer.cs.brown.edu/#publication, Accessed Mar. 5, 2024
by applicant
Imotions, “SMI Eye Tracking Glasses,” https://imotions.com/products/hardware/smi-eye-tracking-glasses/, Accessed Mar. 5, 2024, 7 pages. cited by applicant
M. Kassner et al., “Pupil: An Open Source Platform for Pervasive Eye Tracking and Mobile Gaze-based Interaction,” arXiv:1405.0006v1, Apr. 30, 2014, 10 p
applicant
K. Krafka et al., “Eye Tracking for Everyone,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2016, pp. 217
applicant
M. Kytö et al., “Pinpointing: Precise Head- and Eye-Based Target Selection for Augmented Reality,” Proceedings of the CHI Conference on Human Factors i
Systems, Paper No. 81, Apr. 2018, 14 pages. cited by applicant
Lowpowerlab, “Introduction Current Ranger,” https://lowpowerlab.com/guide/currentranger/, Sep. 19, 2018, 4 pages. cited by applicant
Pupil Labs, “Pupil Invisible, ” https://pupil-labs.com/products/, Accessed Mar. 5, 2024, 2 pages. cited by applicant
A. Lanata et al., “Robust Head Mounted Wearable Eye Tracking System for Dynamical Calibration,” Journal of Eye Movement Research, vol. 8, No. 5, Nov.
cited by applicant
Y. Lei, “Eye Tracking Calibration on Mobile Devices,” ETRA '21 Adjunct: ACM Symposium on Eye Tracking Research and Applications, May 2021, 4 pages
T. Li et al., “Ultra-Low Power Gaze Tracking for Virtual Reality,” SenSys '17: Proceedings of the 15th ACM Conference on Embedded Network Sensor Syste
pages. cited by applicant
T. Li et al., “Battery-Free Eye Tracker on Glasses,” Proceedings of the 10th Wireless of the Students, by the Students, and for the Students Workshop, Nov. 2,
cited by applicant
B. Mahanama, “Multi-User Eye-Tracking,” Symposium on Eye Tracking Research and Applications, Jun. 2022, 3 pages. cited by applicant
M. Matsubara et al., “Extraction of Read Text Using a Wearable Eye Tracker for Automatic Video Annotation,” UbiComp/ISWC'15 Adjunct: Adjunct Proceed
ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Comp
pp. 849-854. cited by applicant
A. Mayberry et al., “iShadow: Design of a Wearable, Real-Time Mobile Gaze Tracker,” Proceedings of the 12th Annual International Conference on Mobile S
Applications, and Services, Jun. 2014, Author Manuscript, 35 pages. cited by applicant
A. Mayberry et al., “CIDER: Enabling Robustness-Power Tradeoffs on a Computational Eyeglass,” Proceedings of the 21st Annual International Conference o
Computing and Networking, Sep. 2015, Author Manuscript, 36 pages. cited by applicant
CDC, “Occupational Noise Exposure,” https://www.cdc.gov/niosh/docs/98-126/default.html#print, Accessed Mar. 5, 2024, 1 page. cited by applicant
T. Ohno et al., “Just Blink Your Eyes: A Head-Free Gaze Tracking System,” CHI '03 Extended Abstracts on Human Factors in Computing Systems, Apr. 2003
applicant

L. Paletta et al., "Smartphone Eye Tracking Toolbox: Accurate Gaze Recovery on Mobile Displays," Proceedings of the Symposium on Eye Tracking Researc
Applications, Mar. 2014, 2 pages. cited by applicant
A. Papoutsaki et al., "WebGazer: Scalable Webcam Eye Tracking Using User Interactions," Proceedings of the Twenty-Fifth International Joint Conference or
Intelligence, Jul. 2016, 7 pages. cited by applicant
A. Papoutsaki et al., "SearchGazer: Webcam Eye Tracking for Remote Studies of Web Search," Proceedings of the 2017 Conference on Conference Human In
Interaction and Retrieval, Mar. 2017, pp. 17-26. cited by applicant
L Piccardi et al., "WearCam: A Head Mounted Wireless Camera for Monitoring Gaze Attention and for the Diagnosis of Developmental Disorders in Young C
IEEE International Symposium on Robot and Human Interactive Communication, Aug. 2007, 5 pages. cited by applicant
Pjrc, "Audio Adaptor Boards for Teensy 3.x and Teensy 4.x," https://www.pjrc.com/store/teensy3_audio.html, Accessed Mar. 6, 2024, 7 pages. cited by applic
Pjrc, "Teensy 4.1 Development Board," https://www.pjrc.com/store/teensy41.html, Accessed Mar. 6, 2024, 23 pages. cited by applicant
W. J. Ryan et al., "Limbus / Pupil Switching For Wearable Eye Tracking Under Variable Lighting Conditions," Proceedings of the 2008 symposium on Eye Tr
Applications, Mar. 2008, 4 pages. cited by applicant
S. Saxena et al., "Towards Efficient Calibration for Webcam Eye-Tracking in Online Experiments," Symposium on Eye Tracking Research and Applications, J
cited by applicant
RealEye, "RealEye Webcam Eye-Tracking," https://www.realeye.io/, Accessed Mar. 6, 2024, 8 pages. cited by applicant
C. Topal et al., "A Wearable Head-Mounted Sensor-Based Apparatus for Eye Tracking Applications," IEEE International Conference on Virtual Environments
Computer Interfaces, and Measurement Systems, Jul. 2008, 4 pages. cited by applicant
C. Topal et al., "A Head-Mounted Sensor-Based Eye Tracking Device: Eye Touch System," Proceedings of the 2008 Symposium on Eye Tracking Research &
Mar. 2008, 4 pages. cited by applicant
M. Vidal et al., "Wearable Eye Tracking for Mental Health Monitoring," Computer Communications, vol. 35, No. 11, Jun. 15, 2012, pp. 1306-1311. cited by a
Q. Wang et al., "Thermographic Eye Tracking," Proceedings of the Ninth Biennial ACM Symposium on Eye Tracking Research & Applications, Mar. 2016, p
by applicant
E. Whitmire et al., "EyeContact: Scleral Coil Eye Tracking for Virtual Reality," Proceedings of the 2016 ACM International Symposium on Wearable Comput
pages. cited by applicant
K. Wisiecka et al., "Comparison of Webcam and Remote Eye Tracking," 2022 Symposium on Eye Tracking Research and Applications, Jun. 2022, 7 pages. ci
Z. Ye et al., "Detecting Eye Contact using Wearable Eye-Tracking Glasses," Proceedings of the 2012 ACM Conference on Ubiquitous Computing, Sep. 2012,
applicant
Y. Zhang et al., "Discrimination of Gaze Directions Using Low-Level Eye Image Features," Proceedings of the 1st International Workshop on Pervasive Eye T
Eye-based Interaction, Sep. 2011, pp. 9-14. cited by applicant
A. Zhu et al., "MobiET: A New Approach to Eye Tracking for Mobile Device," Proceedings of the 2018 ACM International Joint Conference and 2018 Interna
on Pervasive and Ubiquitous Computing and Wearable Computers, Oct. 2018, 8 pages. cited by applicant
W. Sun et al., "VibroSense: Recognizing Home Activities by Deep Learning Subtle Vibrations on an Interior Surface of a House from a Single Point Using La
Vibrometry," Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 4, No. 3, Sep. 2020, pp. 96:1-96:28. cited by applic
F. Zhang et al., "MediaPipe Hands: On-device Real-time Hand Tracking," arXiv:2006.10214v1, Jun. 18, 2020, 5 pages. cited by applicant
H. Zhou et al., "One Ring to Rule Them All: An Open Source Smartring Platform for Finger Motion Analytics and Healthcare Applications," Proceedings of t
Conference on Internet of Things Design and Implementation, May 2023, pp. 27-38. cited by applicant
A.-I. Siean, "A Set of Smart Ring Gestures for Drone Control," The 12th International Conference on Electronics, Communications and Computing, Oct. 202
cited by applicant
Polytec Gmbh, "VibroGo," https://www.polytec.com/eu/vibrometry/products/single-point-vibrometers/vibrogo, Accessed Oct. 21, 2022, 23 pages. cited by ap
R. Abbaszadeh et al., "Prediction of Watermelon Quality Based on Vibration Spectrum," Postharvest Biology and Technology, vol. 86, Dec. 2013, pp. 291-29
applicant
J. D. Achenbach, "Wave Propagation in Elastic Solids," Elsevier, Jan. 1973, 18 pages. Description and Table of Contents Only. cited by applicant
J. R. M. Aerts et al., "Nonlinearity in Eardrum Vibration as a Function of Frequency and Sound Pressure," Hearing Research, vol. 263, May 2010, pp. 26-32.
Y. Avargel et al., "Speech Measurements Using a Laser Doppler Vibrometer Sensor: Application to Speech Enhancement," 2011 Joint Workshop on Hands-fre
Communication and Microphone Arrays, Jun. 2011, pp. 109-114. cited by applicant
D. Avrahami et al., "Below the Surface: Unobtrusive Activity Recognition for Work Surfaces using RF-radar Sensing," 23rd International Conference on Intel
Interfaces, Mar. 2018, pp. 439-451. cited by applicant
A. Bonde et al., "Demo Abstract: Deskbuddy: an Office Activity Detection System," Proceedings of the 18th International Conference on Information Process
Networks, Apr. 2019, pp. 352-353. cited by applicant
P. Castellini et al., "Laser Doppler Vibrometry: Development of Advanced Solutions Answering to Technology's Needs," Mechanical Systems and Signal Pro
Aug. 2006, pp. 1265-1285. cited by applicant
K.-Y. Chen et al., "DOSE: Detecting User-Driven Operating States of Electronic Devices from a Single Sensing Point," IEEE International Conference on Per
and Communications, Mar. 2015, pp. 46-54. cited by applicant
G. Cohn et al., "GasSense: Appliance-Level, Single-Point Sensing of Gas Activity in the Home," International Conference on Pervasive Computing, Apr. 201
cited by applicant
G. Cohn et al., "An Ultra-Low-Power Human Body Motion Sensor Using Static Electric Field Sensing," Proceedings of the 2012 ACM Conference on Ubiqu
Sep. 2012, pp. 99-102. cited by applicant
A. Davis et al., "Visual Vibrometry: Estimating Material Properties from Small Motions in Video," IEEE Transactions on Pattern Analysis and Machine Intell
4, Apr. 2017, pp. 732-745. cited by applicant
A. Davis et al., "The Visual Microphone: Passive Recovery of Sound from Video," ACM Transactions on Graphics, vol. 33, No. 4, Jul. 2014, 10 pages. cited b
J. Fagert et al., "Monitoring Hand-Washing Practices Using Structural Vibrations," Proceedings of Structural Health Monitoring, May 2017, 10 pages. cited by
J. Fagert et al., "Vibration Source Separation for Multiple People Gait Monitoring Using Footstep-Induced Floor Vibrations," Structural Health Monitoring, D
10.12783/shm2019/32338, Nov. 15, 2019, 1 page. Abstract Only. cited by applicant
J. Fogarty et al., "Sensing from the Basement: A Feasibility Study of Unobtrusive and Low-Cost Home Activity Recognition," Proceedings of the 19th Annua
on User Interface Software and Technology, Oct. 2006, pp. 91-100. cited by applicant
Y. Shen et al., "The Impact of Quantity of Training Data on Recognition of Eating Gestures," arXiv:1812.04513v1, Dec. 11, 2018, 7 pages. cited by applicant
D. Goyal et al., "The Vibration Monitoring Methods and Signal Processing Techniques for Structural Health Monitoring: A Review," Archives of Computation
Engineering, vol. 23, pp. 585-594. cited by applicant
T. Shiratori et al., "Motion Capture from Body-Mounted Cameras," ACM Transactions on Graphics, Jul. 25, 2011, 10 pages. cited by applicant
S. Stankoski et al., "Smartwatch-Based Eating Detection: Data Selection for Machine Learning from Imbalanced Data with Imperfect Labels," Sensors, vol. 2
2021, 25 pages. cited by applicant
A. M. Huber et al., "A New Implantable Middle Ear Hearing Device for Mixed Hearing Loss: A Feasibility Study in Human Temporal Bones," Otology & Ne
Dec. 2006, pp. 1104-1109. cited by applicant
A. M. Huber et al., "Evaluation of Eardrum Laser Doppler Interferometry as a Diagnostic Tool," The Laryngoscope, vol. 111, No. 3, Mar. 2001, pp. 501-507.
B. Tekin et al., "Fusing 2D Uncertainty and 3D Cues for Monocular Body Pose Estimation," arXiv:1611.05708v3, Apr. 10, 2017, 13 pages. cited by applicant

Y. Kim et al., “ViridiScope: Design and Implementation of a Fine Grained Power Monitoring System for Homes,” Proceedings of the 11th ACM International Ubiquitous Computing, Sep. 2009, pp. 245-254. cited by applicant
S. Kuznetsov et al., “UpStream: Motivating Water Conservation with Low-Cost Water Flow Sensing and Persuasive Displays,” Proceedings of the SIGCHI C Human Factors in Computing Systems, Apr. 2010, pp. 1851-1860. cited by applicant
T. He et al., “Bag of Tricks for Image Classification with Convolutional Neural Networks,” IEEE/CVF Conference on Computer Vision and Pattern Recogniti pp. 558-567. cited by applicant
E. Thomaz et al., “A Practical Approach for Recognizing Eating Moments with Wrist-Mounted Inertial Sensing,” In Proceedings of the ACM International Joi Pervasive and Ubiquitous Computing, Sep. 2015, pp. 1029-1040. cited by applicant
G. Laput et al., “Zensors: Adaptive, Rapidly Deployable, Human-Intelligent Sensor Feeds,” Proceedings of the 33rd Annual ACM Conference on Human Fact Systems, Apr. 2015, 10 pages. cited by applicant
E. Thomaz et al., “Feasibility of Identifying Eating Moments from First-Person Images Leveraging Human Computation,” Proceedings of the 4th Internationa Pervasive Imaging Conference, Nov. 2013, 8 pages. cited by applicant
H. Li et al., “IDSense: A Human Object Interaction Detection System Based on Passive UHF RFID,” Proceedings of the 33rd Annual ACM Conference on Hu Computing Systems, Apr. 2015, 10 pages. cited by applicant
S. Li et al., “WiFit: Ubiquitous Bodyweight Exercise Monitoring with Commodity Wi-Fi Devices,” IEEE SmartWorld, Ubiquitous Intelligence & Computing, Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation, Oct. 2018, pp. 53 applicant
X. Liu et al., “Wi-Sleep: Contactless Sleep Monitoring via WiFi Signals,” IEEE Real-Time Systems Symposium, Dec. 2014, pp. 346-355. cited by applicant
G. Luo et al., “Wavelet Denoising,” Chapter 4, Advances in Wavelet Theory and Their Applications in Engineering, Physics and Technology, Apr. 2012, 23 pa applicant
S. Mashiyama et al., “Activity Recognition Using Low Resolution Infrared Array Sensor,” IEEE International Conference on Communications, Jun. 2015, 6 p applicant
M. Mirshekari et al., “Human Gait Monitoring Using Footstep-Induced Floor Vibrations Across Different Structures,” Proceedings of the 2018 ACM Internati Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers, Oct. 2018, pp. 1382-1391. cited by applica
M. Mirshekari et al., “Obstruction-Invariant Occupant Localization Using Footstep-induced Structural Vibrations,” Mechanical Systems and Signal Processin 15, 2021, 19 pages. cited by applicant
M. Mirshekari et al., “Poster Abstract: Non-intrusive Occupant Localization Using Floor Vibrations in Dispersive Structure,” Proceedings of the 14th ACM C Embedded Network Sensor Systems, Nov. 2016, pp. 378-379. cited by applicant
P. G. G. Muyshondt et al., “A Single-Ossicle Ear: Acoustic Response and Mechanical Properties Measured in Duck,” Hearing Research, vol. 340, Oct. 2016, 1 applicant
K. Ohara et al., “Detecting State Changes Of Indoor Everyday Objects Using Wi-Fi Channel State Information,” Proceedings of the ACM on Interactive, Mob Ubiquitous Technologies, vol. 1, No. 3, Sep. 2017, 28 pages. cited by applicant
W. Ostachowicz et al., “50th Anniversary Article: Comparison Studies of Full Wavefield Signal Processing for Crack Detection,” Strain, vol. 50, Jun. 25, 201 cited by applicant
S. Pan et al., “Fine-Grained Recognition of Activities of Daily Living through Structural Vibration and Electrical Sensing,” Proceedings of the 6th ACM Inter Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Nov. 2019, pp. 149-158. cited by applicant
S. Pan et al., “FootprintID: Indoor Pedestrian Identification through Ambient Structural Vibration Sensing,” Proceedings of the ACM on Interactive, Mobile, V Ubiquitous Technologies, vol. 1, No. 3, Sep. 2017, 31 pages. cited by applicant
S. N. Patel et al., “Detecting Human Movement by Differential Air Pressure Sensing in HVAC System Ductwork: An Exploration in Infrastructure Mediated S International Conference on Pervasive Computing, May 2008, 18 pages. cited by applicant
S. N. Patel et al., “At the Flick of a Switch: Detecting and Classifying Unique Electrical Events on the Residential Power Line,” International Conference on U Computing, Sep. 2007, pp. 271-288. cited by applicant
Q. Pu et al., “Whole-Home Gesture Recognition Using Wireless Signals,” Proceedings of the 19th Annual International Conference on Mobile Computing & 2013, 13 pages. cited by applicant
J. Tompson et al., “Joint Training of a Convolutional Network and a Graphical Model for Human Pose Estimation,” arXiv:1406.2984v2, Sep. 17, 2014, 9 page applicant
N. B. Roozen et al., “Determining Radiated Sound Power of Building Structures by Means of Laser Doppler Vibrometry,” Journal of Sound and Vibration, vo 2015, 37 pages. cited by applicant
S. J. Rothberg et al., “An International Review of Laser Doppler Vibrometry: Making Light Work of Vibration Measurement,” Optics and Lasers in Engineeri 2017, 35 pages. cited by applicant
L. Shi et al., “Device-free Multiple People Localization through Floor Vibration,” Proceedings of the 1st ACM International Workshop on Device-Free Huma 2019, pp. 57-61. cited by applicant
L. Shi et al., “Data Quality-Informed Multiple Occupant Localization using Floor Vibration Sensing,” Proceedings of the 21st International Workshop on Mob Systems and Applications, Mar. 2020, p. 98. cited by applicant
K. Li et al., “GazeTrak: Exploring Acoustic-based Eye Tracking on a Glass Frame,” arXiv:2402.14634v2, Feb. 23, 2024, 16 pages. cited by applicant
R. Zhang et al., “EchoSpeech: Continuous Silent Speech Recognition on Minimally-obtrusive Eyewear Powered by Acoustic Sensing,” Proceedings of the 20 on Human Factors in Computing Systems, No. 852, Apr. 2023, 18 pages. cited by applicant
L. Billman et al., “Speech Reading with Deep Neural Networks,” Uppsala Universitet Department of Information Technology, Jul. 2018, 65 pages. cited by ap
N. Alghamdi et al., “A Corpus of Audio-Visual Lombard Speech with Frontal and Profile Views,” The Journal of the Acoustical Society of America, vol. 143, pp. EL523-EL529. cited by applicant
Y. M. Assael et al., “LipNet: Sentence-level Lipreading,” arXiv: 1611.01599v1, Nov. 5, 2016, 12 pages. cited by applicant
Y. M. Assael et al., “LipNet: End-to-End Sentence-level Lipreading,” arXiv:1611.01599v2, Dec. 16, 2016, 13 pages. cited by applicant
J. S. Chung et al., “Lip Reading in Profile,” British Machine Vision Conference, Sep. 2017, 11 pages. cited by applicant
J. S. Chung et al., “Lip Reading in the Wild,” Asian Conference on Computer Vision, Mar. 2017, 17 pages. cited by applicant
M. Cooke et al., “An Audio-Visual Corpus for Speech Perception and Automatic Speech Recognition,” The Journal of the Acoustical Society of America, vol. 2006, pp. 2421-2424. cited by applicant
T. Le Cornu et al., “Reconstructing Intelligible Audio Speech from Visual Speech Feature,” 16th Annual Conference of the International Speech Communicati Sep. 2015, 6 pages. cited by applicant
A. Ephrat et al., “Vid2Speech: Speech Reconstruction from Silent Video,” arXiv:1701.00495v2, Jan. 9, 2017, 5 pages. cited by applicant
I. Fung et al., “End-to-End Low-Resource Lip-Reading with Maxout CNN and LSTM,” IEEE International Conference on Acoustics, Speech and Signal Proc Sep. 2018, 5 pages. cited by applicant
Y. Gao et al., “EchoWhisper: Exploring an Acoustic-Based Silent Speech Interface for Smart-phone Users,” Proceedings of the ACM on Interactive, Mobile, V Ubiquitous Technologies, vol. 4, No. 3, Sep. 2020, pp. 80:1-80:27. cited by applicant
Y. Jin et al., “EarCommand: “Hearing” Your Silent Speech Commands in Ear,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Tec No. 2, Jun. 2022, pp. 57:1-57:28. cited by applicant

A. Koumparoulis et al., “Exploring ROI Size in Deep Learning Based Lipreading,” Auditory Visual Speech Processing Conference Proceedings, Aug. 25, 201
applicant
Y. Kunimi et al., “E-Mask: A Mask-Shaped Interface for Silent Speech Interaction with Flexible Strain Sensors,” Augmented Humans, Mar. 2022, pp. 26-34. c
J. Luo et al., “End-To-End Silent Speech Recognition with Acoustic Sensing,” arXiv:2011.11315v1, Nov. 23, 2020, 7 pages. cited by applicant
D. Michelsanti et al., “Vocoder-Based Speech Synthesis from Silent Videos,” arXiv:2004.02541v2, Aug. 15, 2020, 5 pages. cited by applicant
R. Mira et al., “End-to-End Video-to-Speech Synthesis Using Generative Adversarial Networks,” arXiv:2104.13332v3, Aug. 15, 2022, 12 pages. cited by appl
L. Pandey et al., “LipType: A Silent Speech Recognizer Augmented with an Independent Repair Model,” ACM CHI Virtual Conference on Human Factors in
Systems, May 2021, 19 pages. cited by applicant
L. Pandey et al., “Acceptability of Speech and Silent Speech Input Methods in Private and Public,” ACM CHI Virtual Conference on Human Factors in Comp
May 2021, 13 pages. cited by applicant
L. Pandey et al., “Silent Speech and Emotion Recognition from Vocal Tract Shape Dynamics in Real-Time MRI,” arXiv:2106.08706v1, Jun. 16, 2021, 8 pages
applicant
E. K. Patterson et al., “Cuave: A New Audio-Visual Database for Multimodal Human-Computer Interface,” IEEE International Conference on Acoustics, Spee
Processing, May 2002, 4 pages. cited by applicant
A. Rekik et al., “A New Visual Speech Recognition Approach for RGB-D Cameras,” International Conference Image Analysis and Recognition, Oct. 2014, 8 p
applicant
C. Rzepka, “Examining the Use of Voice Assistants: A Value-Focused Thinking Approach,” Twenty-fifth Americas Conference on Information Systems, Aug.
cited by applicant
G. Saon et al., “English Conversational Telephone Speech Recognition by Humans and Machines,” arXiv:1703.02136v1, Mar. 6, 2017, 7 pages. cited by appli
K. Sun et al., “Lip-Interact: Improving Mobile Device Interaction with Silent Speech Commands,” Proceedings of the 31st Annual ACM Symposium on User
and Technology, Oct. 2018, 13 pages. cited by applicant
H. Akbari et al., “Lip2Audspec: Speech Reconstruction from Silent Lip Movements Video,” arXiv:1710.09798v1, Oct. 26, 2017, 9 pages. cited by applicant
A. Thanda et al., “Audio Visual Speech Recognition using Deep Recurrent Neural Networks,” arXiv: 1611.02879v1, Nov. 9, 2016, 12 pages. cited by applican
D. Vorontsova et al., “Silent EEG-Speech Recognition Using Convolutional and Recurrent Neural Network with 85% Accuracy of 9 Words Classification,” Se
6744, Oct. 11, 2021, 19 pages. cited by applicant
K. Vougioukas et al., “Video-Driven Speech Reconstruction using Generative Adversarial Networks,” arXiv:1906.06301v1, Jun. 14, 2019, 6 pages. cited by ap
M. Wand et al., “Lipreading with Long Short-Term Memory,” arXiv:1601.08188v1, Jan. 29, 2016, 5 pages. cited by applicant
J. Wang et al., “RFID Tattoo: A Wireless Platform for Speech Recognition,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Techno
4, Dec. 2019, pp. 155:1-155:24. cited by applicant
Z. Wang et al., “ToothSonic: Earable Authentication via Acoustic Toothprint,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Tech
No. 2, Jun. 2022, pp. 78:1-78:24. cited by applicant
P. Weber et al., “(Non-)Interacting with Conversational Agents: Perceptions and Motivations of Using Chatbots and Voice Assistants,” Proceedings of Mensch
Sep. 2020, pp. 321-331. cited by applicant
W. Xiong et al., “Achieving Human Parity in Conversational Speech Recognition,” arXiv:1610.05256v2, Feb. 17, 2017, 13 pages. cited by applicant
K. Xu et al., “LCANet: End-to-End Lipreading with Cascaded Attention—CTC,” arXiv:1803.04988v1, Mar. 13, 20218, 10 pages. cited by applicant
S. Yang et al., “LRW-1000: A Naturally-Distributed Large-Scale Benchmark for Lip Reading in the Wild,” arXiv:1810.06990v6, Apr. 24, 2019, 8 pages. cited
Q. Zhang et al., “SoundLip: Enabling Word and Sentence-Level Lip Interaction for Smart Devices,” Proceedings of the ACM on Interactive, Mobile, Wearabl
Technologies, vol. 5, No. 21, Mar. 2021, pp. 44:1-44:28. cited by applicant
G. Zhao et al., “Lipreading With Local Spatiotemporal Descriptors,” IEEE Transactions on Multimedica, vol. 11, No. 7, Nov. 2009, pp. 1254-1265. cited by ap
K. Suzuki et al., “Recognition and Mapping of Facial Expressions to Avatar by Embedded Photo Reflective Sensors in Head Mounted Display,” IEEE Virtual
pp. 177-185. cited by applicant
H.-Y. Lai, “Tracking of Eye Movement Features for Individualized Assessment of Neurocognitive State Using Mobile Devices,” Doctor of Philosophy Thesis,
Institute of Technology, Sep. 2021, 150 pages. cited by applicant
S. F. W. Neggers et al., “Ocular Gaze Is Anchored to the Target of an Ongoing Pointing Movement,” Neurophysiology, vol. 83, No. 2, Feb. 2000, pp. 639-651.
Tobii, “Tobii Pro Fusion,” https://www.tobiipro.com/product-listing/fusion/, Accessed Mar. 5, 2024, 16 pages. cited by applicant
Tobii, “Tobii Pro Glasses 3,” https://www.tobiipro.com/product-listing/tobii-pro-glasses-3/, Accessed Mar. 5, 2024, 12 pages. cited by applicant
A. Ablavatski et al., “Real-time Pupil Tracking from Monocular Video for Digital Puppetry,” arXiv:2006.11341v1, Jun. 19, 2020, 4 pages. cited by applicant
C. Ahlstrom et al., “Comparison of Eye Tracking Systems with One and Three Cameras,” 7th International Conference on Methods and Techniques in Behavi
Aug. 2010, pp. 196-199. cited by applicant
Github, “MAX7800x Power Monitor and Energy Benchmarking Guide,”
https://github.com/MaximIntegratedAI/MaximAI_Documentation/blob/master/Guides/MAX7800x%20Power%20Monitor%20and%20Energy%20Benchmar
Accessed Mar. 5, 2024, 11 pages. cited by applicant
T. Bafna et al., “EyeTell: Tablet-based Calibration-free Eye-typing using Smooth-pursuit Movements,” ETRA '21 Short Papers: ACM Symposium on Eye Trac
Applications, May 2021, 6 pages. cited by applicant
F. H. Borsato et al., “Episcleral Surface Tracking: Challenges and Possibilities for Using Mice Sensors for Wearable Eye Tracking,” Proceedings of the Ninth
Symposium on Eye Tracking Research & Applications, Mar. 2016, pp. 39-46. cited by applicant
S. Mahmud et al., “ActSonic: Recognizing Everyday Activities from Inaudible Acoustic Waves Around the Body,” arXiv:2404.13924v2, May 8, 2024, 29 pag
applicant
C.-J. Lee et al., “EchoWrist: Continuous Hand Pose Tracking and Hand-Object Interaction Recognition Using Low-Power Active Acoustic Sensing on a Wrist
arXiv:2401.17409v2, Mar. 29, 2024, 21 pages. cited by applicant
S. Mahmud et al., “PoseSonic: 3D Upper Body Pose Estimation Through Egocentric Acoustic Sensing on Smartglasses,” Proceedings of the ACM on Interact
Wearable and Ubiquitous Technologies, vol. 7, No. 3, Sep. 2023, pp. 111:1-111:28. cited by applicant
K. Li et al., “EyeEcho: Continuous and Low-power Facial Expression Tracking on Glasses,” arXiv:2402.12388v2, Feb. 23, 2024, 24 pages. cited by applicant
R. Zhang et al., “HPSpeech: Silent Speech Interface for Commodity Headphones,” Proceedings of the 2023 ACM International Symposium on Wearable Com
pp. 60-65. cited by applicant
T. C. Yu et al., “Ring-a-Pose: A Ring for Continuous Hand Pose Tracking,” arXiv:2404.12980v1, Apr. 19, 2024, 26 pages. cited by applicant
International Search Report of PCT/US2021/032511, dated Aug. 13, 2021, one page. cited by applicant
H. Li et al., “Wristwash: Towards Automatic Handwashing Assessment Using a Wrist-worn Device,” Proceedings of the 2018 ACM International Symposium
Computers, Oct. 2018, pp. 132-139. cited by applicant
R. K. Kramer et al., “Soft Curvature Sensors for Joint Angle Proprioception,” IEEE/RSJ International Conference on Intelligent Robots and Systems, Sep. 201
by applicant
J. J. Gibson, “The Theory of Affordances,” Chapter 8, The Ecological Approach to Visual Perception, Hillsdale, Jan. 1, 1979, 15 pages. cited by applicant
Go Pro, “Hero 12 Black,” https://gopro.com/en/us/shop/cameras, Accessed Jul. 18, 2024, 10 pages. cited by applicant
Y. Guan et al., “Ensembles of Deep LSTM Learners for Activity Recognition using Wearables,” arXiv:1703.09370v1, Mar. 28, 2017, 28 pages. cited by applic
S. Gupta et al., “ElectriSense: Single-Point Sensing Using EMI for Electrical Event Detection and Classification in the Home,” Proceedings of the 12th ACM

Conference on Ubiquitous Computing, Sep. 2010, pp. 139-148. cited by applicant
N. Y. Hammerla et al., “Deep, Convolutional, and Recurrent Models for Human Activity Recognition Using Wearables,” arXiv: 1604.08880v1, Apr. 29, 2016,
applicant
H. Haresamudram et al., “On the Role of Features in Human Activity Recognition,” Proceedings of the 2019 ACM International Symposium on Wearable Co
pp. 78-88. cited by applicant
H. Haresamudram et al., “Masked Reconstruction Based Self-Supervision for Human Activity Recognition,” Proceedings of the 2020 ACM International Sym
Wearable Computers, Sep. 2020, pp. 45-49. cited by applicant
H. Haresamudram et al., “Contrastive Predictive Coding for Human Activity Recognition,” arXiv:2012.05333v1, Dec. 9, 2020, 23 pages. cited by applicant
K. He et al., “Deep Residual Learning for Image Recognition,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, arXiv:1512
10, 2015, 12 pages. cited by applicant
D.-A. Huang et al., “Connectionist Temporal Modeling for Weakly Supervised Action Labeling,” arXiv:1607.08584v1, Jul. 28, 2016, 17 pages. cited by appli
S. Ioffe et al., “Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift,” arXiv:1502.03167v3, Mar. 2, 2015, 11 page
applicant
Y. Iravantchi et al., “PrivacyMic: Utilizing Inaudible Frequencies for Privacy Preserving Daily Activity Recognition,” Proceedings of the 2021 CHI Conferen
Factors in Computing Systems, May 2021, No. 198, 13 pages. cited by applicant
Y. Iravantchi et al., “Interferi: Gesture Sensing using On-Body Acoustic Interferometry,” CHI Conference on Human Factors in Computing Systems, No. 276,
pages. cited by applicant
M. Kijima et al., “Poster Abstract: Multiple Door Opening/Closing Detection System Using Infrasound Sensor,” 17th ACM/IEEE International Conference on
Processing in Sensor Networks (IPSN), Apr. 2018, pp. 126-127. cited by applicant
D. Kim et al., “Detector-Free Weakly Supervised Group Activity Recognition,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),
20083-20093. cited by applicant
D. P. Kingma et al., “Adam: A Method for Stochastic Optimization,” arXiv:1412.6980v9, Jan. 30, 2017, 15 pages. cited by applicant
A. Kukleva et al., “Unsupervised Learning of Action Classes with Continuous Temporal Embedding,” Computer Vision and Pattern, Apr. 8, 2019, pp. 12066-1
applicant
J. R. Kwapisz et al., “Activity Recognition using Cell Phone Accelerometers,” ACM SIGKDD Explorations Newsletter, vol. 12, No. 2, Mar. 31, 2011, pp. 74-
applicant
H. Kwon et al., “Adding Structural Characteristics to Distribution-Based Accelerometer Representations for Activity Recognition Using Wearables, ” Proceed
ACM International Symposium on Wearable Computers, Oct. 2018, pp. 72-75. cited by applicant
N. D. Lane et al., “DeepEar: Robust Smartphone Audio Sensing in Unconstrained Acoustic Environments using Deep Learning,” Proceedings of the 2015 AC
Joint Conference on Pervasive and Ubiquitous Computing, Sep. 2015, pp. 283-294. cited by applicant
G. Laput et al., “Ubicoustics: Plug-and-Play Acoustic Activity Recognition,” Proceedings of the 31st Annual ACM Symposium on User Interface Software an
2018, pp. 213-224. cited by applicant
G. Laput et al., “Sensing Fine-Grained Hand Activity with Smartwatches,” Proceedings of the 2019 CHI Conference on Human Factors in Computing System
338, 13 pages. cited by applicant
G. Laput et al., “ViBand: High-Fidelity Bio-Acoustic Sensing Using Commodity Smartwatch Accelerometers,” Proceedings of the 29th Annual Symposium o
Software and Technology, Oct. 16, 2016, 13 pages. cited by applicant
G. Laput et al., “Synthetic Sensors: Towards General-Purpose Sensing,” Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems,
pages. cited by applicant
C. Lea et al., “Temporal Convolutional Networks for Action Segmentation and Detection,” IEEE Conference on Computer Vision and Pattern Recognition (C
pp. 156-165. cited by applicant
B. Li et al., “Extraction and Interpretation of Deep Autoencoder-based Temporal Features from Wearables for Forecasting Personalized Mood, Health, and Str
of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 4, No. 2, Jun. 2020, pp. 49:1-49:26. cited by applicant
K. Li et al., “EarlO: A Low-power Acoustic Sensing Earable for Continuously Tracking Detailed Facial Movements,” Proceedings of the ACM on Interactive,
and Ubiquitous Technologies, vol. 6, No. 2, Jun. 2022, pp. 62:1-62:24. cited by applicant
D. Liang et al., “Audio-Based Activities of Daily Living (ADL) Recognition with Large-Scale Acoustic Embeddings from Online Videos,” Proceedings of the
Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 3, No. 1, Mar. 2019, pp. 17:1-17:18. cited by applicant
T.-Y. Lin et al., “Focal Loss for Dense Object Detection,” IEEE Transactions on Pattern Analysis and Machine Intelligence, Jul. 2018, pp. 2980-2988. cited by
D. Liu et al., “Diffusion Action Segmentation,” arXiv:2303.17959v2. Aug. 12, 2023, 15 pages. cited by applicant
P. Lukowicz et al., “Recognizing Workshop Activity Using Body Worn Microphones and Accelerometers,” Pervasive Computing: Proceedings of the 2nd Inte
Conference, Apr. 2004, pp. 18-32. cited by applicant
H. Ma et al., “AttnSense: Multi-level Attention Mechanism For Multimodal Human Activity Recognition,” Proceedings of the Twenty-Eighth International Jo
Artificial Intelligence, Aug. 2019, pp. 3109-3115. cited by applicant
S. Mahmud et al., “Human Activity Recognition from Wearable Sensor Data Using Self-Attention,” European Conference on Artificial Intelligence, Mar. 17,
1339. cited by applicant
J. Meyer et al., “A CNN-based Human Activity Recognition System Combining a Laser Feedback Interferometry Eye Movement Sensor and an IMU for Con
Glasses,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 5, No. 4, Dec. 2021, pp. 172:1-172:24. cited by applica
V. Mollyn et al., “SAMoSA: Sensing Activities with Motion and Subsampled Audio,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquito
vol. 6, No. 3, Sep. 2022, pp. 132:1-132:19. cited by applicant
D. B. Moyano et al., “Possible Effects on Health of Ultrasound Exposure, Risk Factors in the Work Environment and Occupational Safety Review,” Healthca
2022, 19 pages. cited by applicant
V. S. Murahari et al., “On Attention Models for Human Activity Recognition,” arXiv:1805.07648v1, May 19, 2018, 4 pages. cited by applicant
W. J. Murphy et al., “Revisiting the NIOSH Criteria for a Recommended Standard: Occupational Noise Exposure,” The Journal of the Acoustical Society of A
No. 5, May 2002, 2 pages, Abstract Only. cited by applicant
R. Nandakumar et al., “Contactless Sleep Apnea Detection on Smartphones,” Proceedings of the 13th Annual International Conference on Mobile Systems, A
Services, May 2015, 13 pages. cited by applicant
T. Ohkawa et al., “AssemblyHands: Towards Egocentric Activity Understanding via 3D Hand Pose Estimation,” Conference on Computer Vision and Pattern
2023, pp. 12999-13008. cited by applicant
F. J. Ordonez et al., “Deep Convolutional and LSTM Recurrent Neural Networks for MultimodalWearable Activity Recognition,” Sensors, vol. 16, No. 115, J
pages. cited by applicant
A. Waghmare et al., “Z-Pose: Continuous 3D Hand Pose Tracking Using Single-Point Bio-Impedance Sensing on a Ring,” SmartWear '23: Proceedings of the
Smart Wearable Systems and Applications, Oct. 2023, 6 pages. cited by applicant
M. Wilhelm et al., “PeriSense: Ring-Based Multi-Finger Gesture Interaction Utilizing Capacitive Proximity Sensing,” Sensors, vol. 20, No. 3990, Jul. 17, 202
by applicant
C. Zhang et al., “FingOrbits: Interaction with Wearables Using Synchronized Thumb Movements,” ACM International Symposium on Wearable Computers, S
65. cited by applicant

Apple Inc., “Apple Vision Pro,” https://www.apple.com/apple-vision-pro/, Accessed Oct. 18, 2024, 8 pages. cited by applicant

Apple Inc., “Apple Watch Series 10,” https://www.apple.com/shop/buy-watch/apple-watch, Accessed Oct. 18, 2024, 11 pages. cited by applicant

Y. Bi et al., “Autodietary: A Wearable Acoustic Sensor System for Food Intake Recognition in Daily Life,” IEEE Sensors Journal, vol. 16, No. 3, Feb. 1, 2016 by applicant

R. Boldu et al., “Thumb-in-Motion: Evaluating Thumb-to-Ring Microgestures for Athletic Activity,” Proceedings of the 2018 ACM Symposium on Spatial Us 2018, pp. 150-157. cited by applicant

G. Cao et al., “EarphoneTrack: Involving Earphones into the Ecosystem of Acoustic Motion Tracking,” Proceedings of the 18th Conference on Embedded Ne Systems, Nov. 2020, 14 pages. cited by applicant

E. Chan et al., “User Elicitation on Single-hand Microgestures,” Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems, May 201 cited by applicant

L. Chan et al., “CyclopsRing: Enabling Whole-Hand and Context-Aware Interactions Through a Fisheye Ring,” Proceedings of the 28th Annual ACM Sympos Interface Software & Technology, Nov. 5, 2015, 8 pages. cited by applicant

L. Chan et al., “FingerPad: Private and Subtle Interaction Using Fingertips,” Proceedings of the 26th Annual ACM Symposium on User Interface Software an 2013, 6 pages. cited by applicant

T. Chen et al., “EFRing: Enabling Thumb-to-Index-Finger Microgesture Interaction through Electric Field Sensing Using Single Smart Ring,” Proceedings of Interactive, Mobile, Wearable and Ubiquitous Technologies, Jan. 2023, vol. 6, No. 4, pp. 161:1-161:31. cited by applicant

J. Connolly et al., “IMU Sensor-Based Electronic Goniometric Glove for Clinical Finger Movement Analysis,” IEEE Sensors Journal, vol. 18, No. 3, Feb. 1, 2 by applicant

B.-F. Gheran et al., “Gestures for Smart Rings: Empirical Results, Insights, and Design Implications,” Proceedings of the 2018 Designing Interactive Systems 2018, pp. 623-635. cited by applicant

O. Glauser et al., “Interactive Hand Pose Estimation using a Stretch-Sensing Soft Glove,” ACM Transactions on Graphics (TOG), vol. 38, No. 4, Jul. 2019, 15 applicant

J. Gong et al., “Pyro: Thumb-Tip Gesture Recognition Using Pyroelectric Infrared Sensing,” Proceedings of the 30th Annual ACM Symposium on User Interf Technology, Oct. 2017, pp. 553-563. cited by applicant

Y. Gu et al., “QwertyRing: Text Entry on Physical Surfaces Using a Ring,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technol 4, Aug. 2020, pp. 128:1-128:29. cited by applicant

J. Gummeson et al., “An Energy Harvesting Wearable Ring Platform for Gesture Input on Surfaces,” Proceedings of the 12th Annual International Conference systems, Applications, and Services, Jun. 2014, 14 pages. cited by applicant

A. Handa et al., “DexPilot: Vision Based Teleoperation of Dexterous Robotic Hand-Arm System,” arXiv:1910.03135v2, Oct. 14, 2019, 18 pages. cited by app

Y. Jian et al., “HandAvatar: Embodying Non-Humanoid Virtual Avatars through Hands,” Proceedings of the 2023 CHI Conference on Human Factors in Comp Apr. 2023, 17 pages. cited by applicant

L. Jing et al., “Magic Ring: A Self-contained Gesture Input Device on Finger,” Proceedings of the 12th International Conference on Mobile and Ubiquitous M 2013, 4 pages. cited by applicant

W. Kienzle et al., “ElectroRing: Subtle Pinch and Touch Detection with a Ring,” Proceedings of the 2021 CHI Conference on Human Factors in Computing S 12 pages. cited by applicant

Meta, “This is Meta Quest,” https://www.meta.com/quest/, Accessed Oct. 22, 2024, 17 pages. cited by applicant

S. Nanayakkara et al., “EyeRing: A Finger-Worn Input Device for Seamless Interactions with our Surroundings,” Proceedings of the 4th Augmented Human I Conference, Mar. 2013, 8 pages. cited by applicant

V. Nguyen et al., “HandSense: Capacitive coupling-based Dynamic, Micro Finger Gesture Recognition,” Proceedings of the 17th Conference on Embedded N Systems, Nov. 2019, 13 pages. cited by applicant

S. Nirjon et al., “TypingRing: A Wearable Ring Platform for Text Input,” Proceedings of the 13th Annual International Conference on Mobile Systems, Applic Services, Mar. 2015, 14 pages. cited by applicant

M. Ogata et al., “iRing: Intelligent Ring Using Infrared Reflection,” Proceedings of the 25th Annual ACM Symposium on User Interface Software and Techno pp. 131-136. cited by applicant

Oura Health Oy, “Oura Ring,” https://ouraring.com, Accessed Oct. 23, 2024, 7 pages. cited by applicant

S. Pei et al., “Hand Interfaces: Using Hands to Imitate Objects in AR/VR for Expressive Interactions,” Proceedings of the 2022 CHI Conference on Human Fa Computing Systems, Apr. 2022, 16 pages. cited by applicant

M. Poongodi et al., “Diagnosis and Combating COVID-19 Using Wearable Oura Smart Ring with Deep Learning Methods,” Personal and Ubiquitous Compu pages. cited by applicant

F. Chen et al., “Audio Privacy: Reducing Speech Intelligibility while Preserving Environmental Sounds,” Proceedings of the 16th ACM international confere Oct. 2008, pp. 733-736. cited by applicant

Analog Devices, Inc. “MAX78002 Evaluation Kit,” https://www.analog.com/media/en/technical-documentation/data-sheets/MAX78002EVKIT.pdf, May 202 by applicant

H. Duan et al., “Revisiting Skeleton-based Action Recognition,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2022, pp. by applicant

P. F. Edemekong et al., “Activities of Daily Living,” StatPearls, https://digitalcollections.dordt.edu/faculty_work/1222, Jun. 26, 2020, 12 pages. cited by applic

Y. A. Farha et al., “MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation,” IEEE/CVF Conference on Computer Vision and Patte (CVPR), Jun. 2019, pp. 3575-3584. cited by applicant

J. Froehlich et al., “HydroSense: Infrastructure-Mediated Single-Point Sensing of Whole-Home Water Activity,” Proceedings of the 11th International Confere Computing, Sep. 2009, pp. 235-244. cited by applicant

C. Qian et al., “Realtime and Robust Hand Tracking from Depth,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Jun. 201 by applicant

Ringly, “Ringly, Smart Jewelry and Accessories,” https://ringly.com/, Accessed Oct. 23, 2024, 3 pages. cited by applicant

McLEAR Ltd., “RingPay—McLEAR,” https://mclear.com/product/payment-ring/., Accessed Oct. 23, 2024, 7 pages. cited by applicant

J. Romero et al., “Embodied Hands: Modeling and Capturing Hands and Bodies Together,” ACM Transactions on Graphics, arXiv:2201.02610v1, Jan. 7, 2022 by applicant

S. Sen et al., “VibeRing: Using Vibrations from a Smart Ring as an Out-of-band Channel for Sharing Secret Keys,” Pervasive and Mobile Computing, https://www.sciencedirect.com/science/article/pii/S1574119221001309, Dec. 2021, 37 pages. cited by applicant

A. Sharma et al., “SparseIMU: Computational Design of Sparse IMU Layouts for Sensing Fine-Grained Finger Microgestures,” ACM Transactions on Compu Interaction, vol. 30, No. 3, Jun. 2023, pp. 39:1-39:40. cited by applicant

T. Starner et al., “The Gesture Pendant: A Self-illuminating, Wearable, Infrared Computer Vision System for Home Automation Control and Medical Monitori International Symposium on Wearable Computers, Oct. 2000, 8 pages. cited by applicant

D. J. Sturman et al., “A Survey of Glove-based Input,” IEEE Computer Graphics and Applications, Jan. 1994, pp. 30-39. cited by applicant

R.-D. Vatavu, “iFAD Gestures: Understanding Users' Gesture Input Performance with Index-Finger Augmentation Devices,” Proceedings of the 2023 CHI Co Human Factors in Computing Systems, Apr. 2023, 17 pages. cited by applicant

Vicon Motion Systems Ltd., “Award Winning Motion Capture Systems,” https://www.vicon.com/, Accessed Oct. 23, 2024, 4 pages. cited by applicant
A. Wang et al., “MilliSonic: Pushing the Limits of Acoustic Motion Tracking,” arXiv: 1901.06601v1, Jan. 20, 2019, 11 pages. cited by applicant
S. Wange et al., “Interacting with Soli: Exploring Fine-Grained Dynamic Gesture Recognition in the Radio-Frequency Spectrum,” Proceedings of the 29th An on User Interface Software and Technology, Oct. 2016, pp. 851-860. cited by applicant
K. Yatani et al., “BodyScope: A Wearable Acoustic Sensor for Activity Recognition,” Proceedings of the 2012 ACM Conference on Ubiquitous Computing, S 350. cited by applicant
B. Zhang et al., “A Ring-Shaped Interactive Device for Large Remote Display and Mobile Device Control,” Proceedings of the 13th International Conference Computing, Sep. 2011, 2 pages. cited by applicant
Amazon, “SanDisk 32GB Extreme for Mobile Gaming microSD UHS-I Card—C10, U3, V30, 4K, A1, Micro SD—SDSQXAF-032G-GN6GN,” https://www.amazon.com/SanDisk-Extreme-Mobile-Gaming-microSD/dp/B089M5KV4Y/2C66&sr=8-1&th=1, Accessed Oct. 30, 2024, 9 pages. cited by app
C. A. Kardous et al., “So How Accurate Are These Smartphone Sound Measurement Apps?” https://blogs.cdc.gov/niosh-science-blog/2014/04/09/sound-apps, pages. cited by applicant
Android Developers, “Profile Your App Performance,” https://developer.android.com/studio/profile, Acessed Oct. 30, 2024, 5 pages. cited by applicant
Humane, “Ai Pin,” https://humane.com/, Accessed Oct. 30, 2024, 23 pages. cited by applicant
D. Kimm et al., “Hand Speed Measurements in Boxing,” Procedia Engineering, vol. 112, Sep. 2015, pp. 502-506. cited by applicant
H. Lim et al., “C-Auth: Exploring the Feasibility of Using Egocentric View of Face Contour for User Authentication on Glasses,” Proceedings of the 2023 AC Symposium on Wearable Computers, Oct. 2023, pp. 6-10. cited by applicant
M. Loper et al., “SMPL: A Skinned Multi-Person,” ACM Transactions on Graphics (TOG), Oct. 2015, 16 pages. cited by applicant
Maxim Integrated, “MAX98357A/ MAX98357B Tiny, Low-Cost, PCM Class D Amplifier with Class AB Performance,” https://www.analog.com/media/en/te documentation/data-sheets/MAX98357A-MAX98357B.pdf, Jul. 2019, 38 pages. cited by applicant
Nordic Semiconductor, “nRF52840 Multiprotocol Bluetooth 5.4 SoC supporting Bluetooth Low Energy, Bluetooth mesh, NFC, Thread and Zigbee,” https://www.nordicsemi.com/products/nrf52840, Accessed Oct. 30, 2024, 2 pages. cited by applicant
SG Wireless, “SGW111X BLE Modules,” http://www.sgwireless.com, Accessed Oct. 30, 2024, 2 pages. cited by applicant
J. Shin et al., “MyDJ: Sensing Food Intakes with an Attachable on Your Eyeglass Frame,” Proceedings of the 2022 CHI Conference on Human Factors in Con Apr. 2022, 17 pages. cited by applicant
R. Sun et al., “EchoNose: Sensing Mouth, Breathing and Tongue Gestures inside Oral Cavity using a Non-contact Nose Interface,” Proceedings of the 2023 A Symposium on Wearable Computers, Oct. 2023, pp. 22-26. cited by applicant
W. Sun et al., “TeethTap: Recognizing Discrete Teeth Gestures Using Motion and Acoustic Sensing on an Earpiece,” 26th International Conference on Intellig Interfaces, arXiv:2102.12548v1, Feb. 24, 2021, 16 pages. cited by applicant
W.-J. Yan et al., “How Fast are the Leaked Facial Expressions: The Duration of Micro-Expressions,” Journal of Nonverbal Behavior, vol. 37, Jul. 16, 2013, pp applicant
R. Zhang et al., “Monitoring Chewing and Eating in Free-Living Using Smart Eyeglasses,” IEEE Journal of Biomedical and Health Informatics, vol. 22, No. 1 23-32. cited by applicant
R. Zhang et al., “Free-living Eating Event Spotting using EMG-monitoring Eyeglasses,” IEEE EMBS International Conference on Biomedical & Health Infor pp. 128-132. cited by applicant
S. Zhang et al., “I Am an Earphone and I Can Hear My User's Face: Facial Landmark Tracking Using Smart Earphones,” ACM Transactions on Internet of Th 28 pages. cited by applicant
T. Denning et al., “In Situ with Bystanders of Augmented Reality Glasses: Perspectives on Recording and Privacy-Mediating Technologies,” Proceedings of t Conference on Human Factors in Computing Systems, Apr. 2014, 10 pages. cited by applicant
J. H. Huh et al., “WristAcoustic: Through-Wrist Acoustic Response Based Authentication for Smartwatches,” Proceedings of the ACM on Interactive, Mobile, Ubiquitous Technologies, vol. 6, No. 4, Jan. 2023, pp. 167:1-167:34. cited by applicant
J. Kim et al., “SonarID: Using Sonar to Identify Fingers on a Smartwatch,” Proceedings of the 2022 CHI Conference on Human Factors in Computing Syster pages. cited by applicant
O. Kudina et al., “Ethics from Within: Google Glass, the Collingridge Dilemma, and the Mediated Value of Privacy,” Science, Technology, & Human Values, Aug. 21, 2018, pp. 291-314. cited by applicant
G. Laput et al., “Acoustruments: Passive, Acoustically-Driven, Interactive Controls for Handheld Devices,” ACM SIGGRAPH 2015 Emerging Technologies, pages. cited by applicant
D. Li et al., “Room-scale Hand Gesture Recognition Using Smart Speakers,” Proceedings of the 20th ACM Conference on Embedded Networked Sensor Syst 14 pages. cited by applicant
J. O'Hagan et al., “Privacy-Enhancing Technology and Everyday Augmented Reality: Understanding Bystanders' Varying Needs for Awareness and Consent,” ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 6, No. 4, Jan. 2023, 35 pages. cited by applicant
M. Ono et al., “Touch & Activate: Adding Interactivity to Existing Objects Using Active Acoustic Sensing,” Proceedings of the 26th Annual ACM Symposiun Software and Technology, Oct. 2013, pp. 31-40. cited by applicant
Y. Gao et al., “EarEcho: Using Ear Canal Echo for Wearable Authentication,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Techr No. 3, Sep. 2019, pp. 81:1-81:24. cited by applicant
A. Hayashida et al., “Estimation of Room Temperature Based on Acoustic Frequency Response,” Acoustical Science and Technology, vol. 41, No. 4, Jul. 2020 cited by applicant
D. Kim et al., “Pantoenna: Mouth Pose Estimation for VR/AR Headsets Using Low-Profile Antenna and Impedance Characteristic Sensing,” Proceedings of t ACM Symposium on User Interface Software and Technology, Oct.-Nov. 2023, 12 pages. cited by applicant
Gras Sound & Vibration, “What are Microphone Environmental Coefficients?” https://www.grasacoustics.com/blog/working-with-environmental-coefficients pages. cited by applicant
Xreal, Inc., “Air 2 Ultra,” https://developer.xreal.com/?lang=en, Accessed Oct. 30, 2024, 20 pages. cited by applicant
C. Xu et al., “Finger-writing with Smartwatch: A Case for Finger and Hand Gesture Recognition using Smartwatch,” Proceedings of the 16th International Wo Computing Systems and Applications, Feb. 2015, pp. 9-14. cited by applicant
C. Zhang et al., “FingerPing: Recognizing Fine-grained Hand Poses using Active Acoustic On-body Sensing,” Proceedings of the 2018 CHI Conference on Hu Computing Systems, Apr. 2018, 10 pages. cited by applicant
Y. Ji et al., “Updating the Silent Speech Challenge Benchmark with Deep Learning,” Speech Communication, Sep. 2017, 25 pages. cited by applicant
D. Verma et al., “ExpressEar: Sensing Fine-Grained Facial Expressions with Earables,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiqui vol. 5, No. 3, Sep. 2021, pp. 129:1-129:28. cited by applicant
L. Liu et al., “Deep Learning for Generic Object Detection: A Survey,” International Journal of Computer Vision, Oct. 31, 2019, 58 pages. cited by applicant
W. Liu et al., “A Survey of Deep Neural Network Architectures and Their Applications,” Neurocomputing, Authors Manuscript, Nov. 2016, 31 pages. cited by
S. Minaee et al., “Image Segmentation Using Deep Learning: A Survey,” arXiv:2001.05566v5, Nov. 15, 2020, 22 pages. cited by applicant
Y. Nakajima et al., “Non-audible Murmur Recognition,” 8th European Conference on Speech Communication and Technology, Sep. 2003, pp. 220-223. cited
Y. Nakajima et al., “Non-audible Murmur Recognition Input Interface Using Stethoscopic Microphone Attached to the Skin,” IEEE International Conference Speech, and Signal Processing, Apr. 2003, pp. 127-130. cited by applicant

O. R. Nasir et al., "Text2FaceGAN: Face Generation from Fine Grained Textual Descriptions," arXiv:1911.11378v1, Nov. 26, 2019, 10 pages. cited by applica
N. Otsu, "A Threshold Selection Method from Gray-Level Histograms," IEEE Transactions on Systems, Man, and Cybernetics, vol. SMC-9, No. 1, Jan. 1979, by applicant
A. Porbadnigk et al., "EEG-Based Speech Recognition Impact of Temporal Effects," Proceedings of the International Conference on Bio-inspired Systems and Processing, Jan. 2009, pp. 376-381. cited by applicant
B. Shi et al., "An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition," arXiv:1507.0 2015, 9 pages. cited by applicant
K. Simonyan et al., "Very Deep Convolutional Networks for Large-Scale Image Recognition," arXiv:1409.1556v6, Apr. 10, 2015, 14 pages. cited by applicant
T. Stamer, "The Challenges of Wearable Computing: Part 1," IEEE Micro, vol. 21, No. 4, Jul.-Aug. 2001, pp. 44-52. cited by applicant
Y. Tanaka et al., "Long-Term Histological Comparison Between Near-Infrared Irradiated Skin and Scar Tissues," Clinical, Cosmetic and Investigational Derm Nov. 25, 2010, pp. 143-149. cited by applicant
G. Tian et al., "Audio2face: Generating Speech/Face Animation from Single Audio With Attention-Based Bidirectional LSTM Networks," arXiv: 1905.11142 6 pages. cited by applicant
A. Wunsch et al., "A Controlled Trial to Determine the Efficacy of Red and Near-Infrared Light Treatment in Patient Satisfaction, Reduction of Fine Lines, W Roughness, and Intradermal Collagen Density Increase," Photomedicine and Laser Surgery, vol. 32, No. 2, Feb. 2014, pp. 93-100. cited by applicant
W. Yin et al., "Comparative Study of CNN and RNN for Natural Language Processing," arXiv:1702.01923v1, Feb. 7, 2017, 7 pages. cited by applicant
S. Shen et al., "I am a Smartwatch and I Can Track My User's Arm," In Proceedings of the 14th Annual International Conference on Mobile Systems, Applica Services, Jun. 25-30, 2016, pp. 85-96. cited by applicant
S. Oney et al., "ZoomBoard: A Diminutive QWERTY Soft Keyboard Using Iterative Zooming for Ultra-Small Devices," Proceedings of the SIGCHI Confere Factors in Computing Systems, Apr. 2013, pp. 2799-2802. cited by applicant
T. S. Saponas et al., "Enabling Always-Available Input with Muscle-Computer Interfaces," Proceedings of the 22nd Annual ACM Symposium on User Interfa Technology, Oct. 2009, pp. 167-176. cited by applicant
X. Zhou et al., "Sparseness Meets Deepness: 3D Human Pose Estimation from Monocular Video," IEEE Conference on Computer Vision and Pattern Recogni 2016, pp. 4966-4975. cited by applicant
D. Way et al., "A Usability User Study Concerning Free-Hand Microgesture and Wrist-Worn Sensors," 11th International Conference on Wearable and Implan Networks, Jun. 16, 2014, pp. 138-142. cited by applicant
H. Wen et al., "Serendipity: Finger Gesture Recognition using an Off-the-Shelf Smartwatch," Proceedings of the 2016 CHI Conference on Human Factors in Systems, May 7, 2016, pp. 3847-3851. cited by applicant
T. Von Marcard et al., "Sparse Inertial Poser: Automatic 3D Human Pose Estimation from Sparse IMUs," Eurographics, vol. 36, No. 2, May 2017, pp. 349-36 applicant
S.-E. Wei et al., "Convolutional Pose Machines," IEEE Conference on Computer Vision and Pattern Recognition, Jun. 2016, pp. 4724-4732. cited by applican
Y. Zhang et al., "Skin Track: Using the Body as an Electrical Waveguide for Continuous Finger Tracking on the Skin," Proceedings of the 2016 CHI Conferen Factors in Computing Systems, May 2016, pp. 1491-1503. cited by applicant
I. Akhter et al., "Pose-Conditioned Joint Angle Limits for 3D Human Pose Reconstruction," IEEE Conference on Computer Vision and Pattern Recognition, 2 1455. cited by applicant
O. Amft et al., "Detection of Eating and Drinking Arm Gestures Using Inertial Body-Worn Sensors," IEEE International Symposium on Wearable Computers, 160-163. cited by applicant
O. Amft et al., "Analysis of Chewing Sounds for Dietary Monitoring," Proceedings of the 7th International Conference on Ubiquitous Computing, Sep. 2005, applicant
M. Andriluka et al., "Pictorial Structures Revisited: People Detection and Articulated Pose Estimation," IEEE Conference on Computer Vision and Pattern Re 2009, pp. 1014-1021. cited by applicant
A. Bedri et al., "FitByte: Automatic Diet Monitoring in Unconstrained Situations Using Multimodal Sensing on Eyeglasses," Proceedings of the 2020 CHI Co Human Factors in Computing Systems, Apr. 2020, 12 pages. cited by applicant
A. Bedri et al., "EarBit: Using Wearable Sensors to Detect Eating Episodes in Unconstrained Environments," Proceedings of the ACM on Interactive, Mobile, Ubiquitous Technologies, vol. 1, No. 3, Sep. 2017, 20 pages. cited by applicant
A. Bedri et al., "Detecting Mastication: A Wearable Approach," Proceedings of the 2015 ACM on International Conference on Multimodal Interaction, Nov. 2 cited by applicant
A. Bedri et al., "A Wearable System for Detecting Eating Activities with Proximity Sensors in the Outer Ear," Proceedings of the 2015 ACM International Sy Wearable Computers, Sep. 7-11, 2015, pp. 91-92. cited by applicant
S. Bi et al., "Auracle: Detecting Eating Episodes with an Ear-mounted Sensor," Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Tec No. 3, Sep. 2018, 27 pages. cited by applicant
W. Wei et al., "Real-Time 3D Arm Motion Tracking Using the 6-Axis IMU Sensor of a Smartwatch," 17th International Conference on Wearable and Implanta Networks, Jul. 2021,4 pages. cited by applicant
F. Bogo et al., "Keep it SMPL: Automatic Estimation of 3D Human Pose and Shape from a Single Image," arXiv:1607.08128v1, Jul. 27, 2016, 18 pages. cited
L. Xie et al., "Data-driven Motion Estimation with Low-Cost Sensors," 5th International Conference on Visual Information Engineering, Jul. 2008, 6 pages. c
X. Chen et al., "Articulated Pose Estimation by a Graphical Model with Image Dependent Pairwise Relations," arXiv:1407.3399v2, Nov. 4, 2014, 9 pages. cit
J. Chung et al., "A Glasses-type Wearable Device for Monitoring the Patterns of Food Intake and Facial Activity," Scientific Reports, vol. 7, No. 41690, Jan. 3 cited by applicant
E. Cippitelli et al., "A Human Activity Recognition System Using Skeleton Data from RGBD Sensors," Computational Intelligence and Neuroscience, Mar. 2 cited by applicant
F. Cordeiro et al., "Barriers and Negative Nudges: Exploring Challenges in Food Journaling," Proceedings of the 33rd Annual ACM Conference on Human Fa Computing Systems, Authors Manuscript, Feb. 16, 2016, 8 pages. cited by applicant
Y. Dong et al., "Detecting Periods of Eating During Free-Living by Tracking Wrist Motion," IEEE Journal of Biomedical and Health Informatics, vol. 18, No. 1253-1260. cited by applicant
M. Farooq et al., "Segmentation and Characterization of Chewing Bouts by Monitoring Temporalis Muscle Using Smart Glasses with Piezoelectric Sensor," I Biomedical and Health Informatics, Authors Manuscript, Nov. 1, 2018, 24 pages. cited by applicant
N. Fawaz et al., "Deep Learning for Time Series Classification: A Review," arXiv:1809.04356v4, May 14, 2019, 44 pages. cited by applicant
A. Filippeschi et al., "Survey of Motion Tracking Methods Based on Inertial Sensors: A Focus on Upper Limb Human Motion," Sensors, vol. 17, No. 6, Jun. 2 cited by applicant
K. He et al., "Mask R-CNN," arXiv:1703.06870v3, Jan. 24, 2018, 12 pages. cited by applicant
H. Junker et al., "Gesture Spotting with Body-Worn Inertial Sensors to Detect User Activities," Pattern Recognition vol. 41, 2008, pp. 2010-2024. cited by app
K. Kyritsis et al., "End-to-end Learning for Measuring in-meal Eating Behavior from a Smartwatch," 40th Annual International Conference of the IEEE Engi Medicine and Biology Society, Oct. 2018, 4 pages. cited by applicant
K. Kyritsis et al., "Modeling Wrist Micromovements to Measure In-Meal Eating Behavior from Inertial Sensor Data," IEEE Journal of Biomedical and Health 2019, 11 pages. cited by applicant

Y. Liu et al., “Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors,” Proceedings of the 17th Annual International C
Mobile Systems, Applications, and Services, Jun. 2019, pp. 287-299. cited by applicant
Y. Y. Luktuke et al., “Segmentation and Recognition of Eating Gestures from Wrist Motion using Deep Learning,” 2020 IEEE International Conference on Bi
pp. 1368-1373. cited by applicant
D. Mehta et al., “Single-Shot Multi-Person 3D Pose Estimation From Monocular RGB,” arXiv: 1712.03453v3, Aug. 28, 2018, 16 pages. cited by applicant
D. Mehta et al., “VNect: Real-time 3D Human Pose Estimation with a Single RGB Camera,” arXiv:1705.01583v1, May 3, 2017, 13 pages. cited by applicant
M. Mirtchouk et al., “Recognizing Eating from Body-Worn Sensors: Combining Free-Living and Laboratory Data,” Proceedings of the ACM on Interactive, M
and Ubiquitous Technologies, vol. 1, No. 3, Article 85, Sep. 2017, 20 pages. cited by applicant
M. Mirtchouk et al., “Automated Estimation of Food Type from Body-worn Audio and Motion Sensors in Free-Living Environments,” Proceedings of Machin
Research, vol. 106, Aug. 1, 2019, 21 pages. cited by applicant
M. Mirtchouk et al., “Automated Estimation of Food Type and Amount Consumed from Body-Worn Audio and Motion Sensors,” In Proceedings of the 2016 A
Joint Conference on Pervasive and Ubiquitous Computing, Sep. 12-16, 2016, pp. 451-462. cited by applicant
M. B. Morshed et al., “A Real-Time Eating Detection System for Capturing Eating Moments and Triggering Ecological Momentary Assessments to Obtain Fu
System Development and Validation Study,” JMIR Mhealth Uhealth, vol. 8, No. 12, Dec. 2020, 15 pages. cited by applicant
K. Murakami et al., “Eating Frequency Is Positively Associated with Overweight and Central Obesity in US Adults,” The Journal of Nutrition, Dec. 1, 2015, 1
applicant
A. Newell et al., “Stacked Hourglass Networks for Human Pose Estimation,” arXiv:1603.06937v2, Jul. 26, 2016, 17 pages. cited by applicant
J. C. Núñez et al., “Convolutional Neural Networks and Long Short-Term Memory for Skeleton-Based Human Activity and Hand Gesture Recognition,” Patte
vol. 76, Apr. 2018, pp. 80-94. cited by applicant
L. Piyathilaka et al., “Gaussian Mixture Based HMM for Human Daily Activity Recognition Using 3D Skeleton Features,” IEEE 8th Conference on Industrial
Applications, Jul. 2013, pp. 567-572. cited by applicant
S. A. Rahman et al., “Unintrusive Eating Recognition Using Google Glass,” In 9th International Conference on Pervasive Computing Technologies for Health
108-111. cited by applicant
R. I. Ramos-Garcia et al., “Improving the Recognition of Eating Gestures Using Intergesture Sequential Dependencies,” IEEE Journal of Biomedical and Hea
vol. 19, No. 3, May 2015, pp. 825-831. cited by applicant
J. L. Scisco et al., “Examining the Utility of a Bite-Count-Based Measure of Eating Activity in Free-Living Human Beings,” Journal of the Academy of Nutrit
Sep. 3, 2013, 6 pages. cited by applicant
S. Sharma et al., “Automatic Detection of Periods of Eating Using Wrist Motion Tracking,” IEEE First Conference on Connected Health: Applications, Syster
Technologies, Jun. 2016, pp. 362-363. cited by applicant
S. Sharma et al., “The Impact of Walking and Resting on Wrist Motion for Automated Detection of Meals,” ACM Transactions on Computing for Healthcare,
2020, pp. 24:1-24:19. cited by applicant
A. Toshev et al., “DeepPose: Human Pose Estimation via Deep Neural Networks,” arXiv:1312.4659v3, Aug. 20, 2014, 9 pages. cited by applicant
S. Raurale et al., “EMG Acquisition and Hand Pose Classification for Bionic Hands from Randomly-Placed Sensors,” IEEE International Conference on Acou
Signal Processing (ICASSP), Apr. 2018, 5 pages. cited by applicant
J. Rekimoto, “GestureWrist and GesturePad: UnobtrusiveWearable Interaction Devices,” Proceedings Fifth International Symposium on Wearable Computers
pages. cited by applicant
M. Rohrbach et al., “A Database for Fine Grained Activity Detection of Cooking Activities,” IEEE Conference on Computer Vision and Pattern Recognition, J
cited by applicant
J. C. R. Rudolph et al., “Sensing Hand Interactions with Everyday Objects by Profiling Wrist Topography,” Proceedings of the Sixteenth International Confere
Embedded, and Embodied Interaction, Feb. 2022, 14 pages. cited by applicant
A. De Silva et al., “Real-Time Hand Gesture Recognition Using Temporal Muscle Activation Maps of Multi-Channel sEMG Signals,” arXiv:2002.03159v1, F
pages. cited by applicant
I. Sosin et al., “Continuous Gesture Recognition from sEMG Sensor Data with Recurrent Neural Networks and Adversarial Domain Adaptation,” arXiv:2012.
2020, 6 pages. cited by applicant
W. Sun et al., “ThumbTrak: Recognizing Micro-finger Poses Using a Ring with Proximity Sensing,” arXiv:2105.14680v2, Dec. 23, 2021, 16 pages. cited by a
H. Truong et al., “DEMO: Low-Power Capacitive Sensing Wristband for Hand Gesture Recognition,” Proceedings of the 9th ACM Workshop on Wireless of t
Students, and for the Students, Oct. 2017, p. 21. cited by applicant
H. Truong et al., “Capacitive Sensing 3D-printed Wristband for Enriched Hand Gesture Recognition,” Proceedings of the 2017 Workshop on Wearable System
Applications, Jun. 2017, 5 pages. cited by applicant
H. Truong et al., “CapBand: Battery-Free Successive Capacitance Sensing Wristband for Hand Gesture Recognition,” Proceedings of the 16th ACM Conferen
Networked Sensor Systems, Nov. 2018, pp. 54-67. cited by applicant
H.-R. Tsai et al., “ThumbRing: Private Interactions Using One-Handed Thumb Motion Input on Finger Segments,” Proceedings of the 18th International Conf
Computer Interaction with Mobile Devices and Services Adjunct, Sep. 2016, 8 pages. cited by applicant
Ultraleap, “The World's Most Advanced Hand Tracking,” https://www.ultraleap.com/tracking/, Accessed Oct. 7, 2024, 7 pages. cited by applicant
A. Waghmare et al., “Z-Ring: Single-Point Bio-Impedance Sensing for Gesture, Touch, Object and User Recognition,” Proceedings of the 2023 CHI Conferen
Factors in Computing Systems, Apr. 2023, 18 pages. cited by applicant
E. Wu et al., “Back-Hand-Pose: 3D Hand Pose Estimation for a Wrist-Worn Camera via Dorsum Deformation Network,” Proceedings of the 33rd Annual ACM
User Interface Software and Technology, Oct. 2020, pp. 1147-1160. cited by applicant
X. Xu et al., “Enabling Hand Gesture Customization on Wrist-Worn Devices,” Proceedings of the 2022 CHI Conference on Human Factors in Computing Sys
19 pages. cited by applicant
Yeo et al., “Opisthenar: Hand Poses and Finger Tapping Recognition by Observing Back of Hand Using Embedded Wrist Camera,” UIST'19, Association of C
Machinery, Oct. 20-23, 2019, pp. 963-971. cited by applicant
Y. Zhang et al., “Tomo: Wearable, Low-Cost Electrical Impedance Tomography for Hand Gesture Recognition,” Proceedings of the 28th Annual ACM Sympo
Interface Software & Technology, Nov. 5, 2015, pp. 167-173. cited by applicant
Y. Zhang et al., “Advancing Hand Gesture Recognition with High Resolution Electrical Impedance Tomography,” Proceedings of the 29th Annual Symposium
Software and Technology, Oct. 2016, pp. 843-850. cited by applicant
H. Zhou et al., “Learning on the Rings: Self-Supervised 3D Finger Motion Tracking Using Wearable Sensors,” Proceedings of the ACM on Interactive, Mobil
Ubiquitous Technologies, vol. 6, No. 2, Jun. 2022, pp. 90:1-90:31. cited by applicant
L. Zhou et al., “Towards Automatic Learning of Procedures From Web Instructional Videos,” The Thirty-Second AAAI Conference on Artificial Intelligence,
7590-7598. cited by applicant
C. Zimmermann et al., “Learning to Estimate 3D Hand Pose from Single RGB Images,” Proceedings of the IEEE International Conference on Computer Visio
4903-4911. cited by applicant
R.-D. Vatavu et al., “GestuRING: A Web-based Tool for Designing Gesture Input with Rings, Ring-Like, and Ring-Ready Devices,” The 34th Annual ACM Sy
Interface Software and Technology, Oct. 2021, pp. 710-723. cited by applicant
A. Bedri et al., “Toward Silent-Speech Control of Consumer Wearables,” Computer, vol. 48, No. 10, Oct. 2015, pp. 54-62. cited by applicant

L. A. Cheah et al., “A Wearable Silent Speech Interface based on Magnetic Sensors with Motion-Artefact Removal,” 11th International Conference on Biomed and Devices, Jan. 2018, pp. 56-62. cited by applicant
Bose, “QuietComfort 45 Noise Cancelling Smart Headphones,” https://www.bose.com/pxp/bose-quietcomfort-45-headphones?srsltid=AfmBOopTn36boc58vSfai3X5lsoxGJnZO4zTa5tzKMcTldXaj_EZY2bh, Accessed Oct. 7, 2024, 16 pages. cited by applicant
T. G. Csapo et al., “DNN-Based Ultrasound-to-Speech Conversion for a Silent Speech Interface,” Interspeech, Aug. 2017, pp. 3672-3676. cited by applicant
B. Denby et al., “Prospects for a Silent Speech Interface using Ultrasound Imaging,” IEEE International Conference on Acoustics Speech and Signal Processin May 2006, 4 pages. cited by applicant
J. A. Gonzalez et al., “Direct Speech Reconstruction From Articulatory Sensor Data by Machine Learning,” IEEE/ACM Transactions on Audio, Speech, and L Processing, Dec. 2017, 13 pages. cited by applicant
R. Hofe et al., “Small-vocabulary Speech Recognition using a Silent Speech Interface Based on Magnetic Sensing,” Speech Communication, vol. Feb. 1, 555, cited by applicant
C. Q. Howard et al., “A Review of Current Ultrasound Exposure Limits,” The Journal of Occupational Health and Safety of Australia and New Zealand, Sep. cited by applicant
Adesso, “Xtream G1 Stereo Gaming Headset with Microphone,” https://www.adesso.com/products/stereo-gaming-headphones-with-microphone/, Datasheet, 2024, 1 page. cited by applicant
Razer Inc., “Razer Kraken,” https://www.razer.com/gaming-headsets/razer-kraken-v3, Accessed Oct. 7, 2024, 9 pages. cited by applicant
E. M. Juanpere et al., “Ultrasound-Based Silent Speech Interface Using Convolutional and Recurrent Neural Networks,” vol. 105, Acta Acustica United with 2019, pp. 587-590. cited by applicant
A. Kapur et al., “AlterEgo: A Personalized Wearable Silent Speech Interface,” International Conference on Intelligent User Interfaces, Mar. 2018, pp. 43-53. c
A. Kapur et al., “Non-Invasive Silent Speech Recognition in Multiple Sclerosis with Dysphonia,” Proceedings of Machine Learning Research, vol. 116, Dec. cited by applicant
M. Kim et al., “Speaker-Independent Silent Speech Recognition from Flesh-Point Articulatory Movements Using an LSTM Neural Network,” IEEE/ACM Tra Audio, Speech, and Language Processing , vol. 25, No. 12, Dec. 2017, 32 pages. cited by applicant
M. Kim et al., “Preliminary Test of a Wireless Magnetic Tongue Tracking System for Silent Speech Interface,” IEEE Biomedical Circuits and Systems Confer Oct. 2018, 4 pages. cited by applicant
N. Kimura et al., “Mobile, Hands-Free, Silent Speech Texting Using SilentSpeller,” Extended Abstracts of the 2021 CHI Conference on Human Factors in Co May 2021, 4 pages. cited by applicant
N. Kimura et al., “SilentSpeller: Towards Mobile, Hands-Free, Silent Speech Text Entry Using Electropalatography,” Proceedings of the 2022 CHI Conferenc Factors in Computing Systems, Apr. 2022, 19 pages. cited by applicant
N. Kimura et al., “SottoVoce: An Ultrasound Imaging-Based Silent Speech Interaction Using Deep Neural Networks,” arXiv:2303.01758v1, Mar. 3, 2023, 10 applicant
R. Li et al., “TongueBoard: An Oral Interface for Subtle Input,” Proceedings of the 10th Augmented Human International Conference, Mar. 2019, 9 pages. cite
G. S. Meltzner et al., “Development of sEMG Sensors and Algorithms for Silent Speech Recognition,” Journal of Neural Engineering—Author Manuscript, A pages. cited by applicant
H. Sahni et al., “The Tongue and Ear Interface: A Wearable System for Silent Speech Recognition,” Proceedings of the 2014 ACM International Symposium o Computers, Sep. 2014, pp. 47-54. cited by applicant
T. Srivastava et al., “Mutelt: Jaw Motion Based Unvoiced Command Recognition Using Earable,” Proceedings of the ACM on Interactive, Mobile, Wearable Technologies, vol. 6, No. 3, Sep. 2022, pp. 140:1-140:26. cited by applicant
L. Tóth et al., “Multi-Task Learning of Speech Recognition and Speech Synthesis Parameters for Ultrasound-based Silent Speech Interfaces,” Interspeech 201 3172-3176. cited by applicant
Y. Wang et al., “Silent Speech Decoding Using Spectrogram Features Based on Neuromuscular Activities,” Brain Sciences, vol. 10, No. 7, Jul. 11, 2020, 14 pa applicant
R. Zhang et al., “SpeeChin: A Smart Necklace for Silent Speech Recognition,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Tec No. 4, Dec. 2021, p. 192:1-192:23. cited by applicant
Y. Zhang et al., “CELIP: Ultrasonic-based Lip Reading with Channel Estimation Approach for Virtual Reality Systems,” Adjunct Proceedings of the 2021 AC Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers, Sep. 2021, p applicant
A. Abedin et al., “Attend and Discriminate: Beyond the State-of-the-Art for Human Activity Recognition Using Wearable Sensors,” arXiv:2007.07172v1, Jul. pages. cited by applicant
L. Bao et al., “Activity Recognition from User-Annotated Acceleration Data,” International Conference on Pervasive Computing, Apr. 2004, 17 pages. cited b
W. Price et al., “UnweaveNet: Unweaving Activity Stories,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2022, pp. 137 applicant
H. Qian et al., “A Novel Distribution-Embedded Neural Network for Sensor-Based Activity Recognition,” Proceedings of the Twenty-Eighth International Joi Artificial Intelligence, Aug. 2019, 7 pages. cited by applicant
V. Radu et al., “Multimodal Deep Learning for Activity and Context Recognition,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous 1, No. 4, Dec. 2017, pp. 157:1-157:27. cited by applicant
A. Saeed et al., “Federated Self-Supervised Learning of Multi-Sensor Representations for Embedded Intelligence,” IEEE Internet of Things Journal, arXiv:20 25, 2020, 11 pages. cited by applicant
M. Saquib Sarfraz et al., “Temporally-Weighted Hierarchical Clustering for Unsupervised Action Segmentation,” IEEE/CVF Conference on Computer Vision Recognition (CVPR), Jun. 2021, pp. 11225-11234. cited by applicant
R. R. Selvaraju et al., “Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,” arXiv:1610.02391v4, Dec. 3, 2019, 23 pages.
N. Srivastava et al., “Dropout: A Simple Way to Prevent Neural Networks from Overfitting,” Journal of Machine Learning Research, vol. 15, Jun. 2014, pp. 19 applicant
D. Strömbäck et al., “MM-Fit: Multimodal Deep Learning for Automatic Exercise Logging across Sensing Devices,” Proceedings of the ACM on Interactive, and Ubiquitous Technologies, vol. 4, No. 4, Dec. 2020, pp. 168:1-168:22. cited by applicant
C. Tang et al., “Are Accelerometers for Activity Recognition a Dead-end?” arXiv:2001.08111v2, Jan. 30, 2020, 6 pages. cited by applicant
M. Tanjid Hasan Tonmoy et al., “Hierarchical Self Attention Based Autoencoder for Open-Set Human Activity Recognition,” arXiv:2103.04279v1, Mar. 7, 20 by applicant
Y. Vaizman et al., “Context Recognition In-the-Wild: Unified Model for Multi-Modal Sensors and Multi-Label Classification,” Proceedings of the ACM on In Wearable and Ubiquitous Technologies, vol. 1, No. 4, Dec. 2017, pp. 1:1-1:27. cited by applicant
T. Wang et al., “C-FMCW Based Contactless Respiration Detection Using Acoustic Signal,” Proceedings of the ACM on Interactive, Mobile, Wearable and U Technologies, vol. 1, No. 4, Dec. 2017, pp. 170:1-170:20. cited by applicant
J. A. Ward et al., “Activity Recognition of Assembly Tasks Using Body-Worn Microphones and Accelerometers,” IEEE Transactions on Pattern Analysis and Intelligence, Mar. 20, 2006, Draft, 37 pages. cited by applicant
J. Wu et al., “Automatic Class Discovery and One-Shot Interactions for Acoustic Activity Recognition,” Proceedings of the 2020 CHI Conference on Human

Computing Systems, Apr. 2020, pp. 1-14. cited by applicant
B. Xu et al., “Empirical Evaluation of Rectified Activations in Convolution Network,” arXiv:1505.00853v2 , Nov. 27, 2015, 5 pages. cited by applicant
S. Yao et al., “SADeepSense: Self-Attention Deep Learning Framework for Heterogeneous On-Device Sensors in Internet of Things Applications,” IEEE Con
Computer Communications, May 2019, 9 pages. cited by applicant
C. Zhang et al., “FingerSound: Recognizing Unistroke Thumb Gestures Using a Ring,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiqui
vol. 1, No. 3, Sep. 2017, pp. 120:1-120:19. cited by applicant
D. Zhang et al., “anucvml/vidat—Video Annotation Tool,” https://github.com/anucvml/vidat, Accessed Jul. 25, 2024, 7 pages. cited by applicant
R. Zhang et al., “EatingTrak: Detecting Fine-grained Eating Moments in the Wild Using a Wrist-mounted IMU,” Proceedings of the ACM on Human-Comput
6, No. MHCI, Sep. 2022, 22 pages. cited by applicant
Microsoft, “Kinect for Windows,” https://learn.microsoft. com/en-us/windows/apps/design/devices/kinect-for-windows, Aug. 30, 2022, 4 pages. cited by appli
K. Ahuja et al., “Vid2Doppler: Synthesizing Doppler Radar Data from Videos for Training Privacy-Preserving Activity Recognition,” Proceedings of the 2021
on Human Factors in Computing Systems, May 7, 2021, 10 pages. cited by applicant
K. Ahuja et al., “Pose-on-the-Go: Approximating User Pose with Smartphone Sensor Fusion and Inverse Kinematics,” Proceedings of the 2021 CHI Conferen
Factors in Computing Systems, May 7, 2021, 12 pages. cited by applicant
K. Ahuja et al., “ControllerPose: Inside-Out Body Capture with VR Controller Cameras,” Proceedings of the 2022 CHI Conference on Human Factors in Com
Apr. 2022, 13 pages. cited by applicant
T. Babic et al., “Simo: Interactions with Distant Displays by Smartphones with Simultaneous Face and World Tracking,” Extended Abstracts of the 2020 CHI
Human Factors in Computing Systems, Apr. 25, 2020, 12 pages. cited by applicant
Z. Cao et al., “Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields,” IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
7291-7299. cited by applicant
K. He et al., “Mask R-CNN,” IEEE International Conference on Computer Vision, Oct. 2017, pp. 2961-2969. cited by applicant
R. Hori et al., “Silhouette-Based Synthetic Data Generation for 3D Human Pose Estimation With a Single Wrist-Mounted 360° Camera,” IEEE International C
Image Processing (ICIP), Sep. 2021, 5 pages. cited by applicant
D.-Y. Hsiao et al., “Proactive Sensing for Improving Hand Pose Estimation,” Proceedings of the 2016 CHI Conference on Human Factors in Computing Syste
pp. 2348-2352. cited by applicant
D.-H. Hwang et al., “MonoEye: Monocular Fisheye Camera-based 3D Human Pose Estimation,” IEEE Conference on Virtual Reality and 3D User Interfaces
pages. cited by applicant
S. S. Intille et al., “Acquiring In Situ Training Data for Context-Aware Ubiquitous Computing Applications,” Proceedings of the SIGCHI Conference on Hum
Computing Systems, Apr. 25, 2004, 8 pages. cited by applicant
W. Jiang et al., “Coherent Reconstruction of Multiple Humans from a Single Image,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (C
pp. 82-91. cited by applicant
W. Jiang et al., “Towards3DHumanPoseConstructionUsingWiFi,” Proceedings of the 26th Annual International Conference on Mobile Computing and Netwo
14 pages. cited by applicant
A. Kanazawa et al., “End-to-end Recovery of Human Shape and Pose,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 20
cited by applicant
H. Kong et al., “m3Track: mmWave-based Multi-User 3D Posture Tracking,” Proceedings of the 20th Annual International Conference on Mobile Systems, A
Services, Jun. 2022, pp. 491-503. cited by applicant
W. Li et al., “MHFormer: Multi-Hypothesis Transformer for 3D Human Pose Estimation,” IEEE/CVF Conference on Computer Vision and Pattern Recognitio
2022, pp. 13147-13156. cited by applicant
Z. Liu et al., “Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition,” IEEE/CVF Conference on Computer Vision and Patt
(CVPR), Jun. 2020, pp. 143-152. cited by applicant
D. Mehta et al., “Monocular 3D Human Pose Estimation in the Wild Using Improved CNN Supervision,” arXiv:1611.09813v5, Oct. 4, 2017, 16 pages. cited b
V. Mollyn et al., “IMUPoser: Full-Body Pose Estimation using IMUs in Phones, Watches, and Earbuds,” Proceedings of the 2023 CHI Conference on Human
Computing Systems, Apr. 2023, 12 pages. cited by applicant
G. Moon et al., “V2V-PoseNet: Voxel-to-Voxel Prediction Network for Accurate 3D Hand and Human Pose Estimation from a Single Depth Map,” IEEE/CVF
Computer Vision and Pattern Recognition, Jun. 2018, pp. 5079-5088. cited by applicant
R. Nandakumar et al., “FingerIO: Using Active Sonar for Fine-Grained Finger Tracking,” Proceedings of the 2016 CHI Conference on Human Factors in Com
May 2016, pp. 1515-1525. cited by applicant
E. Ng et al., “You2Me: Inferring Body Pose in Egocentric Video via First and Second Person Interactions,” arXiv:1904.09882v2, Mar. 28, 2020, 14 pages. cite
G. Papandreou et al., “Towards Accurate Multi-person Pose Estimation in the Wild,” IEEE Conference on Computer Vision and Pattern Recognition (CVPR),
4903-4911. cited by applicant
D. Pavllo et al. “3D Human Pose Estimation in Video with Temporal Convolutions and Semi-supervised Training,” IEEE/CVF Conference on Computer Visio
Recognition (CVPR), Jun. 2019, pp. 7753-7762. cited by applicant
Y. Ren et al., “Winect: 3D Human Pose Tracking for Free-Form Activity Using Commodity WiFi,” Proceedings of the ACM on Interactive, Mobile, Wearable
Technologies, vol. 5, No. 4, Dec. 2021, pp. 172:1-172:29. cited by applicant
Y. Ren et al., “GoPose: 3D Human Pose Estimation Using WiFi,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vo
2022, pp. 69:1-69:25. cited by applicant
M. Sandler et al., “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),
4510-4520. cited by applicant
P. H. Schonemann, “A Generalized Solution of the Orthogonal Procrustes Problem,” Psychometrika, vol. 31, No. 1, Mar. 1966, 10 pages. cited by applicant
D. Tome et al., “xR-EgoPose: Egocentric 3D Human Pose from an HMD Camera,” IEEE/CVF International Conference on Computer Vision (ICCV), Oct. 20
cited by applicant
Ultraleap, “Leap Motion Controller 2,” https://www.ultraleap.com/product/leap-motion-controller/, Accessed Oct. 3, 2024, 6 pages. cited by applicant
B. Wandt et al., “ElePose: Unsupervised 3D Human Pose Estimation by Predicting Camera Elevation and Learning Normalizing Flows on 2D Poses,” arXiv:2
14, 2021, 11 pages. cited by applicant
W. Wang et al., “Device-Free Gesture Tracking Using Acoustic Signals,” Proceedings of the 22nd Annual International Conference on Mobile Computing and
2016, pp. 82-94. cited by applicant
W. Xie et al., “Acoustic-based Upper Facial Action Recognition for Smart Eyewear,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquito
vol. 5, No. 2, Jun. 2021, pp. 41:1-41:28. cited by applicant
F. Xiong et al., “A2J: Anchor-to-Joint Regression Network for 3D Articulated Pose Estimation from a Single Depth Image,” IEEE/CVF International Confere
Vision (ICCV), Oct. 2019, pp. 793-802. cited by applicant
H. Xu et al., “GHUM & GHUML: Generative 3D Human Shape and Articulated Pose Models,” IEEE/CVF Conference on Computer Vision and Pattern Reco
Jun. 2020, pp. 6184-6193. cited by applicant
W. Xu et al., “Mo2Cap2: Real-time Mobile 3D Motion Capture with a Cap-mounted Fisheye Camera,” arXiv:1803.05959v2, Jan. 23, 2019, 9 pages. cited by a
J. Yang et al., “. HybridTrak: Adding Full-Body Tracking to VR Using an Off-the-Shelf Webcam,” Proceedings of the 2022 CHI Conference on Human Factor

Systems, Apr. 2022, 13 pages. cited by applicant
F. Yin et al., "Accurate Estimation of Body Height From a Single Depth Image via a Four-Stage Developing Network," IEEE/CVF Conference on Computer V
Recognition (CVPR), Jun. 2020, 10 pages. cited by applicant
Y. Zhan et al., "Ray3D: Ray-based 3D Human Pose Estimation for Monocular Absolute 3D Localization," IEEE/CVF Conference on Computer Vision and Pa
(CVPR), Jun. 2022, pp. 13116-13125. cited by applicant
L. Zhao et al., "Semantic Graph Convolutional Networks for 3D Human Pose Regression," IEEE/CVF Conference on Computer Vision and Pattern Recogniti
2019, pp. 3425-3435. cited by applicant
M. Zhao et al., "Through-Wall Human Pose Estimation Using Radio Signals," IEEE/CVF Conference on Computer Vision and Pattern Recognition, Jun. 2018
cited by applicant
M. Zhao et al., "RF-Based 3D Skeletons," Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication, Aug. 2018, pp. 2
applicant
W. Zhao et al., "GraFormer: Graph Convolution Transformer for 3D Pose Estimation," IEEE/CVF Conference on Computer Vision and Pattern Recognition (
pp. 20438-20447. cited by applicant
Pytorch, "PyTorch Mobile," https://pytorch.org/mobile/home/, Accessed Oct. 3, 2024, 3 pages. cited by applicant
T. Ando et al., "CanalSense: Face-Related Movement Recognition System based on Sensing Air Pressure in Ear Canals," Proceedings of the 30th Annual ACM
User Interface Software and Technology, Oct. 2017, 11 pages. cited by applicant
M. T. I. Aumi et al., "DopLink: Using the Doppler Effect for Multi-Device Interaction," Proceedings of the 2013 ACM international Joint Conference on Perv
Ubiquitous Computing, Sep. 2013, 4 pages. cited by applicant
Bitfoic Electronics, "OWR-05049T-38D," https://www.bitfoic.com/detail/owr05049t38d-14578, Accessed Oct. 3, 2024, 4 pages. cited by applicant
T. Chen et al., "C-Face: Continuously Reconstructing Facial Expressions by Deep Learning Contours of the Face with Ear-mounted Miniature Cameras," Proc
33rd Annual ACM Symposium on User Interface Software and Technology, Oct. 2020, 14 pages. cited by applicant
Bose, "Bose Frames Tempo," https://support.bose.com/s/product/bose-frames-tempo/01t8c00000Oyd9mAAB?language=en_US, Accessed Oct. 3, 2024, 37 pa
applicant
R. Cowie et al., "Emotion Recognition in Human-Computer Interaction," IEEE Signal Processing Magazine, vol. 18, No. 1, Jan. 2001, pp. 33-80. cited by app
A. Dementyev et al., "DualBlink: A Wearable Device to Continuously Detect, Track, and Actuate Blinking For Alleviating Dry Eyes and Computer Vision Syr
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, Mar. 2017, vol. 1, No. 1, pp. 1:1-1:19. cited by applicant
Epson, "Moverio BT-35ES Smart Glasses," https://epson.com/For-Work/Wearables/Smart-Glasses/Moverio-BT-35ES-Smart-Glasses/p/V11H935020AL, Acce
5 pages. cited by applicant
P.-L. Hsieh et al., "Unconstrained Realtime Facial Performance Capture," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2015,
cited by applicant
Y. Hu et al., "Demo Abstract: Wireless Glasses for Non-contact Facial Expression Monitoring," 19th ACM/IEEE International Conference on Information Pro
Networks, Apr. 2020, pp. 367-368. cited by applicant
E. P. Ijjina et al., "Facial Expression Recognition using Kinect Depth Sensor and Convolutional Neural Networks," 13th International Conference on Machine
Applications, Dec. 2014, pp. 392-396. cited by applicant
Apple Inc., "Tracking and Visualizing Faces," https://developer.apple.com/documentation/arkit/arkit_in_ios/content_anchors/tracking_and_visualizing_faces,
2024, 8 pages. cited by applicant
TDK Corporation, "ICS-43434," https://invensense.tdk.com/products/ics-43434/, Accessed Oct. 3, 2024, 7 pages. cited by applicant
S. A. Kahou et al., "Combining Modality Specific Deep Neural Networks for Emotion Recognition in Video," Proceedings of the 15th ACM on International
Multimodal Interaction, Dec. 2013, 8 pages. cited by applicant
D. E. King, "Dlib-ml: A Machine Learning Toolkit," Journal of Machine Learning Research, vol. 10, Jul. 2009, pp. 1755-1758. cited by applicant
Knowles Electronics, "SR6438NWS-000," https://www.knowles.com/docs/default-source/model-downloads/sr6438nws-000.pdf, Accessed Oct. 3, 2024, 2 pag
applicant
J. Kwon et al., "Emotion Recognition Using a Glasses-Type Wearable Device via Multi-Channel Facial Responses," IEEE Access, vol. 9, Oct. 19, 2021, pp. 1
cited by applicant
Y.-H. Lai et al., "Emotion-Preserving Representation Learning via Generative Adversarial Network for Multi-view Facial Expression Recognition," 13th IEEE
Conference on Automatic Face & Gesture Recognition, May 2018, pp. 263-270. cited by applicant
M. Lankes et al., "Facial Expressions as Game Input with Different Emotional Feedback Conditions," Proceedings of the 2008 International Conference on Ad
Computer Entertainment Technology, Dec. 2008, pp. 253-256. cited by applicant
Lenovo, "ThinkReality A3 Smart Glasses," https://www.lenovo.com/US/en/p/smart-devices/virtual-reality/thinkreality-a3-pc-edition/wmd00000500, Accesse
pages. cited by applicant
J. Lian et al., "EchoSpot: Spotting Your Locations via Acoustic Sensing," Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technolo
pp. 113:1-113:21. cited by applicant
J. Liu et al., "BlinkListener: "Listen" to Your Eye Blink Using Your Smartphone," Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous T
5, No. 2, Jun. 2021, pp. 73:1-73:27. cited by applicant
M. Liu et al., "Non-Contact, Real-Time Eye Blink Detection with Capacitive Sensing," arXiv:2211.05541v1, Nov. 10, 2022, 5 pages. cited by applicant
M. Liu et al., "Learning Expressionlets on Spatio-Temporal Manifold for Dynamic Facial Expression Recognition," IEEE Conference on Computer Vision an
Recognition, Jun. 2014, pp. 4321-4328. cited by applicant
P. Liu et al., "Facial Expression Recognition via a Boosted Deep Belief Network," IEEE Conference on Computer Vision and Pattern Recognition, Jun. 2014,
applicant
L. Lu et al., "Lip Reading-Based User Authentication Through Acoustic Sensing on Smartphones," IEEE/ACM Transactions on Networking, vol. 27, No. 1, F
460. cited by applicant
D. J. C. Matthies et al., "EarFieldSensing: A Novel In-Ear Electric Field Sensing to Enrich Wearable Gesture Input through Facial Expressions," Proceedings o
Conference on Human Factors in Computing Systems, May 2017, pp. 1911-1922. cited by applicant
Niora, "Microsoft HoloLens," https://www.niora.net/en/p/microsoft_hololens, Accessed Oct. 3, 2024, 18 pages. cited by applicant
amazon.com, Inc., "Echo Frames and Carrera Smart Glasses Battery Life and Testing Information," https://www.amazon.com/gp/help/customer/display.html?
nodeId=GSVK3ZY3G43K435E, Accessed Oct. 3, 2024, 3 pages. cited by applicant
J. Park et al., "Dry Eye Syndrome in Thyroid Eye Disease Patients: The Role of iIncreased Incomplete Blinking and Meibomian Gland Loss," Acta Ophthalm
Aug. 2019, pp. e800-e806. cited by applicant
V. Rantanen et al., "Capacitive Facial Movement Detection for Human-Computer Interaction to Click By Frowning and Lifting Eyebrows," Medical and Biolo
and Computing, Dec. 17, 2009, 9 pages. cited by applicant
M. A. Ranzato et al., "On Deep Generative Models with Applications to Recognition," Conference on Computer Vision and Pattern Recognition (CVPR), Jun
2864. cited by applicant
S. Rifai et al., "Disentangling Factors of Variation for Facial Expression Recognition," European Conference on Computer Vision, Oct. 2012, pp. 802-822. cite
J. Scheirer et al., "Expression Glasses: A Wearable Device for Facial Expression Recognition," Proceedings of the SIGCHI conference on Human Factors in C
Systems, May 1999, pp. 262-263. cited by applicant

J. Scott-Thomas, “Ray-Ban Stories Smart Glasses use Two Frame Mounted Cameras to Capture Images in a First Step towards Augmented Reality,” https://www.techinsights.com/blog/ray-ban-stories-smart-glasses-cameras, Accessed Oct. 3, 2024, 6 pages. cited by applicant
N. Sebe et al., “Authentic Facial Expression Analysis,” Image and Vision Computing, vol. 25, No. 12, Dec. 3, 2007, 8 pages. cited by applicant
Nordic Semiconductor, “The Complete Guide to Bluetooth Low Energy,” https://response.nordicsemi.com/the-complete-guide-to-bluetooth-low-energy, Acce
21 pages. cited by applicant
B.-S. Lin et al., “Design of an Inertial-Sensor-Based Data Glove for Hand Function Evaluation,” Sensors, vol. 18, No. 1545, May 13, 2018, 17 pages. cited by
I. Loshchilov et al., “SGDR: Stochastic Gradient Descent with Warm Restarts,” Proceedings of the International Conference on Learning Representations, arX
May 3, 2017, 16 pages. cited by applicant
A. Krizhevsky et al., “ImageNet Classification with Deep Convolutional Neural Networks,” Advances in Neural Information Processing Systems, vol. 25, Dec
cited by applicant
I. Oikonomidis et al., “Efficient Model-based 3D Tracking of Hand Articulations using Kinect,” Proceedings of the British Machine Vision Conference, Sep. 2
cited by applicant
M. Z. Khan et al., “A Realistic Image Generation of Face From Text Description Using the Fully Trained Generative Adversarial Networks,” IEEE Access, vo
pp. 1250-1260. cited by applicant
C. R. Pittman et al., “Multiwave: Complex Hand Gesture Recognition Using the Doppler Effect,” Proceedings of the 43rd Graphics Interface Conference, Jun
cited by applicant
T. Hirahara et al., “Silent-speech Enhancement using Body-conducted Vocal-tract Resonance Signals,” Speech Communication, Apr. 2010, Author Manuscrip
by applicant
G. Rogez et al., “3D Hand Pose Detection in Egocentric RGB-D Images,” arXiv:1412.0065v1, Nov. 29, 2014, 14 pages. cited by applicant
A. Garg et al., “Lip Reading using CNN and LSTM,” Technical report, Stanford University, CS231, Mar. 2016, 9 pages. cited by applicant
K. Seo et al., “AirPincher: A Handheld Device for Recognizing Delicate Mid-Air Hand Gestures,” Proceedings of the Adjunct Publication of the 27th Annual
on User Interface Software and Technology, Oct. 2014, pp. 83-84. cited by applicant
T. Sharp et al., “Accurate, Robust, and Flexible Realtime Hand Tracking,” Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing
2015, 10 pages. cited by applicant
T. Simon et al., “Hand Keypoint Detection in Single Images using Multiview Bootstrapping,” Proceedings of the IEEE conference on Computer Vision and Pa
Jul. 2017, pp. 1145-1153. cited by applicant
A. Spurr et al., “Cross-modal Deep Variational Hand Pose Estimation,” Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Ju
cited by applicant
S. Sridhar et al., “WatchSense: On- and Above-Skin Input Sensing through a Wearable Depth Sensor,” CHI Conference on Human Factors in Computing Syst
pp. 3891-3902. cited by applicant
T. Stamer et al., “Real-Time American Sign Language Recognition Using Desk and Wearable Computer Based Video,” IEEE Transactions on Pattern Analysis
Intelligence, vol. 20, No. 12, Dec. 1998, pp. 1371-1375. cited by applicant
H. Su et al., “Multi-view Convolutional Neural Networks for 3D Shape Recognition,” Proceedings of IEEE International Conference on Computer Vision, De
953. cited by applicant
L. Sun et al., “WiDraw: Enabling Hands-free Drawing in the Air on Commodity WiFi Devices,” Proceedings of the 21st Annual International Conference on M
and Networking, Sep. 2015, 13 pages. cited by applicant
J. Taylor et al. “Efficient and Precise Interactive Hand Tracking Through Joint, Continuous Optimization of Pose and Correspondences,” ACM Transactions o
35, No. 4, Jul. 2016, 12 pages. cited by applicant
J. Tompson et al., “Real-Time Continuous Pose Recovery of Human Hands Using Convolutional Networks,” ACM Transactions on Graphics, vol. 33, No. 5, A
pages. cited by applicant
A. Vardy et al., “The WristCam as Input Device,” International Symposium on Wearable Computers, Oct. 1999, 4 pages. cited by applicant
T. Li et al., “Reconstructing Hand Poses Using Visible Light,” PACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 1, No. 3, Article 71
pages. cited by applicant
Y. Fan et al., “Video-Based Emotion Recognition using CNN-RNN and C3D Hybrid Networks,” Proceedings of the 18th ACM International Conference on M
Interaction, Oct. 2016, 6 pages. cited by applicant
A. Bozhurt et al., “Safety Assessment of Near Infrared Light Emitting Diodes for Diffuse Optical Measurements,” BioMedical Engineering OnLine, Mar. 22,
cited by applicant
Y. Zhang et al., “Wearing-independent Hand Gesture Recognition Method Based on EMG Armband,” Personal and Ubiquitous Computing, vol. 22, May 11, 2
cited by applicant
S. Hickson et al., “Eyemotion: Classifying Facial Expressions in VR Using Eye-tracking Cameras,” arXiv:1707.07204v2, Jul. 28, 2017, 10 pages. cited by app
U.S. Environment Protection Agency Office of Noise Abatement and Control, “Information on Levels of Environmental Noise Requisite to Protect Public Hea
with Adequate Margin of Safety,” EPA/ONAC 550/9-74-004, Mar. 1974, 242 pages. cited by applicant
Y. Zhao et al., “Demo: Finger and Hand Gesture Recognition using Smartwatch,” 13th Annual International Conference on Mobile Systems, Applications, and
2015, p. 471. cited by applicant
L. E. Emokpae et al., “UREAL: Underwater Reflection Enabled Acoustic-based Localization,” IEEE Sensors Journal, Nov. 2014, 11 pages. cited by applicant
D. Ashbrook et al., “Nenya: Subtle and Eyes-Free Mobile Input with a Magnetically-Tracked Finger Ring,” Proceedings of the SIGCHI Conference on Huma
Computing Systems, May 2011, pp. 2043-2046. cited by applicant
D. A. Bowman et al., “Design and Evaluation of Menu Systems for Immersive Virtual Environments,” Proceedings IEEE Virtual Reality, Mar. 2001, pp. 149-1
applicant
D. Barolet et al., “Infrared and Skin: Friend or Foe,” Journal of Photochemistry & Photobiology, B: Biology, vol. 155, Feb. 2016, pp. 78-85. cited by applican
K.-Y. Chen et al., “uTrack: 3D Input Using Two Magnetic Sensors,” Proceedings of the 26th Annual ACM Symposium on User Interface Software & Technol
237-244. cited by applicant
H. Liang et al., “AR in Hand: Egocentric Palm Pose Tracking and Gesture Recognition for Augmented Reality Applications,” Proceedings of the 23rd ACM i
conference on Multimedia, Oct. 2015, pp. 743-744. cited by applicant
M. Fan et al., “Eyelid Gestures on Mobile Devices for People with Motor Impairments,” Proceedings of the 22nd International Acm Sigaccess Conference on
Accessibility, Oct. 2020, No. 15, 8 pages. cited by applicant
J. S. Brumberg et al., “Brain-Computer Interfaces for Speech Communication,” Speech Communication, Apr. 2010, Author Manuscript, 22 pages. cited by app
W. Chan et al., “Listen, Attend and Spell,” arXiv:1508.01211v2, Aug. 20, 2015, 16 pages. cited by applicant
A. Gupta et al., “RotoSwype: Word-Gesture Typing using a Ring,” Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, May 2
cited by applicant
J. Chorowski et al., “Attention-Based Models for Speech Recognition,” arXiv:1506.07503v1, Jun. 24, 2015, 19 pages. cited by applicant
D.-Y. Huang et al., “DigitSpace: Designing Thumb-to-Fingers Touch Interfaces for One-Handed and Eyes-Free Interactions,” Proceedings of the 2016 CHI Co
Human Factors in Computing Systems, May 2016, pp. 1526-1537. cited by applicant
M. Kianpisheh et al., “Face Recognition Assistant for People with Visual Impairments,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiqu
Sep. 9, 2019, 26 pages. cited by applicant

W. Kienzle et al., "LightRing: Always-Available 2D Input on Any Surface," Proceedings of the 27th Annual ACM Symposium on User Interface Software and
2014, 4 pages. cited by applicant
Y. Kubo et al., "Audio Touch: Minimally Invasive Sensing of Micro-Gestures via Active Bio-Acoustic Sensing," Proceedings of the 21st International Confer
Computer Interaction with Mobile Devices and Services, Oct. 2019, No. 36, 13 pages. cited by applicant
A. Kulshreshth et al., "Exploring the Usefulness of Finger-Based 3D Gesture Menu Selection," Proceedings of the SIGCHI Conference on Human Factors in
Systems, Apr. 2014, pp. 1093-1102. cited by applicant
V. Cuculo et al., "OpenFACS : An Open Source FACS-based 3D Face Animation System," International Conference on Image and Graphics, Aug. 2019, 11 pa
applicant
F. M. Li et al, "FMT: A Wearable Camera-Based Object Tracking Memory Aid for Older Adults, " Proceedings of the ACM on Interactive, Mobile, Wearable a
Technologies, Sep. 2019, 25 pages. cited by applicant
M. Li et al., "BrailleSketch: A Gesture-based Text Input Method for People with Visual Impairments," Proceedings of the 19th International ACM SIGACCES
Computers and Accessibility, Oct. 2017, pp. 12-21. cited by applicant
C. Loclair et al., "PinchWatch: A Wearable Device for One-Handed Microinteractions," 12th International Conference on Human Computer Interaction with M
Services, Sep. 2010, 4 pages. cited by applicant
Z. Lu et al., "A Hand Gesture Recognition Framework and Wearable Gesture-Based Interaction Prototype for Mobile Devices," IEEE Transactions on Human
vol. 44, No. 2, Apr. 2014, pp. 293-299. cited by applicant
A. Mujibiya et al., "The Sound of Touch: On-body Touch and Gesture Sensing Based on Transdermal Ultrasound Propagation," Proceedings of the 2013 ACM
Conference on Interactive Tabletops and Surfaces, Oct. 2013, pp. 189-198. cited by applicant
B. Denby et al., "Silent Speech Interfaces," Speech Communication, Apr. 2011, Accepted Manuscript, 64 pages. cited by applicant
M. El-Gohary et al., "Shoulder and Elbow Joint Angle Tracking with Inertial Sensors," IEEE Transactions on Biomedical Engineering, vol. 59, No. 9, Sep. 20
cited by applicant
H. Lim et al., "BodyTrak: Inferring Full-Body Poses from Body Silhouettes Using a Miniature Camera on a Wristband," Proceedings of the ACM on Interacti
Wearable and Ubiquitous Technologies, vol. 6, No. 3, Sep. 2022, p. 154:1-21. cited by applicant
J. Roggendorf et al., "Arm Swing Asymmetry in Parkinson's Disease Measured with Ultrasound Based Motion Analysis during Treadmill Gait," Gait Posture,
pp. 116-120. cited by applicant
Y. Gao et al., "SonicFace: Tracking Facial Expressions Using a Commodity Microphone Array," . Proceedings of the ACM on Interactive, Mobile, Wearable a
Technologies, vol. 5, No. 4, Dec. 2021, pp. 156:1-156:33. cited by applicant
A. Gruebler et al., "Measurement of Distal EMG Signals Using a Wearable Device for Reading Facial Expressions," 32nd Annual International Conference of
Engineering in Medicine and Biology, Aug.-Sep. 2010, pp. 4594-4597. cited by applicant
S. He et al., "Facial Expression Recognition using Deep Boltzmann Machine from Thermal Infrared Images," 2013 Humaine Association Conference on Affe
and Intelligent Interaction, Sep. 2013, pp. 239-244. cited by applicant
K. Masai et al., "Facial Expression Recognition in Daily Life by Embedded Photo Reflective Sensors on Smart Eyewear," Proceedings of the International Co
Intelligent User Interfaces, Mar. 2016, pp. 317-326. cited by applicant
D. B. M. Yin et al., "A Proposed Approach for Biometric-Based Authentication Using of Face and Facial Expression Recognition," IEEE International Confer
Communication and Information Systems, Dec. 2018, pp. 28-33. cited by applicant
J. A. M. Paro et al., "Video-Based Self-Review: Comparing Google Glass and GoPro Technologies," Annals of Plastic Surgery, vol. 74, Supplement 1, May 20
cited by applicant
J.-W. Lin et al., "BackHand: Sensing Hand Gestures via Back of the Hand," Proceedings of the 28th Annual ACM Symposium on User Interface Software & T
2015, pp. 557-564. cited by applicant
J. Lai et al., "AcousticPose: Acoustic-Based Human Pose Estimation," Wireless Sensor Networks, Jan. 2022, pp. 59-62. cited by applicant
D. Purves est al, "The Audible Spectrum," Neuroscience, 2001, Abstract Only. cited by applicant
T. Chen et al., "NeckFace: Continuously Tracking Full Facial Expressions on Neck-mounted Wearables," Proceedings of the ACM on Interactive, Mobile, We
Ubiquitous Technologies, vol. 5, No. 2, Abstract Only. cited by applicant
J. McIntosh et al., "SensIR: Detecting Hand Gestures with a Wearable Bracelet using Infrared Transmission and Reflection," Proceedings of the 30th Annual A
on User Interface Software and Technology, Oct. 2017, pp. 593-597. cited by applicant
S. Oh et al., "VibEye: Vibration-Mediated Object Recognition for Tangible Interactive Applications," Proceedings of the 2019 CHI Conference on Human Fac
Systems, No. 676, May 2019, pp. 1-12. cited by applicant
J. Xie et al., "mm3DFace: Nonintrusive 3D Facial Reconstruction Leveraging mmWave Signals," Proceedings of the Annual International Conference on Mob
Applications and Services, Jun. 2023, pp. 462-474. cited by applicant
N. Harte et al., "TCD-TIMIT: An Audio-Visual Corpus of Continuous Speech," IEEE Transactions on Multimedia, vol. 17, No. 5, May 2015, pp. 603-615. cite
N. Kimura et al., "TieLent: A Casual Neck-Mounted Mouth Capturing Device for Silent Speech Interaction," Proceedings of the International Conference on A
Interfaces, Sep.-Oct. 2020, 8 pages. cited by applicant
H. Manabe et al., "Unvoiced Speech Recognition using EMG—Mime Speech Recognition," Conference on Human Factors in Computing Systems, Apr. 2003
cited by applicant
T. Schultz, "ICCHP Keynote: Recognizing Silent and Weak Speech Based on Electromyography," International Conference of Computers Helping People wit
Jul. 2010, pp. 595-604. cited by applicant
K. Xu et al., "Ultrasound-Based Silent Speech Interface Using Sequential Convolutional Auto-Encoder," Proceedings of the 27th ACM International Conferen
Oct. 2019, pp. 2194-2195. cited by applicant
M. Hosseini et al., "Towards a Consensus Gesture Set: A Survey of Mid-Air Gestures in HCI for Maximized Agreement Across Domains," Proceedings of the
Conference on Human Factors in Computing Systems, Apr. 2023, 24 pages. cited by applicant
W. Mao et al., "CAT: High-Precision Acoustic Motion Tracking," Proceedings of the 22nd Annual International Conference on Mobile Computing and Netwo
13 pages. cited by applicant
R. Takahashi et al., "TelemetRing: A Batteryless and Wireless Ring-Shaped Keyboard Using Passive Inductive Telemetry," Proceedings of the 33rd Annual AC
User Interface Software and Technology, Oct. 2020, pp. 1161-1168. cited by applicant
Y. Jin et al., "SonicASL: An Acoustic-based Sign Language Gesture Recognizer Using Earphones," Proceedings of the ACM on Interactive, Mobile, Wearable
Technologies, vol. 5, No. 2, Jun. 2021, pp. 67:1-67:30. cited by applicant
W. Ruan et al., "AudioGest: Enabling Fine-Grained Hand Gesture," Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitou
2016, pp. 474-485. cited by applicant
H. Hiraki et al., "SilentMask: Mask-Type Silent Speech Interface with Measurement of Mouth Movement," Augmented Humans Conference, Feb. 2021, pp. 8
applicant
Y. Igarashi et al., "Silent Speech Eyewear Interface: Silent Speech Recognition Method Using Eyewear and an Ear-Mounted Microphone with Infrared Distan
Sensors, vol. 24, No. 7368, Nov. 19, 2024, 16 pages. cited by applicant
J. Rekimoto et al., "Derma: Silent Speech Interaction Using Transcutaneous Motion Sensing," Augmented Humans Conference, Feb. 2021, 10 pages. cited by
C. Hennessey et al., "Long Range Eye Tracking: Bringing Eye Tracking into the Living Room," Proceedings of the Symposium on Eye Tracking Research and
Mar. 2012, pp. 249-252. cited by applicant

B. Noris et al., "A Wearable Gaze Tracking System for Children in Unconstrained Environments," Computer Vision and Image Understanding, vol. 115, No. 4
476-486. cited by applicant
S. Masood et al., "Real-Time Sign Language Gesture (Word) Recognition from Video Sequences Using CNN and RNN," Intelligent Engineering Informatics,
623-632. cited by applicant
J. Cha et al., "An IR-based Facial Expression Tracking Sensor for Head-Mounted Displays," IEEE Sensors, Oct. 2016, 3 pages. cited by applicant
H. R. Schiffman, "Sensation and Perception: An Integrated Approach," John Wiley & Sons, 1990, Excerpts Only. cited by applicant
Nordic Semiconductor, "nRF52840," Product Specification, vol. 1.10, https://www.nordicsemi.com/Products/nRF52840, Jul. 4, 2024 980 pages. cited by appl
K. Sun et al., "VSkin: Sensing Touch Gestures on Surfaces of Mobile Devices Using Acoustic Signals," Proceedings of the 24th Annual International Confere
Computing and Networking, Oct. 2018, 15 pages. cited by applicant
J. Thies et al., "Real-time Expression Transfer for Facial Reenactment," ACM Transactions on Graphics (TOG), vol. 34, No. 6, 14 pages. cited by applicant
U. Von Agris et al., "The Significance of Facial Features for Automatic Sign Language Recognition," 8th IEEE International Conference on Automatic Face &
Recognition, Sep. 2008, 6 pages. cited by applicant
S.-E. Wei et al., "VR Facial Animation via Multiview Image Translation," ACM Transactions on Graphics (TOG), vol. 38, No. 4, Jul. 2019, pp. 67:1-67:16. ci
Wikipedia, "Google Glass," https://en.wikipedia.org/wiki/Google_Glass, Sep. 9, 2024, 28 pages. cited by applicant
C. Wu et al., "An Anatomically-constrained Local Deformation Model for Monocular Face Capture," ACM Transactions on Graphics (TOG), vol. 35, No. 4, J
cited by applicant
W. Wu et al., "Look at Boundary: A Boundary-Aware Face Alignment Algorithm," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVP
2129-2138. cited by applicant
Y. Wu et al., "BioFace-3D: Continuous 3d Facial Reconstruction through Lightweight Single-Ear Biosensors," Proceedings of the 27th Annual International C
Mobile Computing and Networking, Jan. 2022, pp. 350-363. cited by applicant
S. Xiong et al., "iBlink: Smart Glasses for Facial Paralysis Patients," Proceedings of the 15th Annual International Conference on Mobile Systems, Applicatio
Jun. 2017, pp. 359-370. cited by applicant
X. Xu et al., "EarBuddy: Enabling On-Face Interaction via Wireless Earbuds," Proceedings of the 2020 CHI Conference on Human Factors in Computing Sys
14 pages. cited by applicant
S. Yun et al., "Strata: Fine-Grained Acoustic-based Device-Free Tracking," Proceedings of the 15th Annual International Conference on Mobile Systems, App
Services, Jun. 2017, 14 pages. cited by applicant
Y. Zhang et al., "Endophasia: Utilizing Acoustic-Based Imaging for Issuing Contact-Free Silent Speech Commands," Proceedings of the ACM on Interactive,
and Ubiquitous Technologies, vol. 4, No. 1, Mar. 2020, pp. 37:1-37:26. cited by applicant
Y. Zhang et al., "Vernier: Accurate and Fast Acoustic Motion Tracking Using Mobile Devices, " IEEE Conference on Computer Communications, Apr. 2018,
cited by applicant
B. Amento et al., "The Sound of One Hand: A Wrist-Mounted Bio-Acoustic Fingertip Gesture Interface," CHI '02 Extended Abstracts on Human Factors in Co
Apr. 2002, pp. 724-725. cited by applicant
S. Bhattacharya et al., "Leveraging Sound and Wrist Motion to Detect Activities of Daily Living with Commodity Smartwatches," Proceedings of the ACM o
Mobile, Wearable and Ubiquitous Technologies, vol. 6, No. 2, Jun. 2022, pp. 42:1-42:28. cited by applicant
W. Chen et al., "ViFin: Harness Passive Vibration to Continuous Micro Finger Writing with a Commodity Smartwatch," Proceedings of the ACM on Interactiv
Wearable and Ubiquitous Technologies, vol. 5, No. 1, Jun. 2022, pp. 46:1-46:25. cited by applicant
D. Damen et al., "Scaling Egocentric Vision: The Epic-Kitchens Dataset," European Conference on Computer Vision, Sep. 2018, 17 pages. cited by applicant
D. Damen et al., "Rescaling Egocentric Vision: Collection, Pipeline and Challenges for Epic-Kitchens-100," International Journal of Computer Vision, Oct. 2
pp. 33-55. cited by applicant
P. Das et al., "A Thousand Frames in Just a Few Words: Lingual Description of Videos through Latent Topics and Sparse Object Stitching," IEEE Conference
Vision and Pattern Recognition, Oct. 2013, pp. 2634-2641. cited by applicant
A. Dementyev et al., "WristFlex: Low-Power Gesture Input with Wrist-Worn Pressure Sensors," Proceedings of the 27th Annual ACM Symposium on User In
and Technology, Oct. 2014, 6 pages. cited by applicant
N. Devrio et al., "DiscoBand: Multiview Depth-Sensing Smartwatch Strap for Hand, Body and Environment Tracking," The 35th Annual ACM Symposium o
Software and Technology, Oct. 2022, 13 pages. cited by applicant
T. Deyle et al., "Hambone: A Bio-Acoustic Gesture Interface," 11th IEEE International Symposium on Wearable Computers, Oct. 2007, 8 pages. cited by appl
Y. Du et al., "Semi-Supervised Learning for Surface EMG-based Gesture Recognition," Proceedings of the Twenty-Sixth International Joint Conference on Ar
Intelligence, Aug. 2017, pp. 1624-1630. cited by applicant
J. Fan et al., "What is That in Your Hand? Recognizing Grasped Objects via Forearm Electromyography Sensing," Proceedings of the ACM on Interactive, M
and Ubiquitous Technologies, vol. 2, No. 4, Dec. 2018, pp. 161:1-161:24. cited by applicant
L. Ge et al., "3D Hand Shape and Pose Estimation from a Single RGB Image," IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR),
10833-10842. cited by applicant
J. Gong et al., "Acustico: Surface Tap Detection and Localization Using Wrist-Based Acoustic TDOA Sensing," Proceedings of the 33rd Annual ACM Sympo
Interface Software and Technology, Oct. 2020, pp. 406-441. cited by applicant
J. Gong et al., "WristWhirl: One-Handed Continuous Smartwatch Input Using Wrist Gestures," Proceedings of the 29th Annual Symposium on User Interface
Technology, Oct. 2016, 12 pages. cited by applicant
C. Harrison et al., "Skinput: Appropriating the Body as an Input Surface," Proceedings of the SIGCHI Conference on Human Factors in Computing Systems,
462. cited by applicant
F. Hu et al., "FingerTrak: Continuous 3D Hand Pose Tracking by Deep Learning Hand Silhouettes Captured by Miniature Thermal Cameras on Wrist," Procee
on Interactive, Mobile, Wearable and Ubiquitous Technologies, vol. 4, No. 2, Jun. 2020, pp. 71:1-71:24. cited by applicant
Tap Systems, Inc., "Tap Strap 2," https://www.tapwithus.com/product/tap-strap-2/, Accessed Oct. 7, 2024, 3 pages. cited by applicant
Intel, "Intel RealSense Technology," https://www.intel.com/content/www/us/en/architecture-and-technology/realsense-overview.html, Accessed Oct. 7, 2024,
applicant
Y. Iravantchi et al., "BeamBand: Hand Gesture Sensing with Ultrasonic Beamforming," Proceedings of the 2019 CHI Conference on Human Factors in Comp
May 2019, 10 pages. cited by applicant
F. Kerber et al., "User-Independent Real-Time Hand Gesture Recognition Based on Surface Electromyography," Proceedings of the 19th International Confere
Computer Interaction with Mobile Devices and Services, Sep. 2017, 7 pages. cited by applicant
D. Kim et al., "EtherPose: Continuous Hand Pose Tracking with Wrist-Worn Antenna Impedance Characteristic Sensing," The 35th Annual ACM Symposium
Software and Technology, Oct. 2022, 12 pages. cited by applicant
D. Kim et al., "Digits: Freehand 3D Interactions Anywhere Using a Wrist-Worn Gloveless Sensor," Proceedings of the 25th Annual ACM Symposium on User
Software and Technology, Oct. 2012, pp. 167-176. cited by applicant
J. Kim et al., "The Gesture Watch: A Wireless Contact-free Gesture based Wrist Interface," 11th IEEE International Symposium on Wearable Computers, Oct.
cited by applicant
H. Lim et al., "D-Touch: Recognizing and Predicting Fine-Grained Hand-Face Touching Activities Using a Neck-Mounted Wearable," Proceedings of the 28th
Conference on Intelligent User Interfaces, Mar. 2023, pp. 569-583. cited by applicant

Y. Liu et al., “WR-Hand: Wearable Armband Can Track User's Hand,” Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologie Sep. 2021, pp. 118:1-118:27. cited by applicant
Y. Liu et al., “NeuroPose: 3D Hand Pose Tracking Using EMG Wearables,” Proceedings of the Web Conference 2021, Apr. 2021, 12 pages. cited by applicant
C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” arXiv:1906.08172v1, Jun. 14, 2019, 9 pages. cited by applicant
J. McIntosh et al., “EchoFlex: Hand Gesture Recognition Using Ultrasound Imaging,” Proceedings of the 2017 CHI Conference on Human Factors in Comput 2017, pp. 1923-1934. cited by applicant
Microsoft, “Kinect for Windows,” https://learn.microsoft.com/en-us/windows/apps/design/devices/kinect-for-windows, Aug. 30, 2022, 3 pages. cited by appli
F. Mueller et al., “GANerated Hands for Real-Time 3D Hand Tracking from Monocular Rgb,” IEEE/CVF Conference on Computer Vision and Pattern Recog pp. 49-59. cited by applicant
K. Ohnishi et al., “Recognizing Activities of Daily Living with a Wrist-Mounted Camera,” IEEE Conference on Computer Vision and Pattern Recognition (C pp. 3103-3111. cited by applicant
I. Oikonomidis et al., “Full DOF Tracking of a Hand Interacting with an Object by Modeling Occlusions and Physical Constraints,” International Conference Vision, Nov. 2011, 8 pages. cited by applicant
F. S. Parizi et al., “AuraRing: Precise Electromagnetic Finger Tracking,” Communications of the ACM, vol. 65, No. 10, Oct. 2022, pp. 85-92. cited by applica
J. Park et al., “HandOccNet: Occlusion-Robust 3D Hand Mesh Estimation Network,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (C pp. 1496-1505. cited by applicant
J. K. Perng et al., “Acceleration Sensing Glove,” Third International Symposium on Wearable Computers, Oct. 1999, 3 pages. cited by applicant
F. Quivira et al., “Translating sEMG Signals to Continuous Hand Poses using Recurrent Neural Networks,” IEEE Embs International Conference on Biomedi Informatics (BHI), Mar. 2018, 4 pages. cited by applicant
S. Shi et al., “Passive Detection of Situations from Ambient FM-radio Signals,” Proceedings of the 2012 ACM Conference on Ubiquitous Computing, Sep. 20 cited by applicant
J. R. Smith et al., “RFID-Based Techniques for Human-Activity Detection,” Communications of the ACM, vol. 28, No. 9, Sep. 2005, pp. 39-44. cited by appli
A. Spielberg et al., “RapID: A Framework for Fabricating Low-Latency Interactive Objects with RFID Tags,” Proceedings of the 2016 CHI Conference on Hu Computing Systems, May 2016, 12 pages. cited by applicant
Y. Yokokohji et al., “Motion Capture from Demonstrator's Viewpoint and Its Application to Robot Teaching,” IEEE International Conference on Robotics and 2002, pp. 1551-1558. cited by applicant
W. J. Staszewski et al., “Structural Health Monitoring using Scanning Laser Vibrometry: I. Lamb Wave Sensing,” Smart Materials and Structures, vol. 13, No pp. 251-260. cited by applicant
J. Sung et al., “Unstructured Human Activity Detection from RGBD Images,” arXiv:1107.0169v2, Feb. 14, 2012, 8 pages. cited by applicant
H. Tabatabai et al., “Novel Applications of Laser Doppler Vibration Measurements to Medical Imaging,” Sensing and Imaging: An International Journal, vol. pp. 13-28. cited by applicant
E. M. Tapia et al., “Activity Recognition in the Home Setting Using Simple and Ubiquitous Sensors,” International Conference on Pervasive Computing, Apr. cited by applicant
E. M. Tapia et al., “Portable Wireless Sensors for Object Usage Sensing in the Home: Challenges and Practicalities,” European Conference on Ambient Intelli 18 pages. cited by applicant
A. A. Veber et al, “Laser Vibrometry Based on Analysis of the Speckle Pattern from a Remote Object,” Applied Physics B, vol. 105, Jun. 4, 2011, pp. 613-617
J. F. Vignola et al., “Characterization of Silicon Micro-Oscillators by Scanning Laser Vibrometry,” Review of Scientific Instruments, vol. 73, No. 10, Oct. 200 cited by applicant
H. Wang et al., “Human Respiration Detection with Commodity WiFi Devices: Do User Location and Body Orientation Matter?” Proceedings of the 2016 AC Joint Conference on Pervasive and Ubiquitous Computing, Sep. 2016, pp. 25-36. cited by applicant
H. Wang et al., “RT-Fall: A Real-Time and Contactless Fall Detection System with Commodity WiFi Devices,” IEEE Transactions on Mobile Computing, vol. 2017, pp. 511-526. cited by applicant
Z. Wang et al., “Wi-Fi CSI based Behavior Recognition: From Signals, Actions to Activities,” IEEE Communications Magazine, vol. 56, No. 5, Nov. 2017, 10 applicant
D. Wilson et al., “Simultaneous Tracking and Activity Recognition (STAR) Using Many Anonymous, Binary Sensors,” International Conference on Pervasive 8, 2005, 18 pages. cited by applicant
R. Zhang et al., “Retrieval and Timing Performance of Chewing-Based Eating Event Detection in Wearable Sensors,” Sensors, vol. 20, No. 2, Jan. 20, 2020, 1 applicant
Z. Zalevsky et al., “Simultaneous Remote Extraction of Multiple Speech Sources and Heart Beats from Secondary Speckles Pattern,” Optics Express, vol. 17, 2009, 15 pages. cited by applicant
C. Zhang et al., “RGB-D Camera-based Daily Living Activity Recognition,” Signal & Information Processing Association Annual Summit and Conference, Ja cited by applicant
Y. Zhang et al., “Vibrosight: Long-Range Vibrometry for Smart Environment Sensing,” The 31st Annual ACM Symposium on User Interface Software and Te 2018, 12 pages. cited by applicant
Y. Zhang et al., “Occupant Activity Level Estimation Using Floor Vibration,” Proceedings of the 2018 ACM International Joint Conference and 2018 Internati on Pervasive and Ubiquitous Computing and Wearable Computers, Oct. 2018, pp. 1355-1363. cited by applicant
Y. Zhang et al., “Wall++ Room-Scale Interactive and Context-Aware Sensing,” Proceedings of the 2018 CHI Conference on Human Factors in Computing Sys 15 pages. cited by applicant
Z. Zhou et al., “Activity Analysis, Summarization, and Visualization for Indoor Human Activity Monitoring,” IEEE Transactions on Circuits and Systems for vol. 18, No. 11, Nov. 2008, pp. 1489-1498. cited by applicant
L. Zipser et al., “Laser-scanning Vibrometry for Ultrasonic Transducer Development,” Sensors and Actuators A: Physical, vol. 110, Feb. 2004, pp. 264-268. ci
G. Bailly et al., “ShoeSense: A New Perspective on Hand Gestures and Wearable Applications,” Proceedings of the SIGCHI Conference on Human Factors in Systems, May 2012, pp. 1239-1248. cited by applicant
K.-M. Cheung et al., “Shape-From-Silhouette Across Time Part I: Theory and Algorithms,” International Journal of Computer Vision, vol. 62, Nov. 1, 2004, 5 applicant
K.-M. Cheung et al., “Shape-From-Silhouette Across Time Part II: Applications to Human Modeling and Markerless Motion Tracking,” International Journal Vision, vol. 63, Apr. 1, 2005, 39 pages. cited by applicant
L. Ballan et al., “Motion Capture of Hands in Action using Discriminative Salient Points,” European Conference on Computer Vision, Oct. 2012, 14 pages. cit
B. G. Baumgart, “Geometric Modeling for Computer Vision,” Technical Report, Stanford University, Prepared for Office of Naval Research, Oct. 1974, 144 p applicant
J.-B. Chossat et al., “Wearable Soft Artificial Skin for Hand Motion Detection with Embedded Microfluidic Strain Sensing,” IEEE International Conference o Automation, May 2015, pp. 2568-2573. cited by applicant
S. Ciotti et al., “A Synergy-Based Optimally Designed Sensing Glove for Functional Grasp Recognition,” Sensors, vol. 16, No. 811, Jun. 2, 2016, 17 pages. ci
R. Zhang et al., “Diet Eyeglasses: Recognising Food Chewing Using EMG and Smart Eyeglasses,” 13th International Conference on Wearable and Implantab Networks, Jun. 2016, 6 pages. cited by applicant

M. De La Gorce et al., “Model-Based 3D Hand Pose Estimation from Monocular Video,” IEEE Transactions on Pattern Analysis and Machine Intelligence, Se
pages. cited by applicant
S. Zhang et al., “When Generalized Eating Detection Machine Learning Models Fail in the Field,” Proceedings of the 2017 ACM International Joint Conferen
and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers, Sep. 2017, pp. 613-622. cited by applicant
G. Du et al., “Markerless Kinect-Based Hand Tracking for Robot Teleoperation,” International Journal of Advanced Robotic Systems, vol. 9, Jul. 13, 2012, 10
applicant
R. Fukui et al., “Hand Shape Classification with a Wrist Contour Sensor: Development of a Prototype Device,” Proceedings of the 13th International Conferen
Computing, Sep. 2011, pp. 311-314. cited by applicant
L. Ge et al., “Hand PointNet: 3D Hand Pose Estimation using Point Sets,” Conference on Computer Vision and Pattern Recognition, Jun. 2018, pp. 8417-8426
applicant
S. Zhang et al., “I Sense Overeating: Motif-based Machine Learning Framework to Detect Overeating Using Wrist-worn Sensing,” Procedia Computer Scienc
pp. 1-16. cited by applicant
S. Zhang et al., “NeckSense: A Multi-Sensor Necklace for Detecting Eating Activities in Free-Living Conditions,” Proceedings of the ACM on Interactive, Mo
Ubiquitous Technologies, vol. 4, No. 2, Jun. 2020, 47 pages. cited by applicant
O. Glauser et al., “A Stretch-Sensing Soft Glove for Interactive Hand Pose Estimation,” ACM SIGGRAPH Emerging Technologies, Aug. 2019, 2 pages. cited
M. Holl et al., “Efficient Physics-Based Implementation for Realistic Hand-Object Interaction in Virtual Reality,” IEEE Annual International Symposium Virtu
2018, 8 pages. cited by applicant
P. J. Huber, “Robust Estimation of a Location Parameter,” Annals of Mathematical Statistics, vol. 335, No. 1, Mar. 1964, pp. 73-101. cited by applicant
U. Iqbal et al., “Hand Pose Estimation via Latent 2.5D Heatmap Regression,” European Conference on Computer Vision, Sep. 2018, 17 pages. cited by applic
B. Kellogg et al., “Bringing Gesture Recognition to All Devices,” 11th USENIX Symposium on Networked Systems Design and Implementation, Apr. 2014, p
by applicant
S. Hochreiter et al., “Long Short-term Memory,” Neural Computation, vol. 9, No. 8, Nov. 15, 1997, pp. 1735-1780. cited by applicant
H. Li et al., “Facial Performance Sensing Head-Mounted Display,” ACM Transactions on Graphics, vol. 34, No. 4, Jul. 27, 2015, 9 pages. cited by applicant
J. A. Russell, “Is There Universal Recognition of Emotion From Facial Expression? A Review of the Cross-Cultural Studies,” Psychological Bulletin, vol. 115
pp. 102-141. cited by applicant
T. Maekawa et al., “WristSense: Wrist-worn Sensor Device with Camera for Daily Activity Recognition,” Mar. 2012, pp. 510-512. cited by applicant
J. Li et al., “Identification and Classification of Construction Equipment Operators' Mental Fatigue Using Wearable Eye-tracking Technology,” Automation in
Published Version, Jul. 2020, 36 pages. cited by applicant
Y. Jin et al., “TransASL: A Smart Glass based Comprehensive ASL Recognizer in Daily Life,” Proceedings of the 28th International Conference on Intelligent
Mar. 2023, pp. 802-818. cited by applicant
S. Iwakiri et al., “User Authentication Method for Wearable Ring Devices using Active Acoustic Sensing,” Proceedings of the 2023 ACM International Sympo
Computers, Oct. 2023, pp. 17-21. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/343,023, filed May 17, 2022 and entitled “Wearable Facial Movement Tracking Devices” and is a continuation-in-part of PCT Patent Application No. PCT/US2021/032511, filed May 14, 2021 and entitled “Wearable Devices For Facial Expression Recognition,” which claims the benefit of U.S. Provisional Patent Application No. 63/025,979, filed May 15, 2020 and entitled “C-Face: Continuously Reconstructing Facial Expressions By Deep Learning Contours Of The Face With Ear-Mounted Miniature Cameras.” The entire contents of the above-identified priority applications are hereby fully incorporated herein by reference.

TECHNICAL FIELD
(1) This disclosure relates to systems and methods to reconstruct facial expressions and to track facial movements from head or neck-mounted wearable devices.
BACKGROUND
(2) Humans use facial expressions as a natural mode of communication. The ability to continuously record and understand facial movements can improve interactions between humans and computers in a variety of applications.
(3) Conventional facial reconstruction methods require a camera to be positioned in front of a user's face with a specified position and angle relative to the user's face. To achieve reliable facial reconstruction, the camera needs an entire view of the face without occlusions. Conventional facial reconstruction methods do not perform well if the user is in motion, the camera is not appropriately set up, the camera is not in front of the user, or the user's face is partially occluded or not fully visible to the camera due to the camera's position or angle relative to the user's face.
(4) As an alternative to frontal camera systems, wearable devices for facial expression reconstruction have been developed using sensing techniques, such as acoustic interference, pressure sensors, electrical impedance tomography, and electromyography. These wearable devices use instrumentation that is mounted directly on a user's face. These conventional devices often cover the user's face and only recognize discrete facial expressions. Examples of these conventional wearable devices include face masks with built-in ultrasonic transducers or electrodes secured to a human face with electromyography or capacitive sensing abilities. These wearable devices are attached directly to the user's face or body and may block the field of vision and interfere with normal daily activities, such as eating or socializing.
(5) Another alternative to frontal camera systems is smart eyewear including smart glasses, augmented reality glasses, and virtual reality headsets. However, these smart eyewear devices cannot track high-quality facial movements continuously. For example, virtual reality devices cannot depict 3D avatars in virtual worlds with facial expressions of the user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram depicting a facial expression reconstruction system.
(2) FIGS. **2**A and **2**B are perspective views of head-mounted wearable devices for facial expression reconstruction.
(3) FIGS. **3**A, **3**B, and **3**C are illustrations depicting examples of head-mounted wearable devices on a user.
(4) FIG. **4** is a block diagram depicting a deep-learning model process.
(5) FIG. **5** is a block diagram depicting a classifier process.

(6) FIG. **6** is a diagram depicting the correlation between frontal view camera images, landmark training data, predicted facial expression landmarks, and right and left facial contour images.
(7) FIG. **7** is a diagram depicting the correlation between frontal view camera images, two-dimensional training data, predicted two-dimensional facial expressions, and right and left facial contour images.
(8) FIG. **8** is a block flow diagram depicting a method to reconstruct facial expressions.
(9) FIG. **9** is a block flow diagram depicting a method to create a data training set using frontal view digital images.
(10) FIG. **10** is a block flow diagram depicting a method to reconstruct facial expressions using one or more facial digital image(s).
(11) FIG. **11** is a block flow diagram depicting a method to pre-process each pair of synchronized facial digital images.
(12) FIGS. **12**A, **12**B, and **12**C are perspective views of neck-mounted wearable devices for facial expression reconstruction.
(13) FIGS. **13**A and **13**B are illustrations depicting examples of neck-mounted wearable devices on a user.
(14) FIG. **14** is a block diagram depicting an alternate embodiment of a deep-learning model process.
(15) FIG. **15**A is a diagram depicting the correlation between frontal view camera images, three-dimensional training data, predicted three-dimensional face expressions, and infrared facial images.
(16) FIG. **15**B is a diagram depicting the correlation between frontal view camera images, three-dimensional training data, predicted three-dimensional face expressions, and infrared facial images.
(17) FIG. **16** is a block flow diagram depicting an alternative method to create a data training set using frontal view images.
(18) FIG. **17** is a block flow diagram depicting an alternative method to reconstruct facial expressions using one or more facial digital image(s).
(19) FIG. **18** is a block flow diagram depicting an alternative method to pre-process the one or more digital facial image(s).
(20) FIG. **19** is a block diagram depicting an acoustic sensor system based facial expression reconstruction system.
(21) FIG. **20**A is a block diagram depicting a first embodiment of an acoustic sensor system. FIG. **20**B is a block diagram depicting a second embodiment of an acoustic sensor system. FIG. **20**C is a block diagram depicting a third embodiment of an acoustic sensor system.
(22) FIG. **21** is a block diagram depicting a data processing module.
(23) FIG. **22**A depicts a first embodiment of a head-mounted acoustic sensor system wearable device. FIG. **22**B is an illustration depicting the first embodiment of the head-mounted acoustic sensor system wearable device on a user.
(24) FIG. **23**A depicts a second embodiment of a head-mounted acoustic sensor system wearable device. FIG. **23**B is an illustration depicting the second embodiment of the head-mounted acoustic sensor system wearable device on a user.
(25) FIG. **24**A depicts a third embodiment of a head-mounted acoustic sensor system wearable device. FIG. **24**B is an illustration depicting the third embodiment of the head-mounted acoustic sensor system wearable device on a user.
(26) FIG. **25** is a block flow diagram depicting a method to reconstruct facial expressions based on acoustic signals.
(27) FIG. **26** is a diagram depicting an acoustic sensor system based facial expression reconstruction process.
(28) FIG. **27** is a diagram depicting an acoustic sensor system based facial expression reconstruction output process.
(29) FIG. **28** is a diagram depicting the correlation between frontal view camera images, predicted facial expressions, and left and right differential echo profiles.
(30) FIG. **29** is a diagram depicting the correlation between frontal view camera images, training data, predicted facial expressions, and left and right differential echo profiles.
(31) FIG. **30**A is a diagram depicting the correlation between frontal view camera images of an open mouth, a left view image, a left and right echo profile, and a right view image.
(32) FIG. **30**B is a diagram depicting the correlation between frontal view camera images of a right sneer, a left view image, a left and right echo profile, and a right view image.
(33) FIG. **31** is a block diagram depicting a computing machine and a module.

DETAILED DESCRIPTION

Overview

(34) The present technology allows reconstruction of facial expressions and tracking of facial movements using non-obtrusive, wearable devices that capture optical or acoustical images of facial contours, chin profiles, or skin deformation of a user's face. The wearable devices include head-mounted technology that continuously reconstruct full facial expressions by capturing the positions and shapes of the mouth, eyes, and eyebrows. Miniature cameras capture contours of the sides of the face, which are used to train a deep-learning model to predict facial expressions. An alternate embodiment of this technology includes a neck-mounted technology to continuously reconstruct facial expressions. Infrared cameras capture chin and face shapes underneath the neck, which are used to train a deep-learning model to predict facial expressions.
(35) Additional embodiments include various camera types or acoustic imaging systems for the wearable devices. The acoustic imaging systems may comprise microphones and speakers. For example, the wearable devices may use the microphones and speakers to transmit and receive acoustical signals to determine skin deformation of a user. Full facial movements and expressions can be reconstructed from subtle skin deformations. The wearable devices may comprise ear mounted devices, eye mounted devices, or neck mounted devices.
(36) The systems of this technology include wearable devices configured with miniature cameras or acoustical imaging systems in communication with computing devices to transmit images or acoustical signals from the cameras or acoustical imaging systems to a remote server, data acquisition system, or data processing device for facial expression reconstruction or to track facial movement. The wearable devices include headphone, earbud, necklace, neckband, glasses, virtual reality ("VR") headsets, augmented reality ("AR") headsets, and other form factors. Each of the form factors include miniature cameras or acoustical imaging systems and micro computing devices, such as Raspberry Pi™.
(37) The exemplary headphone device includes two cameras and may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earpieces of the headphone. The cameras are attached to the earpieces of the headphone device and are connected to a computing device to transmit acquired images to a remote server. The headphone device is configured to acquire images of the contours of a user's face by directing the cameras at adjustable angles and positions.
(38) The exemplary earbud devices are constructed for wear as a left-side earbud and a right-side earbud. Both the left and right earbud devices include a camera to acquire images of the contours of a user's face. Each camera is connected to a computing device to transmit acquired images to a remote server.
(39) The exemplary necklace device includes an infrared ("IR") camera with an IR LED and an IR bandpass filter. The necklace device is configured to acquire images of a user's chin profile by directing the camera at the profile of a user's chin. The IR LED projects IR light onto a user's chin to enhance the quality of the image captured by the camera. The IR bandpass filter filters visible light such that the camera captures infrared light reflected by a user's skin. The camera is connected a computing device to transmit acquired images to a remote server.
(40) The exemplary neckband device includes two cameras fashioned to be positioned on the left and right sides of a user's neck. The neckband device includes IR LEDs and IR bandpass filters configured in proximity to each camera, similar to the necklace device. Each camera is connected a computing device to transmit acquired images to a remote server.
(41) Using one of the wearable devices, facial expressions can be reconstructed from images acquired from the cameras within the devices. A data training set is created using frontal view images of a user in a machine learning algorithm. Multiple frontal view images of a user are acquired with a variety of facial expressions. The frontal view images are transmitted to a data processing system to create the data training set.
(42) The wearable device captures one or more facial digital image(s) and transmits the digital images to data acquisition computing devices

connected to each of the cameras of the wearable device. The data acquisition computing devices subsequently transmit the images to a remote server or data processing system. The data processing system reconstructs facial expressions using the images. The data processing system pre-processes the images by reducing the image size, removing noise from the image, extracting skin color from the background of the image, and binarizing each the image. In some embodiments, a wearable imaging system comprises one or more imaging sensor(s), wherein the one or more imaging sensor(s) are positioned to capture one or more image(s) with incomplete side contours of a face (for example, from ear(s) and/or from neck), a processor, and a non-transitory machine-readable storage medium comprising machine-readable instructions executed by the processor to extract a plurality of features (for example, landmarks or parameters) from the one or more image(s), compare the extracted features and/or changes of the extracted features with features from a ground truth, and output one or more recognition or prediction result(s), wherein the results comprise a word or a phrase spoken by a user, an emoji of a facial expression of a user, and/or a real-time avatar of a facial expression of a user.
(43) The data processing system applies a deep-learning model to the pre-processed images for facial expression reconstruction. The reconstructed facial expressions may be used in applications such as silent speech and emoji recognition.
(44) In an alternate embodiment, acoustic sensing technology can be used to reconstruct facial expressions using an array of microphones and speakers and deep-learning models. Acoustic sensing technology is small, lightweight and, therefore, suitable to mount to a variety of wearable devices. Facial movements may be tracked by detecting skin deformations observed from different positions on a user's head. When a user performs facial movements, the skin on the face deforms with unique patterns. Using acoustic sensing technology, skin deformations may be captured without capturing images of the entire face. The acoustic technology comprises a wearable device such as headphones, earbuds, necklaces, neckbands, and any other suitable form factors such that the microphones and speakers can be mounted or attached to the wearable device. The positions of the microphones and speakers may be adjustable to optimize the recording of reflected signals.
(45) The acoustic wearable device actively sends signals from the speakers towards a user's face. The signals are reflected by the user's face and captured by the microphones on the acoustic wearable device. The signals are reflected differently back towards the microphones based on different facial expressions or movements. The acoustic wearable device sends the transmitted and reflected signals to a computing device for further processing. An echo profile is calculated based on the transmitted and reflected acoustic signals. The echo profile is calculated by calculating a cross-correlation between the received acoustic signals and the transmitted acoustic signals, which can display the deformations of the skin in temporal and spatial domains. The echo profile is input into a deep learning module to reconstruct the facial expressions of the user.
(46) While acoustic sensing technology may be used to determine facial expressions by detecting skin deformations, acoustic sensing technology may also be used to directly determine other outputs associated with skin deformation without the need to first determine facial expressions. For example, the acoustic sensing technology may track blinking patterns and eyeball movements of a user. Blinking patterns and eyeball movements can be applied to the diagnosis and treatment processes of eye diseases. The acoustic sensing technology may detect movements of various parts of the face associated with speech, whether silent or voiced. While speaking, different words and/or phrases lead to subtle, yet distinct, skin deformations. The acoustic sensing technology can capture the skin deformation to recognize silent speech or to vocalize sound. The acoustic sensing technology may also be used to recognize and track physical activities. For example, while eating, a user opens the mouth, chews, and swallows. During each of those movements, the user's skin deforms in a certain way such that opening the mouth is distinguishable from chewing and from swallowing. The detected skin deformations may be used to determine the type and quantity of food consumed by a user. Similar to eating, a type and quantity of a consumed drink may be determined by skin deformation. The acoustic sensing technology may also be used to determine an emotional status of a user. An emotional status may be related to skin deformations. By detecting skin deformations, an emotional status can be determined and reported to corresponding applications.
(47) These and other aspects, objects, features, and advantages of the disclosed technology will become apparent to those having ordinary skill in the art upon consideration of the following detailed description of illustrated examples.
Example System Architecture
(48) Turning now to the drawings, in which like numerals indicate like (but not necessarily identical) elements throughout the figures, examples of the technology are described in detail.
(49) FIG. **1** is a block diagram depicting a facial expression reconstruction system **100**, in accordance with certain examples. As depicted in FIG. **1**, the facial expression reconstruction system **100** comprises cameras **110-1** through **110**-*n*, computing devices **120-1** through **120**-*n*, data processing system **130**, and user computing device **150**. Computing devices **120**, data processing system **130**, and user computing device **150** are configured to communicate via a network **140**.
(50) In example embodiments, network **140** includes one or more wired or wireless telecommunications system(s) by which network devices may exchange data. For example, the network **140** may include one or more of a local area network (LAN), a wide area network (WAN), an intranet, an Internet, a storage area network (SAN), a personal area network (PAN), a metropolitan area network (MAN), a wireless local area network (WLAN), a virtual private network (VPN), a cellular or other mobile communication network, a BLUETOOTH® wireless technology connection, a near field communication (NFC) connection, any combination thereof, and any other appropriate architecture or system that facilitates the communication of signals, data, and/or messages.
(51) Facial expression reconstruction system **100** comprises cameras **110** that may be any suitable sensors for capturing images. For example, cameras **110** may include depth cameras, red-green-blue (“RGB”) cameras, infrared (“IR”) sensors, acoustic cameras or sensors, speakers, microphones, thermal imaging sensors, charge-coupled devices (“CCDs”), complementary metal oxide semiconductor (“CMOS”) devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology.
(52) The image frame resolution of the camera **110** may be defined by the number of pixels in a frame. The image resolution of the camera **110** may comprise any suitable resolution, including of the following resolutions, without limitation: 32×24 pixels; 32×48 pixels; 48×64 pixels; 160×120 pixels, 249×250 pixels, 250×250 pixels, 320×240 pixels, 420×352 pixels, 480×320 pixels, 640×480 pixels, 720×480 pixels, 1280×720 pixels, 1440×1080 pixels, 1920×1080 pixels, 2048×1080 pixels, 3840×2160 pixels, 4096×2160 pixels, 7680×4320 pixels, or 15360×8640 pixels. The resolution of the camera **110** may comprise a resolution within a range defined by any two of the preceding pixel resolutions, for example, within a range from 32×24 pixels to 250×250 pixels (for example, 249×250 pixels). In some embodiments, at least one dimension (the height and/or the width) of the image resolution of the camera **110** can be any of the following, including but not limited to 8 pixels, 16 pixels, 24 pixels, 32 pixels, 48 pixels, 72 pixels, 96 pixels, 108 pixels, 128 pixels, 256 pixels, 360 pixels, 480 pixels, 720 pixels, 1080 pixels, 1280 pixels, 1536 pixels, or 2048 pixels. The camera **110** may have a pixel size smaller than 1 micron, 2 microns, 3 microns, 5 microns, 10 microns, 20 microns, and the like. The camera **110** may have a footprint (for example, a dimension in a plane parallel to a lens) on the order of 10 mm×10 mm, 8 mm×8 mm, 5 mm×5 mm, 4 mm×4 mm, 2 mm×2 mm, or 1 mm×1 mm, 0.8 mm×0.8 mm, or smaller.
(53) Each camera **110** is in communication with a computing device **120**. Each camera **110** is configured to transmit images or data to a computing device **120**. Each camera **110** may be communicatively coupled to a computing device **120**. In an alternate example, each camera **110** may communicate wirelessly with a computing device **120**, such as via near field communication (“NFC”) or other wireless communication technology, such as Bluetooth, Wi-Fi, infrared, or any other suitable technology.
(54) Computing devices **120** comprise a central processing unit **121**, a graphics processing unit **122**, a memory **123**, and a communication application **124**. In an example, computing devices **120** may be small, single-board computing devices, such as a Raspberry Pi™ device. Computing devices **120** function to receive images or data from cameras **110** and to transmit the images via network **140** to a data processing system **130**.
(55) Computing device **120** comprises a central processing unit **121** configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and perform calculations and generate commands. Central processing

unit **121** may be configured to monitor and control the operation of the components in the computing device **120**. Central processing unit **121** may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a graphics processing unit ("GPU"), a field programmable gate array ("FPGA"), a programmable logic device ("PLD"), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. Central processing unit **121** may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. Central processing unit **121** along with other components of the computing device **120** may be a virtualized computing machine executing within one or more other computing machine(s).

(56) Computing device **120** comprises a graphics processing unit **122** that serves to accelerate rendering of graphics and images in two- and three-dimensional spaces. Graphics processing unit **122** can process multiple images, or data, simultaneously for use in machine learning and high-performance computing.

(57) Computing device **120** comprises a memory **123**. Memory **123** may include non-volatile memories, such as read-only memory ("ROM"), programmable read-only memory ("PROM"), erasable programmable read-only memory ("EPROM"), flash memory, or any other device capable of storing program instructions or data with or without applied power. Memory **123** may also include volatile memories, such as random access memory ("RAM"), static random access memory ("SRAM"), dynamic random access memory ("DRAM"), and synchronous dynamic random access memory ("SDRAM"). Other types of RAM also may be used to implement memory **123**. Memory **123** may be implemented using a single memory module or multiple memory modules. While memory **123** is depicted as being part of the computing device **120**, memory **123** may be separate from the computing device **120** without departing from the scope of the subject technology.

(58) Computing device **120** comprises a communication application **124**. Communication application **124** interacts with web servers or other computing devices or systems connected via network **140**, including data processing system **130**.

(59) Facial expression reconstruction system **100** comprises a data processing system **130**. Data processing system **130** serves to receive images or data from cameras **110** via computing devices **120** and network **140**. Data processing system **130** comprises a central processing unit **131**, a modeling application **132**, a data storage unit **133**, and a communication application **134**.

(60) Data processing system **130** comprises central processing unit **131** configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and perform calculations and generate commands. Central processing unit **131** may be configured to monitor and control the operation of the components in the data processing system **130**. Central processing unit **131** may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a graphics processing unit ("GPU"), a field programmable gate array ("FPGA"), a programmable logic device ("PLD"), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. Central processing unit **131** may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. Central processing unit **131** along with other components of the data processing system **130** may be a virtualized computing machine executing within one or more other computing machine(s).

(61) Data processing system **130** comprises a modeling application **132**. The modeling application **132** employs a variety of tools, applications, and devices for machine learning applications. The modeling application **132** may receive a continuous or periodic feed of images or data from one or more of the computing device(s) **120**, the central processing unit **131**, or the data storage unit **133**. Collecting the data allows the modeling application **132** to leverage a rich dataset to use in the development of a training set of data or ground truth for further use in facial expression reconstructions. The modeling application **132** may use one or more machine learning algorithm(s) to develop facial expression reconstructions, such as a convolution neural network ("CNN"), Naïve Bayes Classifier, K Means Clustering, Support Vector Machine, Apriori, linear regression, logistic regression, decision trees, random forest, or any other suitable machine learning algorithm.

(62) Data processing system **130** comprises a data storage unit **133**. Data storage unit **133** may be accessible by the modeling application **132** and the communication application **134**. The example data storage unit **133** can include one or more tangible computer-readable storage device(s). The data storage unit **133** can be within the data processing system **130** or can be logically coupled to the data processing system **130**. For example, the data storage unit **133** can include on-board flash memory and/or one or more removable memory device(s) or removable flash memory. In certain embodiments, the data storage unit **133** may reside in a cloud-based computing system.

(63) Data processing system **130** comprises a communication application **134**. Communication application **134** interacts with web servers or other computing devices or systems connected via network **140**, include the computing devices **120** and user computing device **150**.

(64) User computing device **150** is a computing device configured to receive and communicate results of facial expression reconstruction or facial movement. The results of the facial expression reconstruction may be displayed as a graphical representation of a facial expression, such as **630** of FIG. **6**, a predicted facial expression with three-dimensional visualization, such as **730** of FIG. **7**, a three-dimensional facial expression, such as **1530**A of FIG. **15**, an emoji, text or audio representing silent speech recognition, or any other suitable display of facial expression reconstructions results. The user computing device **150** also may comprise a speaker to audibly communicate the facial expression reconstructions results directed to silent speech recognition. In an alternate embodiment by tracking facial movement, user computing device **150** may determine skin deformation or stretching and, subsequently, directly determine one or more of facial expression(s), eyeball movement, or eye-related disease. By tracking facial movement, user computing device **150** may recognize, or make audible, silent speech, and recognize and track human activities involving facial movements such as eating or drinking. By tracking facial movement, user computing device **150** may synthesize a user's voice or determine a user's emotional status.

(65) FIGS. **2**A and **2**B are perspective views of head-mounted wearable devices **210**, **220**, respectively, for facial expression reconstruction, in accordance with certain examples.

(66) FIG. **2**A depicts a headphone-style wearable device **210** (herein referred to as headphone device **210**) as a form factor to acquire images or data for use in facial expression reconstruction, in accordance with certain examples. Headphone device **210** comprises two cameras **110-1** and **110-2** to acquire images of a left and a right contour, respectively. Cameras **110** were previously described herein with reference to FIG. **1**. In an example, headphone device **210** is a device constructed of a band **211** suitable to wear across the top of a user's head with a left-side ear piece **212-1** and a right-side ear piece **212-2**. In an example, headphone device **210** may be over-the-ear headphones or outside-the-ear headphones. In an example, headphone device **210** may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earpieces **212**. Camera **110-1** is connected to left-side ear piece **212-1**, and camera **110-2** is connected to right-side ear piece **212-2**. In an alternate example, two cameras **110** are connected to each of the left-side ear piece **212-1** and right-side ear piece **212-2**. Any suitable number of cameras may be connected to each of the ear pieces **212**. Each camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. **1**. In an example, computing device **120** may be connected to each camera **110** and be embedded within the headphone device **210**. In an alternate example, each camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.

(67) Headphone device **210** is configured to acquire images of a left and a right contour of a user's face by directing cameras **110-1** and **110-2** at adjustable angles and positions. Each of the cameras **110** are independently adjustable by a first angle **214**, a slide **216**, and second angle **218**. First angle **214** adjusts a tilt of the position of the camera **110** relative to a plane parallel to an exterior surface of a user's ear. The first angle **214** may be adjusted so that the camera **110**/earpiece **212** assembly is tilted closer to a user's ear, or the first angle **214** may be adjusted so that the camera **110** via the earpiece **212** is tilted farther away or offset from a user's ear. In an example, first angle **214** may be 0° indicating that the camera **110**/earpiece **212** assembly is in a vertical position relative to a plane parallel to the exterior surface of a user's ear. First angle **214** may be adjusted by −10°, −20°, −30°, −40°, or any suitable angle measure relative to the plane parallel to the exterior surface of a user's ear, such that each camera **110** may be

aligned with the left and right contours of a user's face.
(68) Slide **216** adjusts a position of the camera **110** relative to the earpiece **212** in a direction that is perpendicular to a plane parallel to the exterior surface of a user's ear, in other words, the position of the camera **110** may change along the slide **216** while the position of the earpiece **212** is fixed. The position of the camera **110** via the slide **216** may be adjusted such that the earpiece **212** is in close contact with an exterior surface of a user's ear. The position of the camera **110** via the slide **216** may be adjusted such that the camera **110** is extended a distance away from the plane parallel to the exterior surface of a user's ear. In an example, the extended distance may be 1 cm, 2 cm, 3 cm, or any suitable distance away from the plane parallel to the exterior surface of a user's ear. The slide **216** positions the cameras **110** in a manner similar to positioning the earpieces **212** of the headphone device **210**. In some embodiments, the position of an imaging sensor or a camera (for example, the optical center of the lens in the camera) is less than 5 cm, less than 4 cm, less than 3 cm, less than 2 cm, less than 1 cm, or less than 0.5 cm away from the surface plane of a user's ear. In some embodiments, the position of an imaging sensor or a camera projected to the closest skin surface is within the region of a user's ear or less than 5 cm, less than 4 cm, less than 3 cm, less than 2 cm, or less than 1 cm away from the nearest contour edge of a user's ear. In some embodiments, the system comprises at least one, at least two, or at least three imaging sensors (for example, cameras) located in different positions as described above. In some embodiments, an imaging sensor, such as a camera, is positioned below a chin of a user and less than 25 cm, less than 20 cm, less than 15 cm, less than 10 cm, or less than 5 cm away from the chin of a user, or 2-30 cm below and away from the chin of the user.
(69) Second angle **218** adjusts a rotational position of the camera **110** along a horizontal axis through earpieces **212-1** and **212-2**. In an example, second angle **218** adjusts an angular position of the camera **110** relative to the horizontal axis while the position of the earpiece **212** remains unchanged. In an alternate example, second angle **218** adjusts an angular position of the camera **110**/earpiece **212** assembly. Relative to a left or right contour of a user's face, second angle **218** may be 0° indicating that the camera **110** is in a horizontal position. A second angle **218** of 10° indicates that the camera **110** is directed 10° upwards. A second angle **218** of −10° indicates that the camera **110** is directed 10° downwards. Any suitable measure of second angle **218** may be used to align the cameras **110** with the contour of a user's face.
(70) Each of the cameras **110** are independently adjustable by the first angle **214** and the second angle **218**, by any suitable mechanism, for example, by mounting the cameras **110** to the headphone device **210** via rotational positioning devices allowing incremental changes of the direction of the cameras **110**.
(71) FIG. **2**B depicts earbud-style wearable devices **220-1** and **220-2** (herein referred to as earbud devices **220**) as a form factor to acquire images or data for use in facial expression reconstruction, in accordance with certain examples. Earbud device **220-1** is constructed for wear as a left-side earbud, and earbud device **220-2** is constructed for wear as a right-side earbud. In an example, earbud device **220** may be ear pods, earphones, or in-the-ear (“ITE”) headphones. Earbud device **220-1** comprises a camera **110-1**, and earbud device **220-2** comprises a camera **110-2** to acquire images of a left and a right contour, respectively, of a user's face by directing cameras **110-1** and **110-2** at the left and right contours of a user's face. Cameras **110-1** and **110-2** were previously described in reference to FIG. **1**. In an example, earbud devices **220** may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the earbuds **220**. Each camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. **1**. In an example, a computing device **120** may be connected to each camera **110** and be embedded within each earbud device **220**. In an alternate example, each camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.
(72) The camera **110** position for each earbud device **220** may be controlled by twisting and/or rotating the earbud device **220** in the user's ear. The earbud device **220** may be rotated such that the camera **110** is angled closer to the contour of a user's face. In an alternate example, the camera **110** may be attached to the earbud device **220** such that the camera **110** may be positioned independently of the earbud device **220**. The camera **110** may be attached to the earbud device **220** with a ball and socket joint or any other suitable attachment method such that the position of the camera **110** may be adjusted independently of the earbud device **220**.
(73) FIGS. **3**A, **3**B, and **3**C are illustrations depicting examples **310-350** of head-mounted, wearable devices **210**, **220** on a user, camera views **330**, captured facial contours **340**, and corresponding facial expressions **350**, in accordance with certain examples. In FIG. **3**A, earbuds **220** are illustrated in a closeup view of the back of a user's ear. Camera **110** of the earbuds **220** is illustrated in this example as being positioned in a forward direction towards a right contour of a user's face. Although not depicted in FIG. **3**A, a second earbud **210** is worn by the user in the opposite, left ear, having a camera **110** positioned in a forward direction towards a left contour of the user's face.
(74) Also, in FIG. **3**A, alternative headphone **210** is illustrated in a closeup view from a front perspective of the user. Cameras **110-1** and **110-2** of the headphone **210** are illustrated in this example as being positioned in a forward direction towards left and right contours of a user's face.
(75) FIG. **3**B depicts a back and side view of a user to illustrate the contour of the user's face from the point of view of a camera **110** in either the earbud device **220** or headphone device **210** depicted in FIG. **3**A.
(76) FIG. **3**C depicts the left and right contours of a user's face as captured by cameras **110** through the use of the headphone **210** or the earbuds **220**. FIG. **3**B also depicts the facial expression of a user a reconstructed through the images captured by cameras **110** through the use of the headphone **210** or the earbuds **220**.
(77) The computing devices **120**, **130**, and **150** and any other network computing devices other computing machines associated with the technology presented herein may be any type of computing machine, such as, but not limited to, those discussed in more detail with respect to FIG. **31**. For example, each device can include any suitable processor-driven device.
(78) Furthermore, any functions, applications, or components associated with any of these computing machines, such as those described herein or others (for example, scripts, web content, software, firmware, hardware, or modules) associated with the technology presented herein may be any of the components discussed in more detail with respect to FIG. **31**.
(79) The network connections illustrated are examples and other means of establishing a communications link between the computers and devices can be used. The computing machines discussed herein may communicate with one another, as well as with other computing machines or communication systems over one or more network(s). Each network may include various types of data or communications networks, including any of the network technology discussed with respect to FIG. **31**.
(80) Additionally, those having ordinary skill in the art having the benefit of the present disclosure will appreciate that the devices illustrated in the figures may have any of several other suitable computer system configurations.
Example Processes
(81) The methods illustrated in FIGS. **4** through **11** are described hereinafter with respect to the components of facial expression reconstruction system **100**, headphone device **210** of FIG. **2**A, and earbud devices **220** of FIG. **2**B. The methods of FIGS. **4** through **11** may also be performed with other systems and in other environments. The operations described with respect to FIGS. **4** through **11** can be implemented as executable code stored on a computer- or machine-readable, non-transitory, tangible storage medium (for example, floppy disk, hard disk, ROM, EEPROM, nonvolatile RAM, CD-ROM, etc.) that are completed based on execution of the code by a processor circuit implemented using one or more integrated circuit(s); the operations described herein also can be implemented as executable logic that is encoded in one or more non-transitory, tangible media for execution (for example, programmable logic arrays or devices, field programmable gate arrays, programmable array logic, application specific integrated circuits, etc.).
(82) FIG. **4** is a block diagram depicting a deep-learning model process **400**, in accordance with certain examples. The deep-learning model process **400** may be performed by the modeling application **132** of the data processing system **130**, previously described herein in reference to FIG. **1**. In an example where the machine learning algorithm is a convolution neural network (“CNN”), the deep-learning model process **400** receives images as an input and assigns importance (learnable weights and biases) to various aspects/objects in the images. The CNN reduces the images into a form that is

easier to process without losing the features of the images that are used to get a good prediction. In the current example, the prediction is a reconstructed facial expression based on the received right and left contour facial images.
(83) The deep-learning model process **400** comprises an image processing phase **410** and a regression phase **450**. In the image processing phase **410**, image processing is divided into two parallel paths. Path A is directed to processing an image of a right facial contour, and path B is directed to processing an image of a left facial contour. The right facial contour images and left facial contour images are processed independently of each other and combined in block **454** of the deep-learning model process **400**.
(84) In block **412**, the data processing system **130** receives an input of an image of a right facial contour and an input of an image of a left facial contour. Example right facial contour images are depicted in row **640** of FIGS. **6** and **7**, and example left facial contour images are depicted in row **650** of FIGS. **6** and **7**. The images are transmitted through blocks **414-1** through **414**-*n*. Blocks **414-1** through **414**-*n* comprise layers of the CNN that include multiple convolution filtering operations. In an example, the CNN may comprise between 32 and 512 layers to extract features from the images. The first convolution layer in block **414-1** captures low-level features, such as edges, color, and gradient orientation. As the process progresses through the additional layers to layer **414**-*n*, higher-level features are captured such that the object of the image is captured. In the current example, the object of the image is a facial contour.
(85) From block **414**-*n*, the process proceeds to block **416**. Block **416** is a pooling layer. The pooling layer may be a Max Pooling that returns a maximum value from the image. The pooling layer may be an Average Pooling that returns an average of all of the values from the image. The pooling layer of block **416** outputs a vector representation of each input image (in other words, a right image vector for the right facial contour and a left image vector for the left facial contour).
(86) The right image vector and the left image vector are received as inputs to the regression phase **450**. Within the regression phase **450**, the right image vector and the left image vector are inputs into two fully connected layers **452** with a rectified linear unit ("ReLU") between the fully connected layer **452**. The fully connected layer **452** learns facial landmarks of the right image vector and the left image vector based on facial landmarks in a training data set, or ground truth, set of facial expressions. The fully connected layer **452** compares features of the right image vector to right side facial landmarks in the training data set of facial expressions to match the features of the right image vector to a right side facial expression in the training data set. Similarly, the fully connected layer **452** compares features of the left image vector to left side facial landmarks in the training data set of facial expressions to match the features of the left image vector to a left side facial expression in the training data set. The fully connected layer **452** outputs landmarks of both the right and left side of a user's face. The output landmarks are inputs to a matching module **454**. The matching module **454** concatenates the landmarks from the right and left sides by aligning key landmarks that are present in both the right and left sides using translation and scaling. The matching module **454** outputs the final facial expression reconstruction, or predicted facial expression, as a set of facial landmarks as the reconstruction result in block **456**. Examples of predicted facial expressions as a set of facial landmarks are depicted in row **630** of FIG. **6**. In an alternate example, the facial expression reconstruction is further modeled to create a three-dimensional visualization of the predicted facial expression as depicted in row **730** of FIG. **7**. The process **400** continues to repeat as additional right facial contour images and left facial contour images are received.
(87) FIG. **5** is a block diagram depicting a classifier process **500**, in accordance with certain examples. The classifier process **500** classifies the facial expression reconstruction output **456** of the deep-learning model process **400** for use in applications such as silent speech recognition, emoji input, real-time avatar facial expressions, or other suitable applications. In block **510**, the reconstruction output **456** from the deep-learning model process **400** (FIG. **4**) is received as input to the classifier process **500**. In block **510**, a stream of landmark positions from the reconstruction output **456** are received as frames. In block **520**, the landmark positions are segmented. Segmentation extracts an event from a stream of data and labels its start and end time. In the example of silent speech recognition, segmentation would identify the initial facial expression associated with the pronunciation of a word and the final facial expression associated with the pronunciation of the same word. For each frame, two indicators are calculated to measure the degree of transformation for the mouth and eyes, known as the Mouth Transformation Degree ("MTD") and Eye Transformation Degree ("ETD"). The MTD and ETD are calculated by summing the differences between the landmarks outlining the mouth or eye/eyebrow at the time and the landmarks of the natural facial expression. Then, a lowpass filter is applied to filter out high-frequency noise. When an event happens, the MTD or ETD raises to a single peak or a primary peak with several secondary peaks. A peaking seeking algorithm is applied to find the peaks in the stream. Thresholds are set for peak height, peak prominence, and peak width to filter peaks that may be associated with noise. After finding peak positions, the first-order difference of the ETD and MTD are calculated. In the first-order difference sequence, the left nearest peak to the event is labeled as the beginning, and the right nearest valley to the event is labeled as the end. The frames associated with the event are placed sequentially on a timeline **525** as frames **530-1** through frames **530**-*n*.
(88) After the sequential placement of frames **530-1** through **530**-*n*, the process **500** proceeds to block **540** where the frames are inputs into a Bidirectional Long Short-Term Memory ("BLSTM") model for classification. Block **540** depicts a two-layer BLSTM model. However, any suitable number of layers may be used. The blocks **541-1** through **541**-*n* depicted within the BLSTM model **540** are recurrently connected memory blocks. Each of the blocks **541** have feedback connections such that sequences of data can be processed as opposed to single images or data points. As depicted in block **540**, blocks **541-1** through **541**-*n* comprise bidirectional connections within each of the blocks **541** in each layer of block **540**. Processing sequences of data allows for speech recognition, emoji recognition, real-time avatar facial expressions, or other suitable applications. The output from block **540** is a vector representation of the input frames.
(89) In block **550**, the vector representation is received by a fully connected layer comprising a SoftMax function. The SoftMax function transforms the vector representation into a probability distribution to predict a facial event. The output from block **550** is an encoding of a facial event class. The facial event classification may be a facial expression. In the example of silent speech recognition, the facial event classification may be the pronunciation of a specific word or phrase, such as "hello" or "how are you." In the example of emoji recognition, the facial event classification may be an emoji indicating a smiling face, a frowning face, a crying face, or any other suitable emoji associated with a facial expression. In the example of real-time avatar facial expressions, the facial event classification may be a three-dimensional visualization of a facial expression.
(90) FIG. **6** is a diagram **600** depicting the correlation between frontal view camera images **610**, landmark training data **620**, predicted facial expressions **630**, and captured right and left facial contour images **640**, **650**, in accordance with certain examples. Row **610** illustrates frontal view camera images of a user. To construct a training data set or "ground truth," frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images **610-1** through **610**-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method **810** of FIG. **9**. Row **620** illustrates a training data set in the form of facial landmarks. The landmark training data images depicted in images **620-1** through **620**-*n* correlate with images **610-1** through **610**-*n*. The landmark training data images **620** are the output of method **810** of FIG. **9**.
(91) Row **640** depicts right facial contour images **640-1** through **640**-*n* captured by a head-mounted, wearable device, such as headphone device **210** or earbud device **220** described herein with reference to FIG. **2**. Row **650** depicts left facial contour images **650-1** through **650**-*n* captured by a head-mounted, wearable device, such as headphone device **210** or earbud device **220** described herein with reference to FIG. **2**. The right and left facial contour images **640**, **650** are synchronized such that the images are representing the same facial expression. The synchronization of the images is described in greater detail herein with reference to block **1020** of FIG. **10**. The synchronized images are used as inputs to the deep-learning process **400**, previously described herein with reference to FIG. **4**.
(92) The right facial contour images **640**, the left facial contour images **650**, and the landmark training data images **620** are used in the deep-learning model process **400** to construct the predicted facial expressions in row **630**. The predicted facial expressions **630** are depicted as landmark images in images **630-1** through **630**-*n*. The predicted facial expressions **630** are illustrated in FIG. **6** beneath the landmark training data images **620** to illustrate

the accuracy of predicting facial expressions using right facial contour images **640** and left facial contour images **650** acquired from headphone device **210** or earbud device **220**.
(93) FIG. **7** is a diagram **700** depicting the correlation between frontal view camera images **610**, training data with three-dimensional visualization **720**, predicted facial expressions with three-dimensional visualization **730**, and right and left facial contour images **640**, **650**, in accordance with certain examples. FIG. **7** is an alternate embodiment of FIG. **6**, wherein training data images with three-dimensional visualization are depicted in row **720** and predicted facial expressions with three-dimensional visualization are depicted in row **730**. The training data images **720-1** through **720**-*n* are correlated with predicted facial expression images **730-1** through **730**-*n*. The three-dimensional visualizations depicted in blocks **720** and **730** are created using the facial landmarks described in rows **620** and **630** in reference to FIG. **6**. The facial landmarks are input into a three-dimensional modeling application to create a three-dimensional visualization of the training data and the predicted facial expressions.
(94) FIG. **6** and FIG. **7** present two-dimensional and three-dimensional correlations between training data and predicted facial expressions. Any suitable correlation can be made based on the type of camera used and the type of data collected. For example, correlations can be made based on imaging data such as thermal gradients or acoustic reflections.
(95) FIG. **8** is a block flow diagram depicting a method **800** to reconstruct facial expressions, in accordance with certain examples. In block **810**, a data training set is created using frontal view digital images. Block **810** is described in greater detail in method **810** of FIG. **9**.
(96) FIG. **9** is a block flow diagram depicting a method **810** to create a data training set using frontal view digital images, in accordance with certain examples. In block **910**, one or more frontal view digital image(s) are acquired of a user. The one or more frontal view digital image(s) depict the user making a variety of facial expressions. For example, the user may smile, frown, squint, close one or both eyes, make an angry facial expression, raise one or both eyebrows, or any other suitable expressions such that the one or more frontal view digital image(s) represent a wide range of potential facial expressions. Example frontal view digital images are depicted in row **610** of FIG. **6**. The user may take the one or more frontal view digital image(s) of himself, another person may take the one or more frontal view digital image(s), or any other suitable method to acquire the one or more frontal view digital image(s) may be used. The one or more frontal view digital image(s) may be taken by any device or sensor capable of capturing a digital image including, but not limited to, depth cameras, red-green-blue (“RGB”) cameras, infrared (“IR”) sensors, thermal imaging sensors, charge-coupled devices (“CCDs”), complementary metal oxide semiconductor (“CMOS”) devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology.
(97) In block **920**, the frontal view digital images are transmitting to data processing system **130**. The frontal view digital images may be transmitted to the data processing system **130** by a user's computing device, such as user computer device **150**, or any other suitable device to transmit data.
(98) In block **930**, the data processing system **130** receives the frontal view digital images.
(99) In block **940**, the data processing system **130** extracts facial landmark positions. To extract the facial landmark positions, the data processing system **130** uses a computer vision library as a ground truth acquisition method. In an example, the computer vision library is a Dlib library. The computer vision library is used to extract key feature points or landmarks from the frontal view digital images. The computer vision library may extract 42, 68, or any suitable number of key feature points.
(100) In block **950**, the data processing system **130** aligns the extracted landmark positions using an affine transformation. Aligning the extracted landmark positions accounts for variations in a user's head position in the frontal view digital images. When a user acquires the frontal view digital images as described in block **910**, the same facial expressions may vary if the user slightly changes his face orientation. To align the extracted landmark positions, a set of landmarks are selected whose relative positions change very little when making facial expressions. For example, the selected landmarks may be one or more of a right canthus, left canthus, or apex of the nose. The selected landmarks are used to calculate an affine matrix for each frontal view digital image. The landmark positions are aligned to the same range to reduce the influence of head position change when the frontal view digital images are acquired.
(101) In block **960**, the data processing system **130** creates the data training set based on the aligned facial landmark positions. In an example, after the extracted landmark positions are aligned, the data processing system **130** selects the most informative feature points from the extracted landmark positions. In an example, the Dlib library may extract 68 facial landmarks from each of the frontal view digital images. When making facial expressions, changes mainly occur in the areas around the mouth, eyes, and eyebrows. The less informative features points may be removed, leaving a smaller set of facial landmarks, such as 42 facial landmarks. Any suitable number of facial landmarks may be used. The data training set is the set of the most informative landmark positions for each facial expression from each of the frontal view digital images. Example data training set images are depicted in row **620** of FIG. **6**. In an alternate example, the facial expression reconstruction is further modeled to create a three-dimensional visualization of the training data images as depicted in row **720** of FIG. **7**. In an example, the data processing system saves the data training set in the data storage unit **133**. In some embodiments, the system may extract information of at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, or at least 100 parameters or features from one or more image(s) captured by one or more imaging sensor(s) for a facial expression, wherein the parameters or features comprise facial landmark positions, shape parameters (for example, blendshape parameters), orientations (for example, head orientations), or any combination thereof.
(102) FIG. **6** depicts frontal view digital images of a user at row **610**, which are used to create the corresponding data training set depicted in row **620**. In an alternate example, training data with three-dimensional visualization, as depicted in row **720** of FIG. **7**, may be utilized. Any suitable method or visualization may be used to create the data training set.
(103) From block **960**, the method **810** returns to block **820** of FIG. **8**.
(104) Referring back to FIG. **8**, in block **820**, a wearable device captures one or more facial digital image(s). In an example, the wearable device may be a head-mounted device, such as headphone device **210** or earbud device **220**, previously described herein with reference to FIG. **2**. The head-mounted device is configured to capture right facial contour images and left facial contour images of a user's face directed towards the right and left cheekbones using two or more cameras **110**. In the current example, the right and left facial contour images are two-dimensional images of the contour of a user's face. Example right and left facial contour images are depicted in rows **640** and **650**, respectively, of FIGS. **6** and **7**.
(105) In an alternate example, the wearable device may be a neck-mounted device such as necklace device **1210** or neckband device **1220**, described hereinafter with reference to FIG. **12**. The neck-mounted device is configured to take digital images of the contour of a user's face directed towards the user's chin using one or more camera(s) **110**.
(106) In block **830**, the wearable device transmits the one or more facial digital image(s) to one or more data acquisition computing device(s) **120**. As depicted and described in reference to FIG. **1**, each camera **110** is in communication with a data acquisition computing device **120**.
(107) In block **840**, the one or more data acquisition computing device(s) **120** transmit the one or more facial digital image(s) to the data processing system **130**.
(108) In block **850**, the data processing system **130** receives the one or more facial digital image(s).
(109) In block **860**, the data processing system **130** reconstructs facial expressions using the one or more facial digital image(s). Block **860** is described in greater detail herein with reference to method **860** of FIG. **10**.
(110) FIG. **10** is a block flow diagram depicting a method **860** to reconstruct facial expressions using one or more facial digital image(s), in accordance with certain examples. The examples of FIG. **10** are directed to the method to reconstruct facial expressions from one or more facial digital image(s) received from a head-mounted wearable device. An alternate method to reconstruct facial expressions from one or more facial digital image(s) received from a neck-mounted wearable device is described herein with reference to method **860**′ of FIG. **17**.
(111) In block **1010**, the data processing system **130** receives the one or more facial digital image(s).
(112) In block **1020**, the data processing system **130** creates one or more pair(s) of synchronized facial digital images from the one or more facial

digital image(s). In the example where the one or more facial digital image(s) are acquired from a head-mounted wearable device, the wearable device captures right facial contour images and left facial contour images. To accurately reconstruct a facial expression of a user, the data processing system **130** synchronizes the images from the left camera **110-1** and the right camera **110-2** such that each pair of right and left facial contour images represents a particular facial expression.

(113) In block **1030**, the data processing system **130** pre-processes each pair of synchronized facial digital images. Block **1030** is described in greater detail herein with reference to method **1030** of FIG. **11**.

(114) FIG. **11** is a block flow diagram depicting a method **1030** to pre-process each pair of synchronized facial digital images, in accordance with certain examples. In block **1110**, the data processing system **130** converts a digital image color space of each pair of synchronized facial digital images from a red-green-blue ("RGB") color space to a luminance, chrominance (red-yellow), chrominance (blue-yellow) ("YCrCb") color space. The YCrCb color space is used to take advantage of a lower resolution with respect to luminosity amount of data stored with respect to each pair of synchronized facial digital images. The conversion from the RGB color space to the YCrCb color space is accomplished using a standard process.

(115) In block **1120**, the data processing system **130** extracts skin color from the background of each converted pair of facial digital images. In an example, the skin color is extracted using Otsu's thresholding method. Otsu's thresholding method determines whether pixels in an image fall into a foreground or a background. In the current example, the foreground represents a facial contour of each facial digital image, and the background represents an area of the image outside of the facial contour.

(116) In block **1130**, the data processing system **130** binarizes each facial digital image after the extraction of the skin color from the background. Image binarization is the process of taking the image in YCrCb color space and converting it to a black and white image. The binarization of the image allows for an object to be extracted from an image, which in this example is a facial contour.

(117) In block **1140**, the data processing system **130** filters the binarized digital images to remove noise from the images. Filtering the binarized digital images produces a smoother image to assist in more accurate facial expression reconstructions.

(118) From block **1140**, the method **1030** returns to block **1040** of FIG. **10**. Referring back to FIG. **10**, in block **1040**, the data processing system **130** applies a deep-learning model to each pair of pre-processed facial digital images for facial expression reconstruction. The deep-learning model for facial expression reconstruction was described in detail herein in the deep-learning model process **400** with reference to FIG. **4** and in the classifier process **500** with reference to FIG. **5**. The results of the facial expression reconstruction from block **1040** may include the reconstruction output from block **456** of FIG. **4** and/or the encoding of a facial event class from block **550** of FIG. **5**. For example, the data processing system **130** may output from block **1040** the landmark images from the process **400** of FIG. **4**. The data processing system **130** also, or alternatively, may further process the facial expression reconstruction such that the output from block **1040** comprises a three-dimensional visualization of the facial expression, silent speech recognition, an emoji associated with the facial expression reconstruction, and/or real-time avatar facial expressions associated with the facial expression reconstruction, such as described by the classifier process **500** of FIG. **5**.

(119) From block **1040**, the method **860** returns to block **870** of FIG. **8**. Referring back to FIG. **8**, in block **870**, the data processing system **130** outputs the results of the facial expression reconstruction. In an example, the results may be sent to user computing device **150** to be displayed visually or audibly to a user. In an example, the results of the facial expression reconstruction may be displayed as a graphical representation of a facial expression, such as row **630** of FIG. **6**; a predicted facial expression with three-dimensional visualization, such as row **730** of FIG. **7**; a three-dimensional facial expression, such as row **1530**A of FIG. **15**; an emoji; text representing silent speech recognition; or any other suitable display of facial expression reconstructions results. User computing device **150** may comprise a speaker to audibly communicate the facial expression reconstructions results directed to silent speech recognition.

(120) From block **870** of FIG. **8**, the method **800** returns to **820** where the method **800** repeats the process of reconstructing facial expressions by capturing additional facial digital images.

Other Example Architectures

(121) FIGS. **12**A, **12**B, and **12**C are perspective views of neck-mounted wearable devices **1210**, **1220** for facial expression reconstruction with example IR images **1230** and **1240**, in accordance with certain examples.

(122) FIG. **12**A depicts a necklace style neck-mounted wearable device **1210** (herein referred to as necklace device **1210**) as a form factor to acquire images or data for use in facial expression reconstruction. Necklace device **1210** comprises a camera **110**, previously described herein in reference to FIG. **1**. In the current example, camera **110** is an infrared ("IR") camera. Necklace device **1210** also comprises an IR LED **1212** and an IR bandpass filter **1214**. Necklace device **1210** is configured to acquire images of a user's chin profile by directing the camera **110** at the profile of a user's chin. The IR LED **1212** projects IR light onto a user's chin to enhance the quality of the image captured by camera **110**. The IR bandpass filter **1214** filters visible light such that camera **110** captures infrared light reflected by the skin of a user.

(123) Necklace device **1210** comprises a chain **1218** or other suitable device for securing the necklace device **1210** about the neck of a user. In an alternate example, necklace device **1210** may attach to a user's clothing instead of being secured about the neck of a user. In the current example, necklace device **1210** may be attached to a user's clothing underneath the user's chin. In an alternate example, multiple necklace devices **1210** may be attached to a user's clothing to capture camera **110** images from multiple viewpoints. In the current example, a necklace device **1210** may be attached on a user's clothing close to each shoulder of the user. The necklace device **1210** may comprise a clip, a pin, a clasp, or any other suitable device to attach necklace device **1210** to a user's clothing.

(124) The camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. **1**. In an example, a computing device **120** may be connected to the camera **110** and be embedded within the necklace device **1210**. In an alternate example, the camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.

(125) FIG. **12**B depicts a neckband style neck-mounted wearable device **1220** (herein referred to as neckband device **1220**) as a form factor to acquire images or data for use in facial expression reconstruction. Neckband device **1220** comprises two cameras **110-1** and **110-2** fashioned to be positioned on the left and right sides of a user's neck. Neckband device **1220** also comprises IR LEDs and IR bandpass filters configured in proximity to each camera **110-1** and **110-2** (not depicted in FIG. **12**). In an example, neckband device **1220** may include functionality with Bluetooth or Wi-Fi connectivity and speakers embedded within the neckband **1220**. Each camera **110** is in communication with a computing device **120**, as previously described with respect to FIG. **1**. In an example, a computing device **120** may be connected to each camera **110** and be embedded within the neckband device **1220**. In an alternate example, each camera **110** may communicate with a computing device **120** via a wireless technology, such as Bluetooth or Wi-Fi.

(126) FIG. **12**C depicts two example IR images **1230** and **1240**. Example **1230** is an IR image acquired from necklace device **1210**. Example **1240** is a right camera **110** IR image acquired from neckband device **1220**. Additional examples of IR images acquired by either necklace device **1210** or neckband device **1220** are depicted in row **1540**A of FIG. **15**A and row **1540**B of FIG. **15**B.

(127) FIGS. **13**A and **13**B are illustrations **1300** depicting examples **1310-1360** of neck-mounted wearable devices **1210**, **1220** on a user, camera views **1320** and **1360**, and corresponding facial expressions **1330**, in accordance with certain examples. In FIG. **13**A, example **1310** illustrates a necklace device **1210** as being worn by a user in a front view of the user. Example **1320** illustrates a view of the user as captured by camera **110** of the necklace device **1210**. The corresponding example **1330** illustrates a reconstructed facial expression based on IR images captured by the necklace device **1210**.

(128) In FIG. **13**B, example **1340** illustrates a neckband device **1220** as being worn by a user in a front view of the user. Example **1350** illustrates a neckband device **1220** as being worn by a user in a right-side view of the user. Example **1360** illustrates two views of the user as captured by cameras **110-1** and **110-2** of the neckband device **1220**. The corresponding example **1330** illustrates a reconstructed facial expression based on IR images

captured by the necklace device **1220**.

Other Example Processes

(129) The methods illustrated in FIGS. **14** through **18** are described hereinafter with respect to the components of facial expression reconstruction system **100**, necklace device **1210** of FIG. **12**A, and neckband device **1220** of FIG. **12**B. The methods of FIGS. **14** through **18** may also be performed with other systems and in other environments. The operations described with respect to FIGS. **14** through **18** can be implemented as executable code stored on a computer- or machine-readable, non-transitory, tangible storage medium (for example, floppy disk, hard disk, ROM, EEPROM, nonvolatile RAM, CD-ROM, etc.) that are completed based on execution of the code by a processor circuit implemented using one or more integrated circuit(s); the operations described herein also can be implemented as executable logic that is encoded in one or more non-transitory, tangible media for execution (for example, programmable logic arrays or devices, field programmable gate arrays, programmable array logic, application specific integrated circuits, etc.).

(130) FIG. **14** is a block diagram depicting an alternate embodiment of a deep-learning model process **1400**, in accordance with certain examples. The deep-learning model process **1400** may be performed by the modeling application **132** of the data processing system **130**, previously described herein with reference to FIG. **1**. In an example where the machine learning algorithm is a convolution neural network ("CNN"), the deep-learning model process **1400** receives images as an input and assigns importance (learnable weights and biases) to various aspects/objects in the images. The CNN reduces the images into a form that is easier to process without losing the features of the images that are critical for getting a good prediction. In the current example, the prediction is a reconstructed facial expression based on received IR images of a user's chin profile.

(131) In block **1410**, the data processing system **130** receives IR images from either the necklace device **1210** or the neckband device **1220**. Example necklace **1210** IR images are depicted at **1410**-**1**. Example neckband **1220** IR images are depicted at **1410**-**2** and **1410**-**3** as the neckband acquires both a right and left side IR image of a user's chin profile. Other example IR images are depicted in row **1540**A of FIG. **15**A and row **1540**B of FIG. **15**B.

(132) In block **1420**, the data processing system **130** pre-processes the IR images. Pre-processing the IR images is described in greater detail in reference to method **1720** of FIG. **18**.

(133) In block **1425**, the data processing system **130** duplicates the pre-processed IR images of the necklace device **1210** into three channels to improve the expressiveness and the ability to extract features of the model. As the neckband device **1220** already comprises two pre-processed images, the images are not duplicated into additional channels.

(134) The pre-processed IR images are input into an image processing phase of the deep-learning model process **1400** depicted at block **1430**. Block **1430** comprises convolution layers, normalization layers, and an averaging pooling layer. The processing of block **1430** is described in greater detail herein with reference to blocks **414** and **416** of FIG. **4**. The output from block **1430** is a vector representation of each of the pre-processed IR images.

(135) The vector representations of each of the pre-processed IR images are input into a regression phase **1440** of the deep-learning model process **1400**. The architecture of the regression phase **1440** is similar to the architecture of regression phase **450**, previously described herein with reference to FIG. **4**. Within the regression phase **1440**, the data processing system **130** extracts details from the vector representations of the pre-processed IR images regarding facial components including the cheek, mouth, eyes, eyebrows, chin, and nose and three-dimensional head rotation angles. The data processing system **130** represents each facial expression with blend shapes that are a three-dimensional representation of the vector representations. The parameters of the blend shapes and the three-dimensional head rotation angles, or Euler angles, are obtained by learning from the IR images captured from the neck. The data processing system **130** compares the blend shapes and Euler angles from the vector representations to the blend shapes and Euler angles of the data training set to match the blend shapes/Euler angles to a facial expression from the data training set.

(136) In block **1450**, the data processing system **130** combines the blend shapes with three-dimensional angles of rotation of the user's head. In an example, the three-dimensional angles of rotation are represented by Euler's angles of roll, yaw, and pitch. In block **1460**, the final facial expression reconstruction is output as a three-dimensional image. Example three-dimensional facial expression reconstructions are depicted in row **1530**A of FIG. **15**A and row **1530**B of FIG. **15**B. In an example, the three-dimensional facial expression reconstructions may be further processed by a classifier process, such as classifier process **500** for applications such as silent speech recognition, emoji input, real-time avatar facial expressions, or any other suitable applications.

(137) FIG. **15**A is a diagram **1500**A depicting the correlation between frontal view camera images **1510**A, three-dimensional training data **1520**A, predicted three-dimensional face expressions **1530**A, and infrared facial images **1540**A, in accordance with certain examples.

(138) Row **1510**A illustrates frontal view camera images of a user. To construct a training data set or ground truth, frontal camera images of a user are acquired with the user making various different facial expressions, as depicted in images **1510**A-**1** through **1510**-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method **810**′ of FIG. **16**. Row **1520**A illustrates three-dimensional training data in three-dimensional blend shapes and head rotation angles. The three-dimensional training data images depicted in images **1520**A-**1** through **1520**A-n correlate with images **1510**A-**1** through **1510**-*n*. The three-dimensional training data images **1520**A are the output of method **810**′ of FIG. **16**.

(139) Row **1540**A depicts IR images **1540**A-**1** through **1540**A-n captured by a neck-mounted wearable device, such as necklace device **1210** or neckband device **1220** described herein in greater detail in reference to FIG. **12**. The IR images **1540**A, and the three-dimensional training data images **1520**A are used in the deep-learning model process **1400** to construct the predicted facial expressions **1530**A illustrated in row **1530**A of FIG. **15**A. The predicted facial expressions **1530**A are depicted as three-dimensional images in images **1530**A-**1** through **1530**A-n. The predicted facial expressions **1530**A are illustrated in FIG. **15**A beneath the three-dimensional training data images **1520**A to illustrate the accuracy of predicting facial expressions using IR images **1540**A acquired from necklace device **1210** or neckband device **1220**.

(140) FIG. **15**B is a diagram **1500**B depicting the correlation between additional frontal view camera images **1510**B, three-dimensional training data **1520**B, predicted three-dimensional face expressions **1530**B, and infrared facial images **1540**B, in accordance with certain examples. Each of the frontal view camera images **1510**B, three-dimensional training data **1520**B, predicted three-dimensional face expressions **1530**B, and infrared facial images **1540**B correspond to the frontal view camera images **1510**A, three-dimensional training data **1520**A, predicted three-dimensional face expressions **1530**A, and infrared facial images **1540**A described herein with reference to FIG. **15**A, except these items correlate to different facial expressions.

(141) In this example, the images captured by the cameras of the neck-mounted devices can be processed for facial reconstruction similarly to the methods discussed previously with reference to FIGS. **8**-**11**. Various alternatives to the methods of FIGS. **8**-**11** will be discussed.

(142) FIG. **16** is a block flow diagram depicting an alternative method **810**′ to create a data training set using frontal view images, in accordance with certain examples.

(143) Blocks **910**, **920**, and **930** of FIG. **16** were previously described herein with reference to FIG. **9**.

(144) In block **1640**, the data processing system **130** extracts a set of facial geometric features from the one or more frontal view digital image(s). In an example, the one or more frontal view digital image(s) are three-dimensional digital images captured from a camera that provides depth data in real time along with visual information. Example frontal view digital images are depicted in rows **1510**A and **1510**B, respectively, of FIGS. **15**A and **15**B. In an example, the data processing system **130** extracts full facial expressions using an augmented reality ("AR") application. The AR application extracts high-quality and fine-grained details of facial expressions/movements of the eyes, eyebrows, cheeks, mouth, nose, and chin. The AR application can extract facial features such as facial geometry and rotation from a depth map associated with the three-dimensional digital images. In an example, the rotation is represented by Euler's angles of roll, yaw, and pitch.

(145) In block **1650**, the data processing system **130** compares the extracted features to pre-defined shape parameters. In an example, the AR

application comprises pre-defined blend shapes as templates for complex facial animations. In the example, the AR application comprises blend shapes with features for left and right eyes, mouth and jaw movement, eyebrows, cheeks, nose, tongue, and any other suitable facial features.
(146) In block **1660**, the data processing system **130** creates a data training set based on the comparison of the extracted features to the pre-defined shape parameters. The data training set comprises a blend shape with a Euler angle of head rotation as depicted in block **1520**A of FIG. **15**A and block **1520**B of FIG. **15**B. In an example, the data processing system **130** stores the data training set in the data storage unit **133**.
(147) From block **1660**, the method **810**′ returns to block **820** of FIG. **8**.
(148) FIG. **17** is a block flow diagram depicting an alternative method **860**′ to reconstruct facial expressions using one or more facial digital image(s), in accordance with certain examples.
(149) Block **1010** of FIG. **17** was previously described herein with reference to FIG. **10**.
(150) In block **1720**, the data processing system **130** pre-processes the one or more facial digital image(s). Example facial digital images are depicted in rows **1540**A and **1540**B of FIGS. **15**A and **15**B, respectively. Block **1720** is described in greater detail herein with reference to method **1720** of FIG. **18**.
(151) FIG. **18** is a block flow diagram depicting an alternative method **1720** to pre-process the one or more digital facial image(s), in accordance with certain examples.
(152) In block **1810**, the data processing system **130** converts each of the one or more digital facial image(s) into gray-scale digital facial images. The one or more digital facial image(s) are converted to gray-scale to remove any potential color variance. As the IR bandpass filter **1214** only allows monochrome light into the camera **110**, any color present in the one or more digital facial image(s) does not represent details related to the facial expression of the user.
(153) In block **1820**, the data processing system **130** separates the facial image from the background image in each of the gray-scale digital facial images. Using the IR technology previously discussed in reference to FIG. **12**, the user's skin in the images appears brighter than the surrounding background. However, some light sources in the background may contain IR light that would introduce noise in the background. To remove the noise, the gray-scale digital facial images are binarized based on a set brightness threshold. The data processing system **130** separates the facial image as the brighter component of the gray-scale digital facial image.
(154) In block **1830**, the data processing system **130** applies data augmentation to each of the separated facial images. As a user wears a necklace device **1210** or a neckband device **1220** and performs activities, the position and angles of the cameras **110** of the necklace device **1210** and neckband device **1220** may not be constant. To mitigate the issue, a probability of 60% to conduct three types of image transformations is set, which can be caused by camera shifting causing translation, rotation, and scaling on the images. In an alternate example, any suitable probability may be used. Three Gaussian Models are deployed to generate the parameters for the translation (μ=0, σ.sup.2=30), rotation (μ=0, σ.sup.2=10), scaling (μ=1, σ.sup.2=0.2) on the synthesized training data. The data augmentation is performed on all the images in the training dataset during each training epoch before feeding the images into deep-learning model process **1400**. Data augmentation improves the deep-learning model's ability to confront camera shifting and avoid over-fitting during model training.
(155) From block **1830**, the method **1720** returns to block **1730** of FIG. **17**.
(156) In block **1730**, the data processing system **130** applies the deep-learning model to each of the pre-processed one or more facial digital image(s) to generate facial expression reconstruction. The deep-learning model for facial expression reconstruction was described in the deep-learning model process **1400** herein with reference to FIG. **14** and in the classifier process **500** with reference to FIG. **5**. The results of the facial expression reconstruction from block **1730** may include the reconstruction output from block **1460** of FIG. **14** and/or the encoding of a facial event class from block **550** of FIG. **5**. For example, the data processing system **130** may output from block **1730** the images from the process **1400** of FIG. **14**. The data processing system **130** also, or alternatively, may further process the facial expression reconstruction such that the output from block **1730** comprises a three-dimensional visualization of the facial expression, silent speech recognition, an emoji associated with the facial expression reconstruction, and/or real-time avatar facial expressions associated with the facial expression reconstruction, such as described by the classifier process **500** of FIG. **5**.
(157) From block **1730**, the method **860**′ returns to block **870** of FIG. **8** to output results of the facial reconstruction.
(158) Cameras **110** were described herein with reference to FIG. **1** as any suitable sensors for capturing images. For example, cameras **110** may include depth cameras, red-green-blue (“RGB”) cameras, infrared (“IR”) sensors, acoustic cameras or sensors, microphones and speakers, thermal imaging sensors, charge-coupled devices (“CCDs”), complementary metal oxide semiconductor (“CMOS”) devices, active-pixel sensors, radar imaging sensors, fiber optics, and any other suitable image device technology. While the examples herein described specific cameras for the headband device **210**, earbud devices **220**, necklace device **1210**, and neckband device **1220**, it should be appreciated that any type of camera can be used with any of the wearable devices. The methods to process the images and recreate the facial expressions are adjusted based on camera/image type and are not dependent on the type of device used to acquire the images.
(159) The embodiments herein describe wearable devices in the form of head-mounted devices headband device **210** and earbud devices **220**, and neck-mounted devices necklace device **1210** and neckband device **1220**. Any suitable wearable device may be used such that the one or more camera(s) may be directed towards a user's face including, but not limited to, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality (“VR”) headset.
(160) The embodiments herein describe head-mounted and neck-mounted wearable devices comprises cameras **110** with adjustable positions and angles. Each camera **110** may be positioned such that at least one of a buccal region, a zygomatic region, and/or a temporal region of a user's face are included in the field of view of the camera **110**.
(161) In an alternate embodiment, the wearable devices previously described herein may be configured for use in a hand-face touching detection system to recognize and predict a time and position that a user's hand touches the user's face. An important step in reducing the risk of infection is avoiding touching the face because a virus, such as COVID-19, may enter the mucous membranes of the eyes, nose, and/or mouth. Touching different areas of the face carries different health related risks. For example, contacting a mucous membrane may introduce a higher risk of transmitting a virus than touching non-mucous areas such as the chin and cheek. In addition, the frequency of touching the face may be an indicator regarding the stress level of a person. Understanding how people touch their face may alleviate multiple health challenges. Accurately recognizing where the hand touches the face is an important step towards alleviating health risks introduced by hand-face touching behaviors. In order to implement behavior intervention technologies, the hand-face touching detection system predicts the behavior in advance rather than simply detecting the touching behavior.
(162) A data training set is created similar to the data training sets described herein with reference to FIGS. **9** and **16**. The difference is that instead of frontal view images of a user capturing just the user's face, the data training set uses frontal view images of a user with different hand-face touching positions. For example, frontal view images of the user may be acquired with the user touching his eye, nose, mouth, cheek, or any other suitable facial feature. In addition to touching faces directly, people move their hands closer to their faces when performing daily activities such as eating and drinking. The daily activities are classified as separate behaviors to avoid false-positive errors of a person touching their face. Additional frontal view images are acquired for the data training set depicting example daily activities. For example, the frontal view images may comprise a user eating with his hand, a fork, a spoon, or chopsticks. The user may be drinking with a straw, a bottle, or a cup. The user may be placing an earbud in or out, putting on or taking off eyeglasses, adjusting eyeglasses, applying lip balm, or adjusting his hair. Any suitable daily activities may be used in the frontal view images to create the data training set.
(163) The frontal view images of the user are sent to a server, such as data processing system **130**, to create the data training set, as previously

described herein with reference to FIGS. **9** and **16**. However, instead of extracting facial landmark positions as described in block **940** of FIG. **9** or facial geometric features as described in block **1640** of FIG. **16**, the data processing system extracts hand/facial landmark positions or hand/geometric features to create the data training set.

(164) In an example, necklace device **1210** may be positioned on a user to acquire images of the user's facial area, as previously described herein, to also include a user's hand if positioned or in motion near the user's face. Any suitable wearable device may be used to acquire the images of the user's face. To predict the time and location of a hand-face touch, camera images are monitored over a period of time. Camera images are sent to a server, such as data processing system **130**, for processing. The data processing system receives the hand/facial images and reconstructs the position of the user's hand relative to the user's face using the data training set and a deep-learning model, such as the models previously described herein with reference to FIGS. **4** and **14**. The data processing system monitors the hand-face position over time to predict where and when a user may touch his face. Alternatively, the data processing system may determine that the user is participating in a daily activity and that the user is not in the process of a hand-face touch. In each case, the system outputs a reconstructed image (or other similar information as described herein) depicting the user's facial expression and a representation of the hand-touch activity. The system also can log each hand-touch to the facial area and output notifications of the activity.

(165) In an alternate embodiment, acoustic sensing technology can be used to reconstruct facial expressions using an array of microphones and speakers and deep-learning models. The acoustic technology comprises a wearable device such as headphones, earbuds, necklaces, neckbands, and any other suitable form factors such that the microphones and speakers can be mounted or attached to the wearable device. In an example, the microphones are Micro-Electro-Mechanical System (“MEMS”) microphones that are placed on a printed circuit board (“PCB”). In an example, each microphone/speaker assembly may comprise four microphones and one speaker. Any suitable number of microphones and speakers may be included. The positions of the microphones and speakers may be adjustable to optimize the recording of reflected signals.

(166) The acoustic wearable device actively sends signals from the speakers towards a user's face. In an example, the speakers transmit inaudible acoustic signals within a frequency range of 16 kHz to 24 kHz towards the user's face. Any suitable frequency range may be used. The signals are reflected by the user's face and captured by the microphones on the acoustic wearable device. The signals are reflected differently back towards the microphones based on different facial expressions or movements. A Channel Impulse Response (“CIR”) is calculated based on the acoustic signals received at each microphone.

(167) Each microphone is in communication with a computing device, such as computing device **120**, such that the CIR images/data can be transmitted for processing to a server, such as data processing system **130**.

(168) The data processing system creates a data training set using frontal view images of a user in a machine learning algorithm, previously described herein with reference to FIGS. **9** and **16**. The data processing system receives the CIR images/data and reconstructs the user's facial expression using the data training set and a deep-learning model, such as the models previously described herein with reference to FIGS. **4** and **14**. The data processing system outputs the resultant facial expression reconstruction, such as previously described herein with reference to block **870** of FIG. **8**.

(169) The wearable devices and systems described herein may be used in applications such as silent speech recognition, emoji input, and real-time avatar facial expressions. Silent speech recognition is a method to recognize speech when vocalization is inappropriate, background noise is excessive, or vocalizing speech is challenging due to a disability. The data training set for silent speech recognition comprises a set of frontal view facial images directed to the utterance of a word or phrase. To recognize silent speech, the wearable device, such as necklace **1220**, captures a series of facial movements from underneath the chin of a user while the user silently utters words or commands and transfers the series of digital facial images to the data processing system **130** for facial expression reconstruction. The results of the facial expression reconstruction, previously described herein with reference to block **456** or block **1460**, are used as inputs to the classifier process **500**, previously described herein with reference to FIG. **5**. Using the series of facial expression reconstruction images, the classifier process **500** detects the start time and end time of an utterance and selects the frames from the series of facial expression reconstruction images associate with the utterance. A vector representation of the utterance is created and input into a probability distribution to predict a word or phrase. In some embodiments for applications, such as in a testing process after the training, or using a trained system, a front view of a user is not needed. In some embodiments for speech recognition, one or more imaging sensor(s), such as camera(s) in a necklace and/or a wearable device worn in or on ear(s) of a user, capture a series of images for speaking a syllable, word, or phrase with or without audio (for example, at least two images of a syllable, a word, or a phrase), wherein each image comprises only a portion of a full face. In some embodiments, one image used in the application only captures less than 50%, less than 40%, less than 30%, less than 25%, less than 20%, or less than 15% of a full face. In some embodiments, the one or more imaging sensor(s) used in the application do not capture direct visual information for eyes, nose, and/or mouth. In some embodiments, the one or more image(s) and the combination thereof used in the application capture incomplete visual information for eyes, nose, and/or mouth. In some embodiments, the system and method are configured to recognize or predict speech, and/or other facial expression(s) at a high accuracy, wherein the accuracy is at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 98% for one or multiple words, phrases, or facial expressions on average. In some embodiments, the application is a speech recognition based on images/videos captured by the system. In other embodiments, the image-based or video-based speech recognition may be combined with an audio-based speech recognition to improve the accuracy of speech recognition. In some embodiments, the image-based or video-based speech recognition may be used to validate a result of an audio-based speech recognition.

Alternate Embodiment

Alternate Embodiment Example System Architecture

(170) FIG. **19** is a block diagram depicting an acoustic sensor system based facial expression reconstruction system **1900**, in accordance with certain examples. As depicted in FIG. **19**, acoustic sensor system based facial expression reconstruction system **1900** comprises user computing device **150**, acoustic sensor systems **1910-1** through **1910**-*n*, and data processing computing device **1920**. User computing device **150** and data processing computing device **1920** are configured to communicate via network **140**. Network **140** and user computing device **150** were previously described herein with reference to FIG. **1**. User computing device **150** may be configured to receive display a facial output, such as a predicted facial expression with three-dimensional visualization with examples depicted at **2830** of FIGS. **28** and **2930** of FIG. **29**. Acoustic sensor systems **1910** are described in greater detail herein with reference to FIGS. **20**A, **20**B, and **20**C.

(171) FIG. **20**A is a block diagram depicting a first embodiment of an acoustic sensor system **1910** in accordance with certain examples. Acoustic sensor system **1910** comprises speaker **1911**, microphone **1912**, Bluetooth module **1913**, and printed circuit boards (“PCB”) **1914**. Speaker **1911** is a small form factor speaker. In an example, speaker **1911** is a speaker configurable to transmit signals in an inaudible frequency range. For example, speaker **1911** may be configured to transmit signals in a frequency range of 16-24 kHz. Any suitable frequency range may be used. In an example, speaker **1911** transmits signals that are frequency modulation continuous wave (“FMCW”) transmissions, Channel Impulse Response (“CIR”) transmissions, or any other suitable type of acoustic signal. In an example, the speakers transmit inaudible acoustic signals within the frequency range of 16 kHz to 24 kHz towards a user's face. As depicted in FIG. **20**A, speaker **1911** is affixed to a PCB **1914**.

(172) Microphone **1912** is a small form factor microphone. In an example, microphone **1912** is a Micro-Electro-Mechanical System (“MEMS”) microphone. Microphone **1912** may be a microphone chip, a silicon microphone, or any other suitable microphone. In an example, microphone **1912** is configured to receive reflected signals transmitted by speaker **1911**. As depicted in FIG. **20**A, microphone **1912** is affixed to a PCB **1914**.

(173) Bluetooth module **1913** may be a Bluetooth low energy (“BLE”) module. Bluetooth module **1913** may be a System-on-Chip (“SoC”) module. In an example, Bluetooth module **1913** may comprise a microcontroller unit (“MCU”) with memory. In an example, the memory may be an on-board SanDisk (“SD”) card. The SD card may function to save acoustic data on the MCU. Bluetooth module **1913** may be configured to control a battery

(not depicted) to provide power to the acoustic sensing system **1910**. In an example, the battery may be a lithium polymer ("LiPo") battery. Bluetooth module **1913** may be configured to power and control speaker **1911** transmissions, microphone **1912** signal receptions, and data transmissions external to a wearable device, for example transmissions to computing device **1920**. As depicted in FIG. **20**A, Bluetooth module **1913** is affixed to a PCB **1914**.
(174) FIG. **20**B is a block diagram depicting a second embodiment of an acoustic sensor system **1910**, in accordance with certain examples. FIG. **20**B depicts speaker **1911** and microphone **1912** affixed to a common PCB **1914** with Bluetooth module **1913** affixed to a separate PCB **1914**.
(175) FIG. **20**C is a block diagram depicting a third embodiment of an acoustic sensor system **1910**, in accordance with certain examples. FIG. **20**C depicts speaker **1911**, microphone **1912**, and Bluetooth module **1913** affixed to a common PCB **1914**. Any suitable configuration of speaker **1911**, microphone **1912**, Bluetooth module **1913** and PCBs **1914** may be used in addition to the configurations depicted in FIGS. **20**A, **20**B, and **20**C.
(176) In an example, a wearable device system may comprise one or more speaker(s) **1911** and one or more microphone(s) **1912**, with a pair of a speaker **1911** and a microphone **1912** affixed on a PCB **1914**. Any suitable number of speakers **1911** and microphones **1912** may be included on a particular wearable device system. The positions and/or orientations of the speakers **1911** and microphones **1912** may be adjustable to optimize the recording of reflected signals.
(177) Returning to FIG. **19**, acoustic sensor system based facial expression reconstruction system **1900** comprises data processing computing device **1920**. Data processing computing device **1920** functions to receive data from Bluetooth module **1913** to construct a facial output. The facial output may comprise one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression. Data processing computing device **1920** may be a smart phone, a mobile device, or any type of computing device described herein with reference to FIG. **31**. Data processing computing device **1920** comprises central processing unit **121**, graphics processing unit **122**, memory **123**, communication application **124**, and data processing module **1930**. Central processing unit **121**, graphics processing unit **122**, memory **123**, and communication application **124** were previously described herein with reference to FIG. **1**. Data processing module **1930** is described in greater detail herein with reference to FIG. **21**.
(178) FIG. **21** is a block diagram depicting a data processing module **1930**, in accordance with certain examples. Data processing module **1930** comprises central processing unit **131**, modeling application **132**, data storage unit **133**, and communication application **134**. Central processing unit **131**, modeling application **132**, data storage unit **133**, and communication application **134** were previously described herein with reference to FIG. **1**.
(179) FIG. **22**A depicts a first embodiment of a head-mounted acoustic sensor system wearable device **2210**. FIG. **22**A depicts a glasses style wearable device **2210** (herein referred to as glasses device **2210**). Glasses device **2210** comprises two speakers **1911-1** and **1911-2**, two microphones **1912-1** and **1912-2**, and a Bluetooth module **1913**. While two speakers **1911** and two microphones **1912** are depicted, any suitable quantity of speakers **1911** and microphones **1912** may be used.
(180) FIG. **22**B is an illustration depicting the first embodiment of the head-mounted acoustic sensor system wearable device **2210** on a user. As illustrated in FIG. **22**B, glasses device **2210** may be positioned on a user such that a pair of speaker **1911-1** and microphone **1912-1** are located on the left side of the user's face and a pair of speaker **1911-2** and microphone **1912-2** are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.
(181) FIG. **23**A depicts a second embodiment of a head-mounted acoustic sensor system wearable device **2310**. FIG. **23**A depicts a headset style wearable device **2310** (herein referred to as headset device **2310**). In an example, headset device **2310** may be a virtual reality headset. Headset device **2310** comprises two speakers **1911-1** and **1911-2**, two microphones **1912-1** and **1912-2**, and a Bluetooth module **1913** (depicted in FIG. **23**B). While two speakers **1911** and two microphones **1912** are depicted, any suitable quantity of speakers **1911** and microphones **1912** may be used.
(182) FIG. **23**B is an illustration depicting the second embodiment of the head-mounted acoustic sensor system wearable device **2310** on a user. As illustrated in FIG. **23**B, headset device **2310** may be positioned on a user such that a pair of speaker **1911-1** and microphone **1912-1** are located on the left side of the user's face and a pair of speaker **1911-2** and microphone **1912-2** are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.
(183) FIG. **24**A depicts a third embodiment of a head-mounted acoustic sensor system wearable device **2410**. FIG. **24**A depicts a headphone style wearable device **2410** (herein referred to as headphone device **2410**). In an example, headphone device **2410** may be a headband style headphone, ear pods, earphones, ITE headphones, over the ear headphones, or any other headphone type device. Headphone device **2410** comprises two speakers **1911-1** and **1911-2**, two microphones **1912-1** and **1912-2**, and a Bluetooth module **1913**. While two speakers **1911** and two microphones **1912** are depicted, any suitable quantity of speakers **1911** and microphones **1912** may be used.
(184) FIG. **24**B is an illustration depicting the third embodiment of the head-mounted acoustic sensor system wearable device **2410** on a user. As illustrated in FIG. **24**B, headphone device **2410** may be positioned on a user such that a pair of speaker **1911-1** and microphone **1912-1** are located on the left side of the user's face and a pair of speaker **1911-2** and microphone **1912-2** are located on the right side of the user's face. The disposition of each pair on the left and right side of the user allows data associated with skin deformations to be collected for the left and right contours of the user's face.
(185) As described herein, FIGS. **22**A, **23**A, and **24**A depict example embodiments of a head-mounted acoustic sensor system wearable device. Other example embodiments may include ear buds, ear pods, in-the-ear ("ITE") headphones, over-the-ear headphones, outside-the-ear ("OTE") headphones, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, another head-borne device, a necklace, a neckband, a garment-attachable system, or any other type of device or system to which at least one acoustic sensor system **1910** may be affixed.
Alternate Embodiment Example Processes
(186) FIG. **25** is a block flow diagram depicting a method **2500** to reconstruct facial expressions based on acoustic signals, in accordance with certain examples. In block **810**, a data training set is created using frontal view digital images. Block **810** was previously described herein with reference to method **810** of FIG. **9**.
(187) In block **2510**, a wearable system transmits at least one acoustic signal. The wearable system may be a system as described herein with reference to glasses device **2210**, headset device **2310**, headphone device **2410**. In an example, the wearable system may include other embodiments such as ear buds, ear pods, ITE headphones, over-the-ear headphones, OTE headphones, glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, another head-borne device, a necklace, a neckband, a garment-attachable system, or any other type of device or system to which at least one acoustic sensor system **1910** may be affixed. In an example, the wearable system is positioned on a wearer such that the distance from the wearable system to the wearer's face is less than 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, or any suitable distance. The at least one acoustic signal is transmitted by a speaker **1911** of at least one acoustic sensor system **1910**. Acoustic sensor system **1910** may be configured to transmit signals in a frequency range of 16-24 kHz. Any suitable frequency range may be used. Acoustic sensor system **1910** may transmit signals that are FMCW transmissions, CIR transmissions, or any other suitable type of acoustic signal. In an example, the transmitted signals have an associated sample length. For example, the sample length may be set to 600 corresponding to a sample length of 0.012 seconds. In this example, approximately 83.3 frames may be collected per second. Any suitable sample length may be used. In an example, acoustic sensor system **1910** transmits inaudible acoustic signals within the frequency range of 16 kHz to 24 kHz towards a wearer's face. In an example, acoustic sensor system **1910** may be positioned on the wearable system to transmit acoustic signals towards a first facial feature on a first side of a sagittal plane of a wearer

and to transmit acoustic signals towards a second facial feature on a second side of the sagittal plane of a wearer. In an example, acoustic sensor system **1910** may be positioned on the wearable system to transmit acoustic signals towards an underside contour of a chin of a wearer.

(188) In block **2515**, the wearable system receives at least one reflected acoustic signal. Subsequent to transmitting the at least one acoustic signal toward a wearer's face as described herein with reference to block **2510**, the signal is reflected by the wearer's face. The at least one reflected acoustic signal is received by microphone **1912** of the at least one acoustic sensor system **1910**. The at least one transmitted signal is reflected differently to microphone **1912** based on skin deformations associated with different facial movements.

(189) In block **2520**, the wearable system transmits the at least one transmitted acoustic signal and the at least one reflected acoustic signal to data processing computing device **1920**. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are transmitted in frames. The at least one transmitted acoustic signal and the at least one reflected acoustic signal are transmitted by the Bluetooth module **1913** of the at least one acoustic sensor system **1910**.

(190) In block **2525**, data processing computing device **1920** receives the at least one transmitted acoustic signal and the at least one reflected acoustic signal.

(191) In block **2530**, data processing computing device **1920** filters the received acoustic signals. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are filtered to remove noise outside of a target frequency range. In an example, the target frequency range is 15.5 kHz to 20.5 kHz. Any suitable target frequency range may be used. In an example, the at least one transmitted acoustic signal and the at least one reflected acoustic signal are filtered using a Butterworth band-pass filter. Any suitable filter may be used.

(192) In block **2535**, data processing computing device **1920** constructs a profile using the filtered acoustic signals. In an example, the profile is an echo profile. The echo profile is determined by calculating a cross-correlation between the filtered at least one transmitted acoustic signal and the filtered at least one reflected acoustic signal. The echo profile depicts deformations of the skin of the wearer's face in temporal and spatial domains. In an example, a differential echo profile may be calculated. A differential echo profile is calculated by subtracting echo profiles between two adjacent echo frames. The differential echo profile removes static objects that may be present in the echo profile. Further, as the wearable system was positioned to be less than 10-15 cm away from a wearer's face, echo profiles comprising distance greater than that range may also be removed. For example, echo profiles with a distance greater than ±16 cm, ±17 cm, ±18 cm, or any other suitable distance may be removed. In an example, the transmitted acoustic signal is a FMCW signal from which the echo profile is constructed. In alternate examples, the transmitted acoustic signal may be a CIR signal with global system for mobiles (“GSM”), characteristic impedance (“ZC”), or Barker sequence encoding; angle of arrival (“AoA”); doppler effect; or phase change detection.

(193) In block **2540**, data processing computing device **1920** applies a deep learning model to the constructed profile. The deep-learning model for facial expression reconstruction was previous described herein with reference to the processes and methods of FIG. **4**, FIG. **5**, and FIG. **14**. In an example with reference to acoustic data, a sliding window of one second may be applied to the differential echo profiles to provide 84 frames of differential echo profiles in each window. In an example, each differential echo profile may have approximately 200 data points providing an input vector of size 200×84. The input vector is used in the deep learning model along with the training data to estimate facial expressions.

(194) In block **2545**, data processing computing device **1920** assigns a deformation to the constructed profile based on the results of the deep learning model. In an example, the assigned deformation has a predetermined degree of correspondence to a selected one of a plurality of deformations in the deep learning model.

(195) In block **2550**, the data processing computing device **1920** communicates a facial output based on the assigned deformation. In an example, the facial output comprises one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression. In an example, the facial output is communicated to user computing device **150**. In an example, the facial output is communicated in real time. In an example, the facial output is continuously updated.

(196) FIG. **26** is a diagram **2600** depicting an acoustic sensor system based facial expression reconstruction process. FIG. **26** is a visual depiction of the method **2500** described herein with reference to FIG. **25**. At **2610**, a wearer is depicted with glasses device **2210** and headset device **2310**. In an example, a wearer may select a wearable device from the various embodiments previously described herein. At **2620**, speaker **1911**, microphone **1912**, and Bluetooth module **1913** of an acoustic sensor system **1910** of the wearable device are illustrated. At **2630**, an acoustic signal is depicted as being transmitted by speaker **1911** and directed to the wearer's face. At **2630**, the acoustic signal is deflected by the wearer's face and received by microphone **1912** at **2640**. At **2650**, the transmitted acoustic signal and the deflected acoustic signal are transmitted to data processing computing device **1920** (not depicted) by Bluetooth module **1913**. At **2660**, the transmitted acoustic signal and the deflected acoustic signal are filtered. At **2670**, differential echo profiles are depicted. At **2680**, the differential echo profiles are input into a deep learning model. At **2690**, predicted facial expressions are depicted.

(197) FIG. **27** is a diagram **2700** depicting an acoustic sensor system based facial expression reconstruction output process. FIG. **27** is a second visual depiction of the method **2500** described herein with reference to FIG. **25**. At **2710**, a wearer is depicted with glasses device **2210** with an acoustic sensor system **1910** comprising speaker **1911**, microphone **1912**, Bluetooth module **1913**, and PCB **1914**. At **2720**, acoustic data, i.e., transmitted acoustic signal and deflected acoustic signal, are transmitted by acoustic sensor system **1910** to data processing computing device **1920**. At **2730**, the transmitted acoustic signal and the deflected acoustic signal are filtered. At **2740**, differential echo profiles are depicted. At **2750**, the differential echo profiles are input into a deep learning model. At **2760**, data processing computing device **1920** transmits the output of the deep learning model to user computing device **150**. At **2770**, example facial outputs are depicted as received by user computing device **150**. Example facial outputs may be a 3D avatar, such as depicted at **2771**, or a personalized avatar, such as depicted at **2772**.

(198) FIG. **28** is a diagram **2800** depicting the correlation between frontal view camera images **2810** and **2820**, predicted facial expressions **2830**, and left and right differential echo profiles, **2840** and **2850** respectively, in accordance with certain examples. Rows **2810** and **2820** illustrate frontal view camera images of a user. To construct a training data set or “ground truth,” frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images **2810-1** through **2810**-*n* and images **2820-1** through **2820**-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method **810** of FIG. **9**. Row **2840** depicts left differential echo profiles **2840-1** through **2840**-*n* captured by a head-mounted, wearable device, such as glasses device **2210** or headset device **2310** described herein with reference to FIG. **22** and FIG. **23**. Row **2850** depicts right differential echo profiles **2850-1** through **2850**-*n* captured by a head-mounted, wearable device, such as glasses device **2210** or headset device **2310** described herein with reference to FIG. **22** and FIG. **23**. Left differential echo profiles **2840** and right differential echo profiles **2850** are input into a deep learning model, as previously described herein, to create the predicted facial expressions depicted at **2830**. Predicted facial expressions **2830-1** through **2830**-*n* are illustrated in FIG. **28** beneath the frontal view camera images in rows **2810-1** through **2810**-*n* and **2820-1** through **2820**-*n* to illustrate the accuracy of the predicted facial expressions **2830**.

(199) FIG. **29** is a diagram **2900** depicting the correlation between frontal view camera images **2910**, training data **2920**, predicted facial expressions **2930**, and left and right differential echo profiles, **2940** and **2950** respectively. Row **2910** illustrates frontal view camera images of a user. To construct a training data set or “ground truth,” frontal view camera images of a user are acquired, via an imaging capturing device, with the user making various different facial expressions as depicted in images **2910-1** through **2910**-*n*. The method to create the training data set from the frontal view camera images is described herein in greater detail with reference to method **810** of FIG. **9**. Row **2920** depicts a training data image or ground truth for the frontal view camera images in row **2910**. Training data images **2920-1** through **2920**-*n* correspond with frontal view camera images

2910-1 through **2910**-*n*. Row **2940** depicts left differential echo profiles **2940-1** through **2940**-*n* captured by a head-mounted, wearable device. Row **2950** depicts right differential echo profiles **2950-1** through **2950**-*n* captured by a head-mounted, wearable device. Left differential echo profiles **2940** and right differential echo profiles **2950** are input into a deep learning model, as previously described herein, to create the predicted facial expressions depicted at **2930**. Predicted facial expressions **2930-1** through **2930**-*n* are illustrated in FIG. **29** beneath the ground truth images **2920-1** through **2920**-*n* to illustrate the accuracy of the predicted facial expressions **2930**.

(200) FIG. **30**A is a diagram depicting the correlation between frontal view camera images of an open mouth **3010**, a left view image **3015**, a left and right echo profile, **3020** and **3025** respectively, and a right view image **3030**. The skin deformations, depicted in left view images **3015-1** through **3015**-*n* and right view images **3030-1** through **3030**-*n*, created by the various facial expressions **3010-1** through **3010**-*n* result in distinctive patterns changes depicted in left echo profiles **3020-1** through **3020**-*n* and right echo profiles **3025-1** through **3025**-*n*. FIG. **30**B is a diagram depicting the correlation between frontal view camera images of a right sneer **3035**, a left view image **3040**, a left and right echo profile, **3045** and **3050** respectively, and a right view image **3055**. The skin deformations, depicted in left view images **3040-1** through **3040**-*n* and right view images **3055-1** through **3055**-*n*, created by the various facial expressions **3035-1** through **3035**-*n* result in distinctive patterns changes depicted in left echo profiles **3045-1** through **3045**-*n* and right echo profiles **3050-1** through **3050**-*n*.

(201) While acoustic sensing technology may be used to determine facial expressions by detecting skin deformations, acoustic sensing technology may be used to directly determine other outputs associated with skin deformation without the need to first determine facial expressions. For example, the acoustic sensing technology may track blinking patterns and eyeball movements of a user. Blinking patterns and eyeball movements can be applied to the diagnosis and treatment processes of eye diseases. The acoustic sensing technology may detect movements of various parts of the face associated with speech, whether silent or voiced. While speaking, different words and/or phrases lead to subtle, yet distinct, skin deformations. The acoustic sensing technology can capture the skin deformation to recognize silent speech or to vocalize sound. By tracking subtle skin deformations, the acoustic sensing technology can be used to synthesize voice. The acoustic sensing technology may also be used to recognize and track physical activities. For example, while eating, a user opens the mouth, chews, and swallows. During each of those movements, the user's skin deforms in a certain way such that opening the mouth is distinguishable from chewing and from swallowing. The detected skin deformations may be used to determine the type and quantity of food consumed by a user. Similar to eating, a type and quantity of a consumed drink may be determined by skin deformation. The acoustic sensing technology may also be used to determine an emotional status of a user. An emotional status may be related to skin deformations. By detecting skin deformations, an emotional status can be determined and reported to corresponding applications. In an example, acoustic sensor system based facial expression reconstruction system **1900**, described herein with reference to FIG. **19**, may be configured to directly determine the other outputs associated with skin deformation without the need to first determine facial expressions, including, but not limited to, facial movements including eyeball movement, speech recognition, physical activities, and emotional status. Acoustic sensor system based facial expression reconstruction system **1900** may be configured to directly determine the other outputs associated with skin deformation using any type acoustic sensor system **1910** as described herein with reference to FIG. **19**.

Other Examples

(202) FIG. **31** depicts a computing machine **4000** and a module **4050** in accordance with certain examples. The computing machine **4000** may correspond to any of the various computers, servers, mobile devices, embedded systems, or computing systems presented herein. The module **4050** may comprise one or more hardware or software element(s) configured to facilitate the computing machine **4000** in performing the various methods and processing functions presented herein. The computing machine **4000** may include various internal or attached components such as a processor **4010**, system bus **4020**, system memory **4030**, storage media **4040**, input/output interface **4060**, and a network interface **4070** for communicating with a network **4080**.

(203) The computing machine **4000** may be implemented as a conventional computer system, an embedded controller, a laptop, a server, a mobile device, a smartphone, a set-top box, a kiosk, a router or other network node, a vehicular information system, one or more processor(s) associated with a television, a customized machine, any other hardware platform, or any combination or multiplicity thereof. The computing machine **4000** may be a distributed system configured to function using multiple computing machines interconnected via a data network or bus system.

(204) The processor **4010** may be configured to execute code or instructions to perform the operations and functionality described herein, manage request flow and address mappings, and to perform calculations and generate commands. The processor **4010** may be configured to monitor and control the operation of the components in the computing machine **4000**. The processor **4010** may be a general purpose processor, a processor core, a multiprocessor, a reconfigurable processor, a microcontroller, a digital signal processor ("DSP"), an application specific integrated circuit ("ASIC"), a graphics processing unit ("GPU"), a field programmable gate array ("FPGA"), a programmable logic device ("PLD"), a controller, a state machine, gated logic, discrete hardware components, any other processing unit, or any combination or multiplicity thereof. The processor **4010** may be a single processing unit, multiple processing units, a single processing core, multiple processing cores, special purpose processing cores, co-processors, or any combination thereof. The processor **4010** along with other components of the computing machine **4000** may be a virtualized computing machine executing within one or more other computing machine(s).

(205) The system memory **4030** may include non-volatile memories such as read-only memory ("ROM"), programmable read-only memory ("PROM"), erasable programmable read-only memory ("EPROM"), flash memory, or any other device capable of storing program instructions or data with or without applied power. The system memory **4030** may also include volatile memories such as random access memory ("RAM"), static random access memory ("SRAM"), dynamic random access memory ("DRAM"), and synchronous dynamic random access memory ("SDRAM"). Other types of RAM also may be used to implement the system memory **4030**. The system memory **4030** may be implemented using a single memory module or multiple memory modules. While the system memory **4030** is depicted as being part of the computing machine **4000**, one skilled in the art will recognize that the system memory **4030** may be separate from the computing machine **4000** without departing from the scope of the subject technology. It should also be appreciated that the system memory **4030** may include, or operate in conjunction with, a non-volatile storage device such as the storage media **4040**.

(206) The storage media **4040** may include a hard disk, a floppy disk, a compact disc read only memory ("CD-ROM"), a digital versatile disc ("DVD"), a Blu-ray disc, a magnetic tape, a flash memory, other non-volatile memory device, a solid state drive ("SSD"), any magnetic storage device, any optical storage device, any electrical storage device, any semiconductor storage device, any physical-based storage device, any other data storage device, or any combination or multiplicity thereof. The storage media **4040** may store one or more operating system(s), application programs and program modules such as module **4050**, data, or any other information. The storage media **4040** may be part of, or connected to, the computing machine **4000**. The storage media **4040** may also be part of one or more other computing machine(s) that are in communication with the computing machine **4000** such as servers, database servers, cloud storage, network attached storage, and so forth.

(207) The module **4050** may comprise one or more hardware or software element(s) configured to facilitate the computing machine **4000** with performing the various methods and processing functions presented herein. The module **4050** may include one or more sequence(s) of instructions stored as software or firmware in association with the system memory **4030**, the storage media **4040**, or both. The storage media **4040** may therefore represent machine or computer readable media on which instructions or code may be stored for execution by the processor **4010**. Machine or computer readable media may generally refer to any medium or media used to provide instructions to the processor **4010**. Such machine or computer readable media associated with the module **4050** may comprise a computer software product. It should be appreciated that a computer software product comprising the module **4050** may also be associated with one or more process(es) or method(s) for delivering the module **4050** to the computing machine **4000** via the network **4080**, any signal-bearing medium, or any other communication or delivery technology. The module **4050** may also comprise hardware circuits or information for configuring hardware circuits such as microcode or configuration information for an FPGA

or other PLD.
(208) The input/output ("I/O") interface **4060** may be configured to couple to one or more external device(s), to receive data from the one or more external device(s), and to send data to the one or more external device(s). Such external devices along with the various internal devices may also be known as peripheral devices. The I/O interface **4060** may include both electrical and physical connections for operably coupling the various peripheral devices to the computing machine **4000** or the processor **4010**. The I/O interface **4060** may be configured to communicate data, addresses, and control signals between the peripheral devices, the computing machine **4000**, or the processor **4010**. The I/O interface **4060** may be configured to implement any standard interface, such as small computer system interface ("SCSI"), serial-attached SCSI ("SAS"), fiber channel, peripheral component interconnect ("PCP"), PCI express (PCIe), serial bus, parallel bus, advanced technology attached ("ATA"), serial ATA ("SATA"), universal serial bus ("USB"), Thunderbolt, FireWire, various video buses, and the like. The I/O interface **4060** may be configured to implement only one interface or bus technology. Alternatively, the I/O interface **4060** may be configured to implement multiple interfaces or bus technologies. The I/O interface **4060** may be configured as part of, all of, or to operate in conjunction with, the system bus **4020**. The I/O interface **4060** may include one or more buffer(s) for buffering transmissions between one or more external device(s), internal device(s), the computing machine **4000**, or the processor **4010**.
(209) The I/O interface **4060** may couple the computing machine **4000** to various input devices including mice, touch-screens, scanners, electronic digitizers, sensors, receivers, touchpads, trackballs, cameras, microphones, keyboards, any other pointing devices, or any combinations thereof. The I/O interface **4060** may couple the computing machine **4000** to various output devices including video displays, speakers, printers, projectors, tactile feedback devices, automation control, robotic components, actuators, motors, fans, solenoids, valves, pumps, transmitters, signal emitters, lights, and so forth.
(210) The computing machine **4000** may operate in a networked environment using logical connections through the network interface **4070** to one or more other system(s) or computing machines across the network **4080**. The network **4080** may include WANs, LANs, intranets, the Internet, wireless access networks, wired networks, mobile networks, telephone networks, optical networks, or combinations thereof. The network **4080** may be packet switched, circuit switched, of any topology, and may use any communication protocol. Communication links within the network **4080** may involve various digital or an analog communication media such as fiber optic cables, free-space optics, waveguides, electrical conductors, wireless links, antennas, radio-frequency communications, and so forth.
(211) The processor **4010** may be connected to the other elements of the computing machine **4000** or the various peripherals discussed herein through the system bus **4020**. It should be appreciated that the system bus **4020** may be within the processor **4010**, outside the processor **4010**, or both. Any of the processor **4010**, the other elements of the computing machine **4000**, or the various peripherals discussed herein may be integrated into a single device such as a system on chip ("SOC"), system on package ("SOP"), or ASIC device.
(212) Examples may comprise a computer program that embodies the functions described and illustrated herein, wherein the computer program is implemented in a computer system that comprises instructions stored in a machine-readable medium and a processor that executes the instructions. However, it should be apparent that there could be many different ways of implementing examples in computer programming, and the examples should not be construed as limited to any one set of computer program instructions. Further, a skilled programmer would be able to write such a computer program to implement an example of the disclosed examples based on the appended flow charts and associated description in the application text. Therefore, disclosure of a particular set of program code instructions is not considered necessary for an adequate understanding of how to make and use examples. Further, those skilled in the art will appreciate that one or more aspect(s) of examples described herein may be performed by hardware, software, or a combination thereof, as may be embodied in one or more computing system(s). Moreover, any reference to an act being performed by a computer should not be construed as being performed by a single computer as more than one computer may perform the act.
(213) The examples described herein can be used with computer hardware and software that perform the methods and processing functions described herein. The systems, methods, and procedures described herein can be embodied in a programmable computer, computer-executable software, or digital circuitry. The software can be stored on computer-readable media. Computer-readable media can include a floppy disk, RAM, ROM, hard disk, removable media, flash memory, memory stick, optical media, magneto-optical media, CD-ROM, etc. Digital circuitry can include integrated circuits, gate arrays, building block logic, field programmable gate arrays ("FPGA"), etc.
(214) The systems, methods, and acts described in the examples presented previously are illustrative, and, alternatively, certain acts can be performed in a different order, in parallel with one another, omitted entirely, and/or combined between different examples, and/or certain additional acts can be performed, without departing from the scope and spirit of various examples. Accordingly, such alternative examples are included in the scope of the following claims, which are to be accorded the broadest interpretation so as to encompass such alternate examples.
(215) Although specific examples have been described above in detail, the description is merely for purposes of illustration. It should be appreciated, therefore, that many aspects described above are not intended as essential elements unless explicitly stated otherwise. Modifications of, and equivalent components or acts corresponding to, the disclosed aspects of the examples, in addition to those described above, can be made by a person of ordinary skill in the art, having the benefit of the present disclosure, without departing from the spirit and scope of examples defined in the following claims, the scope of which is to be accorded the broadest interpretation so as to encompass such modifications and equivalent structures.
(216) Various embodiments are described herein. It should be noted that the specific embodiments are not intended as an exhaustive description or as a limitation to the broader aspects discussed herein. One aspect described in conjunction with a particular embodiment is not necessarily limited to that embodiment and can be practiced with any other embodiment(s). Reference throughout this specification to "one embodiment," "an embodiment," "an example embodiment," or other similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention described herein. Thus, appearances of the phrases "in one embodiment," "in an embodiment," "an example embodiment," or other similar language in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiment(s), as would be apparent to a person having ordinary skill in the art and the benefit of this disclosure. Furthermore, while some embodiments described herein include some, but not other, features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention. For example, in the appended claims, any of the claimed embodiments can be used in any combination.
(217) The following examples are presented to illustrate the present disclosure. The examples are not intended to be limiting in any manner.
(218) Example 1 is a wearable system, comprising: at least one acoustic sensor system configured to transmit at least one acoustic signal, to receive at least one reflected acoustic signal, and to output the at least one transmitted acoustic signal and the at least one received reflected acoustic signal; a processor that receives the at least one transmitted acoustic signal and the at least one received reflected acoustic signal from each of the at least one acoustic sensor system; and a physical memory, the physical memory comprising instructions that when executed by the processor cause the processor to: calculate a profile associated with the at least one transmitted acoustic signal and the at least one received reflected acoustic signal; assign a deformation to the profile, the assigned deformation having a predetermined degree of correspondence to a selected one of a plurality of deformations in a model when compared to the profile; and communicate a facial output based on the assigned deformation.
(219) Example 2 includes the subject matter of Example 1, wherein the at least one acoustic sensor system comprises at least one speaker and at least one microphone.
(220) Example 3 includes the subject matter of Examples 1 and 2, wherein the at least one acoustic sensor system comprises a Bluetooth low energy ("BLE") module to output the at least one transmitted acoustic signal and the at least one received reflected acoustic signal.

(221) Example 4 includes the subject matter of any of Examples 1-3, wherein the at least one transmitted acoustic signal is a frequency modulation continuous wave ("FMCW") signal.
(222) Example 5 includes the subject matter of any of Examples 1-4, wherein the profile is a differential echo profile.
(223) Example 6 includes the subject matter of any of Examples 1-5, wherein the deformation is a skin deformation.
(224) Example 7 includes the subject matter of any of Examples 1-6, wherein the communicated facial output comprises one of a facial movement, an avatar movement associated with the facial movement, a speech recognition, text associated with the speech recognition, a physical activity, an emotional status, a two-dimensional visualization of a facial expression, a three-dimensional visualization of a facial expression, an emoji associated with a facial expression, or an avatar image associated with a facial expression.
(225) Example 8 includes the subject matter of any of Examples 1-7, the at least one acoustic sensor system positioned on the wearable system to transmit acoustic signals towards a first facial feature on a first side of a sagittal plane of a wearer and to transmit acoustic signals towards a second facial feature on a second side of the sagittal plane of a wearer.
(226) Example 9 includes the subject matter of any of Examples 1-8, the at least one acoustic sensor system positioned on the wearable system to transmit acoustic signals towards an underside contour of a chin of a wearer.
(227) Example 10 includes the subject matter of any of Examples 1-9, the wearable system comprising ear buds, ear pods, in-the-ear (ITE) headphones, over-the-ear headphones, or outside-the-ear (OTE) headphones to which the at least one acoustic sensor system is attached.
(228) Example 11 includes the subject matter of any of Examples 1-10, the wearable system comprising glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, or another head-borne device to which the at least one acoustic sensor system is attached.
(229) Example 12 includes the subject matter of any of Examples 1-11, the wearable system comprising a necklace, a neckband, or a garment-attachable system to which the at least one acoustic sensor system is attached.
(230) Example 13 includes the subject matter of any of Examples 1-12, further comprising a computing device that receives and displays the communicated facial output.
(231) Example 14 includes the subject matter of any of Examples 1-13, wherein the model is trained using machine learning.
(232) Example 15 includes the subject matter of any of Examples 1-14, the training comprising: receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a deformation of a plurality of deformations of the subject; receiving one or more transmitted acoustic signal(s) and one or more corresponding reflected acoustic signal(s) associated with the subject, each of the one or more transmitted acoustic signal(s) and the one or more corresponding reflected acoustic signal(s) from the at least one acoustic sensor system also corresponding to a deformation of the plurality of deformations of the subject; and correlating, for each of the deformations, the one or more transmitted acoustic signal(s) and the one or more corresponding reflected acoustic signal(s) from the at least one acoustic sensor system corresponding to a particular deformation to the one or more frontal view facial image(s) corresponding to the particular deformation.

Claims

1. A wearable system, comprising: at least one imaging sensor configured to capture an image of a facial feature of a wearer of the wearable system and to output image data corresponding to the image; a processor that receives the image data from each of the at least one imaging sensor; and a physical memory, the physical memory comprising instructions that when executed by the processor cause the processor to: compare the image data from each of the at least one imaging sensor to a model; assign a contour to the image data, the assigned contour having a predetermined degree of correspondence to a selected one of a plurality of contours in the model when compared to the image data; and communicate a facial expression image based on the assigned contour; wherein the model is trained using machine learning; and wherein the training comprises: receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions of the subject; receiving one or more image(s) of the subject from the at least one imaging sensor, each of the images from the at least one imaging sensor also corresponding to a facial expression of the plurality of facial expressions of the subject; and correlating, for each of the facial expressions, the one or more image(s) from the at least one imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.
2. The wearable system of claim 1, wherein the at least one imaging sensor comprises an acoustic camera or an acoustic sensor configured to: transmit signals towards a wearer's face; and detect reflected signals from the wearer's face, wherein the transmitted signals are reflected differently based on different facial expressions or movements of the wearer.
3. The wearable system of claim 1, wherein the communicated facial expression image comprises one of a two-dimensional visualization of the facial expression image, a three-dimensional visualization of the facial expression image, an emoji associated with the facial expression image, or an avatar image associated with the facial expression image.
4. The wearable system of claim 1, wherein comparing the image data from each of the at least one imaging sensor to the model comprises processing the image data to at least one derivative image data set and comparing the at least one derivative image data set to the model.
5. The wearable system of claim 1, the at least one imaging sensor positioned on the wearable system to capture image data of a first facial feature on a first side of a sagittal plane of the wearer and to capture image data of a second facial feature on a second side of the sagittal plane of the wearer.
6. The wearable system of claim 1, the at least one imaging sensor positioned on the wearable system to capture image data of an underside contour of a chin of the wearer.
7. The wearable system of claim 1, wherein correlating for each of the facial expressions comprises: extracting a plurality of landmark positions from each of the one or more frontal view facial image(s) of the subject; extracting a plurality of landmark positions from each of the one or more image(s) of the subject from the at least one imaging sensor; correlating the landmark positions for a particular frontal view facial image with a particular facial expression; and correlating the landmark positions for a particular one of the images of the subject from the at least one imaging sensor with the particular facial expression.
8. The wearable system of claim 1, the at least one imaging sensor comprising a first imaging sensor and a second imaging sensor, the first imaging sensor positioned on the wearable system to capture first image data of a first facial feature of the wearer, and the second imaging sensor positioned on the wearable system to capture second image data of a second facial feature of the wearer, wherein comparing the image data from each of the at least one imaging sensor to the model comprises comparing the first image data of the first facial feature of the wearer to the model and comparing the second image data of the second facial feature of the wearer to the model, wherein assigning a contour to the image data comprises assigning a first contour to the first image data and a second contour to the second image data, and wherein the facial expression image being based on the assigned first contour and the assigned second contour.
9. The wearable system of claim 8, the first imaging sensor positioned on the wearable system to capture image data of the first facial feature on a first side of a sagittal plane of the wearer, and the second imaging sensor positioned on the wearable system to capture an image of the second facial feature on a second side of the sagittal plane of the wearer.
10. The wearable system of claim 8, the first facial feature comprising a left side contour of a face of the wearer, and the second facial feature comprising a right side contour of the face of the wearer.
11. The wearable system of claim 8, the first facial feature comprising a left underside contour of a chin of the wearer, and the second facial feature comprising a right underside contour of the chin of the wearer.
12. The wearable system of claim 1, the wearable system comprising ear buds, ear pods, in-the-ear (ITE) headphones, over-the-ear headphones, or

outside-the-ear (OTE) headphones to which the at least one imaging sensor is attached.
13. The wearable system of claim 1, the wearable system comprising glasses, smart glasses, a visor, a hat, a helmet, headgear, a virtual reality headset, or another head-borne device to which the at least one imaging sensor is attached.
14. The wearable system of claim 1, the wearable system comprising a necklace, a neckband, or a garment-attachable system to which the at least one imaging sensor is attached.
15. The wearable system of claim 1, the physical memory comprising further instructions that when executed by the processor cause the processor to: compare the assigned contour to a second model; and assign the facial expression image based on the assigned contour, the assigned facial expression image having a predetermined degree of correspondence to a selected one of a plurality of contours in the second model when compared to the assigned contour.
16. The wearable system of claim 15, wherein the second model is trained using machine learning.
17. The wearable system of claim 16, wherein the training of the second model comprises: receiving one or more frontal view facial image(s) of a subject, each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions of the subject; receiving one or more image(s) of the subject from the at least one imaging sensor, each of the images from the at least one imaging sensor also corresponding to a facial expression of the plurality of facial expressions of the subject; and correlating, for each of the facial expressions, the one or more image(s) from the at least one imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.
18. The wearable system of claim 17, wherein correlating for each of the facial expressions in the training of the second model comprises: extracting a plurality of landmark positions from each of the one or more frontal view facial image(s) of the subject; extracting a plurality of landmark positions from each of the one or more image(s) of the subject from the at least one imaging sensor; correlating the landmark positions for a particular frontal view facial image with a particular facial expression; and correlating the landmark positions for a particular one of the images of the subject from the at least one imaging sensor with the particular facial expression.
19. The wearable system of claim 1, further comprising a computing device that receives and displays the communicated facial expression image.
20. The wearable system of claim 1, wherein the at least one imaging sensor comprises a depth camera, a red-green-blue (“RGB”) camera, an infrared (“IR”) sensor, an acoustic camera, an acoustic sensor, a thermal imaging sensor, a charge-coupled device (“CCD”), a complementary metal oxide semiconductor (“CMOS”) device, an active-pixel sensor, or a radar imaging sensor.
21. The wearable system of claim 1, the physical memory comprising further instructions that when executed by the processor cause the processor to: compare the facial expression image to a second model; assign at least one of a letter or a word to the facial expression image based on a predetermined degree of correspondence of the assigned at least one of a letter or a word to a selected one of a plurality of letters or words in the second model when compared to the facial expression image, and output text or speech corresponding to the assigned at least one of a letter or a word.
22. The wearable system of claim 1, wherein the captured image further captures an object in a vicinity of the captured facial feature of the wearer; and wherein the predetermined degree of correspondence of the image data to the selected one of the contours in the model includes correspondence based on the captured object in the image data.
23. A method to determine a facial expression using a wearable system, comprising: positioning a first imaging sensor of a wearable system to image a first facial feature of a wearer; imaging the first facial feature of the wearer using the first imaging sensor; comparing, via a processor, the imaged first facial feature to a first model; and assigning, via the processor, a facial expression corresponding to the imaged first facial feature selected from one of a plurality of facial expressions in the first model; wherein the first model is trained using machine learning; and wherein the training comprises: receiving one or more frontal view facial image(s), each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions; receiving one or more image(s) from the first imaging sensor, each of the images from the first imaging sensor also corresponding to a facial expression of the plurality of facial expressions; and correlating, for each of the facial expressions, the one or more image(s) from the first imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.
24. The method of claim 23, wherein the act of positioning the first imaging sensor of the wearable system comprises positioning the first imaging sensor in a vicinity of an ear of the wearer facing forward to include within a field of view at least one of a buccal region, a zygomatic region, and/or a temporal region of one side of the wearer's face.
25. A method to determine a facial expression using a wearable system, comprising: positioning a first imaging sensor of a wearable system to image a first facial feature of a wearer; positioning a second imaging sensor of the wearable system to image a second facial feature of the wearer; imaging the first facial feature of the wearer using the first imaging sensor; imaging the second facial feature of the wearer using the second imaging sensor; comparing, via a processor, the imaged first facial feature and the imaged second facial feature to a model; and assigning, via the processor, a facial expression selected from one of a plurality of facial expressions in the model corresponding to the imaged first facial feature and the imaged second facial feature; wherein the model is trained using machine learning; and wherein the training comprises: receiving one or more frontal view facial image(s), each of the frontal view facial images corresponding to a facial expression of a plurality of facial expressions; receiving one or more image(s) from at least one of the first imaging sensor and the second imaging sensor, each of the images from the at least one of the first imaging sensor and the second imaging sensor also corresponding to a facial expression of the plurality of facial expressions; and correlating, for each of the facial expressions, the one or more image(s) from the at least one of the first imaging sensor and the second imaging sensor corresponding to a particular facial expression to the one or more frontal view facial image(s) corresponding to the particular facial expression.